

Chapter 7 Series FACTS Controllers: GCSC, TSSC, TCSC and SSSC

Learning Objectives

After working through this chapter, the reader should be able to:

- derive current, transmitted power, capacitor duty and the distinction between the electrical midpoint and physical capacitor terminals, and compare series and shunt compensation under stated operating conditions;
- quantify the influence of series compensation on voltage stability, transient stability and power oscillation damping, and compute the subsynchronous resonance frequencies it introduces;
- derive the capacitor voltage, fundamental reactance, RMS voltage and harmonics of the GTO-controlled series capacitor and exploit its duality with the TCR;
- explain the insertion and bypass rules of the thyristor-switched series capacitor and design stepped banks for voltage or impedance compensation;
- analyse the thyristor-controlled series capacitor with both the simplified parallel-LC model and the exact line-current-driven model, explain why they differ, and use the exact closed-form reactance law for design;
- derive the SSSC power-angle characteristic, distinguish commanded voltage from current-relative capacitive or inductive operation, and explain energy, synchronization and subsynchronous-interaction limits;
- compare the four controllers on range, response, harmonics, SSR behaviour, losses and cost, and place them in the current research landscape.

Table 7.1 summarizes the four controllers treated in this chapter, together with the fixed series capacitor from which they all derive.

Conventions. Network phasors are RMS fundamental quantities. $X_C > 0$ denotes a capacitor-reactance magnitude, whereas a signed device reactance is negative in capacitive operation. Waveform equations state explicitly where a current is a peak value. A per-phase power becomes a balanced three-phase power on multiplication by 3, and per-unit formulas use one common three-phase base. Angles inside derivatives and integrals are in radians even when results are quoted in degrees. Sections 7.1 to 7.3 and 7.5 form the core of the chapter; the exact TCSC waveform, the impedance-based stability tests and the delayed-damping models are intended for advanced and research study.

Table 7.1 Series FACTS controllers at a glance.

Device	Power stage	Control principle	Fundamental output / range	Characteristic behaviour
Fixed series capacitor	Capacitor bank with MOV and bypass	None (switched by breaker)	Fixed $-jX_C$	Raises transfer; introduces SSR risk
GCSC	Capacitor + antiparallel GTO/IGCT valve	Turn-off delay γ from current peak	Continuous capacitive magnitude $0 \leq X_C(\gamma) \leq X_C$	Dual of the TCR; harmonics generated
TSSC	Capacitor stages with thyristor bypass valves	Insert at current zero, bypass at voltage zero	Stepped capacitive	No harmonics; offset transients; SSR as a capacitor
TCSC	Capacitor in parallel with thyristor-controlled reactor	Conduction (firing) angle	Continuous capacitive boost and inductive vernier	Subsynchronous behavior depends on firing/control and network; widely applied
SSSC	VSC behind series transformer	Magnitude (and angle) of injected voltage	$\pm V_q$ independent of current	No literal fixed-capacitor resonance; either voltage polarity; active exchange with DC source/sink

7.1 Fundamentals of Series Compensation

7.1.1 Two ways of looking at a series capacitor

Consider the balanced, lossless two-machine system of Figure 7.1 with equal RMS phase-voltage magnitudes $V_s = V_r = V$, total line reactance X per phase and a central series capacitor of positive reactance magnitude X_C . Shunt charging is neglected. The ideal formulas below assume $0 \leq k < 1$ and do not establish a thermal or stability rating.

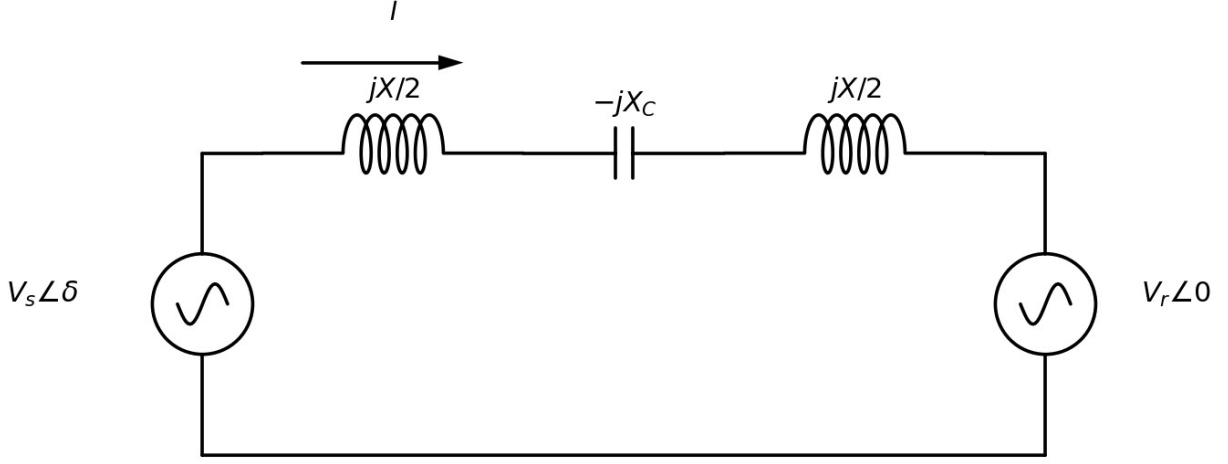


Figure 7.1. Two-machine system with a series capacitor at the centre of the line.

The conventional explanation is that the capacitor cancels part of the line reactance, so the line appears electrically shorter. Defining the *degree of series compensation*

$$k = \frac{X_C}{X}, 0 \leq k < 1 \quad (7.1)$$

where k is dimensionless and $X_C = 1/(\omega C) > 0$. The capacitor impedance is $-jX_C$, so the effective series reactance is

$$X_{eff} = X - X_C = (1 - k)X \quad (7.2)$$

where X_{eff} is the net series reactance. The second explanation, which proves more useful once converters are involved, looks at voltages rather than impedances. To push a larger current through the fixed line reactance X , a larger voltage must appear across it. The capacitor supplies that extra voltage: at a given current it develops a voltage $-jX_C I$ that is opposite in phase to the drop $jX I$, so the voltage available across the line inductance is the sum of the end-voltage difference and the capacitor voltage. Figure 7.2 shows this. Once compensation is seen as a series *voltage* in quadrature with the line current, the physical nature of the element that produces it no longer matters, which is exactly the step from the capacitor to the SSSC.

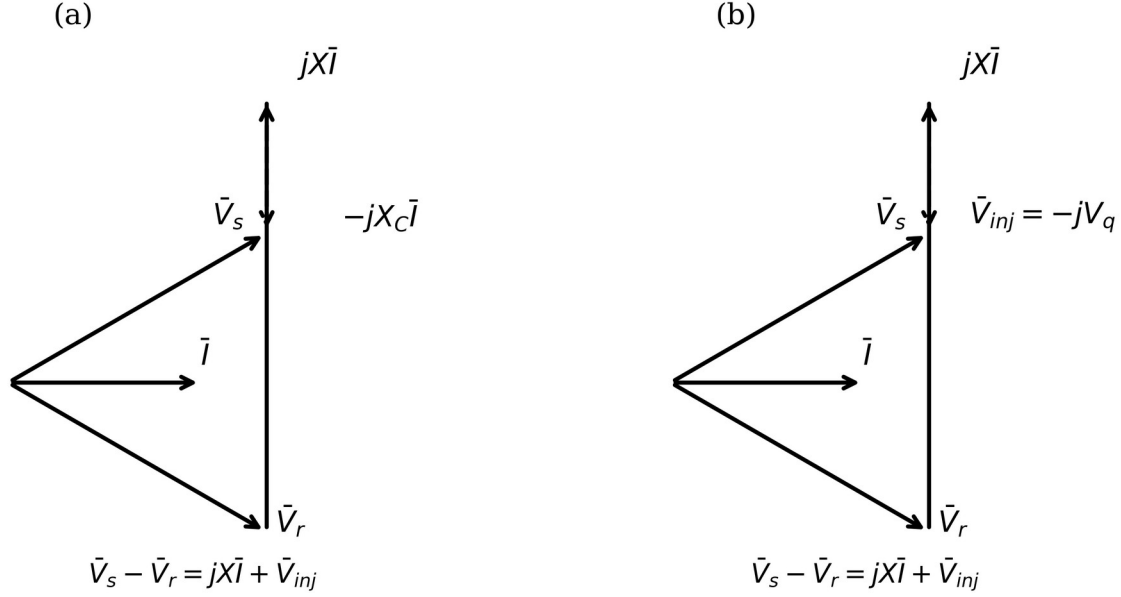


Figure 7.2. Symmetric-reference phasors with current directed from sending to receiving end. The capacitor or SSSC drop satisfies $V_s - V_r = jXI + V_{inj}$. Current arrows use a separate scale; capacitive V_{inj} points downward for forward current.

7.1.2 Current, power and reactive power of the compensated line

Rotate the phasor reference so that $\underline{V}_s = V \angle (\delta/2)$ and $\underline{V}_r = V \angle (-\delta/2)$. For forward transfer with $0 \leq \delta < \pi$, the current is real on this reference axis. The RMS current, voltage of the symmetric electrical midpoint and per-phase transmitted power are

$$I = \frac{2V}{(1-k)X} \sin \frac{\delta}{2}, V_m = V \cos \frac{\delta}{2}, P = \frac{V^2}{(1-k)X} \sin \delta \quad (7.3)$$

Here I is the RMS line-current magnitude. The reference $V_m = V \cos(\delta/2)$ is the midpoint of the **net electrical impedance**, not either physical capacitor terminal. The power-angle curve scales by $1/(1-k)$; $k=1/3$ raises its ideal maximum by 50%, and $k=1/2$ doubles it. At the central capacitor, $\underline{V}_a = \underline{V}_s - j(X/2)\underline{I}$ and $\underline{V}_b = \underline{V}_r + j(X/2)\underline{I}$, hence $\underline{V}_a - \underline{V}_b = -jX_C \underline{I}$. Their average is V_m , but the two terminals have different phasors and can have higher magnitudes. Example 7.16 illustrates the physical voltage stress. The capacitor supplies per-phase reactive power

$$Q_C = I^2 X_C = \frac{2V^2}{X} \cdot \frac{k}{(1-k)^2} (1 - \cos \delta) \quad (7.4)$$

where Q_C is the capacitor's reactive output. Figure 7.3 plots both quantities. The reactive power grows rapidly with k and with δ , because it scales with the square of a current that itself rises with compensation. This sets the economic limit of series compensation, together with the fault-current, insulation and SSR limits discussed below. Examples 7.13 to 7.15 work through these relations numerically.

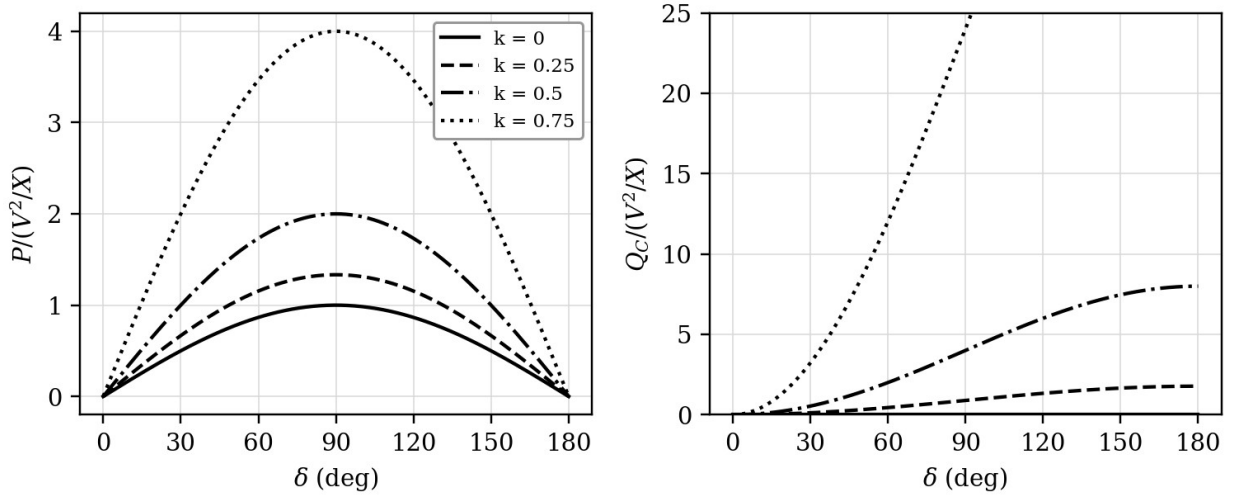


Figure 7.3. Transmitted power (left) and series-capacitor reactive power (right) versus angle for several degrees of compensation.

Series and shunt ratings must be compared for the same transfer objective, angle, voltage and device limits. A series capacitor changes the transfer reactance and supplies $I^2 X_C$ per phase, whereas a shunt device changes the voltage profile by supplying bus current. In the ideal equal-end-voltage model, series compensation achieves a given transfer increase at a moderate angle with less installed var duty than an equally constrained midpoint shunt scheme (Example 7.19). Voltage support, losses, congestion, protection and contingencies can still favour a different solution in a particular network.

7.1.3 Voltage stability

For a radial lossless line feeding a constant-power load, reducing the series reactance moves the nose of the static P-V curve outward (Figure 7.4). With sending voltage E , receiving voltage V_r and load $Q = P \tan \phi$, the power-flow equation is $E^2 V_r^2 = (V_r^2 + X_{eff} Q)^2 + (X_{eff} P)^2$, and the two voltage branches merge where its discriminant vanishes. The capacitor output rises as $I^2 X_C$, which helps exactly when it is needed, but only within the capacitor, MOV, line-current and protection limits. Load recovery and motor stalling are dynamic phenomena, so a larger static nose has to be confirmed by a dynamic voltage-stability study.

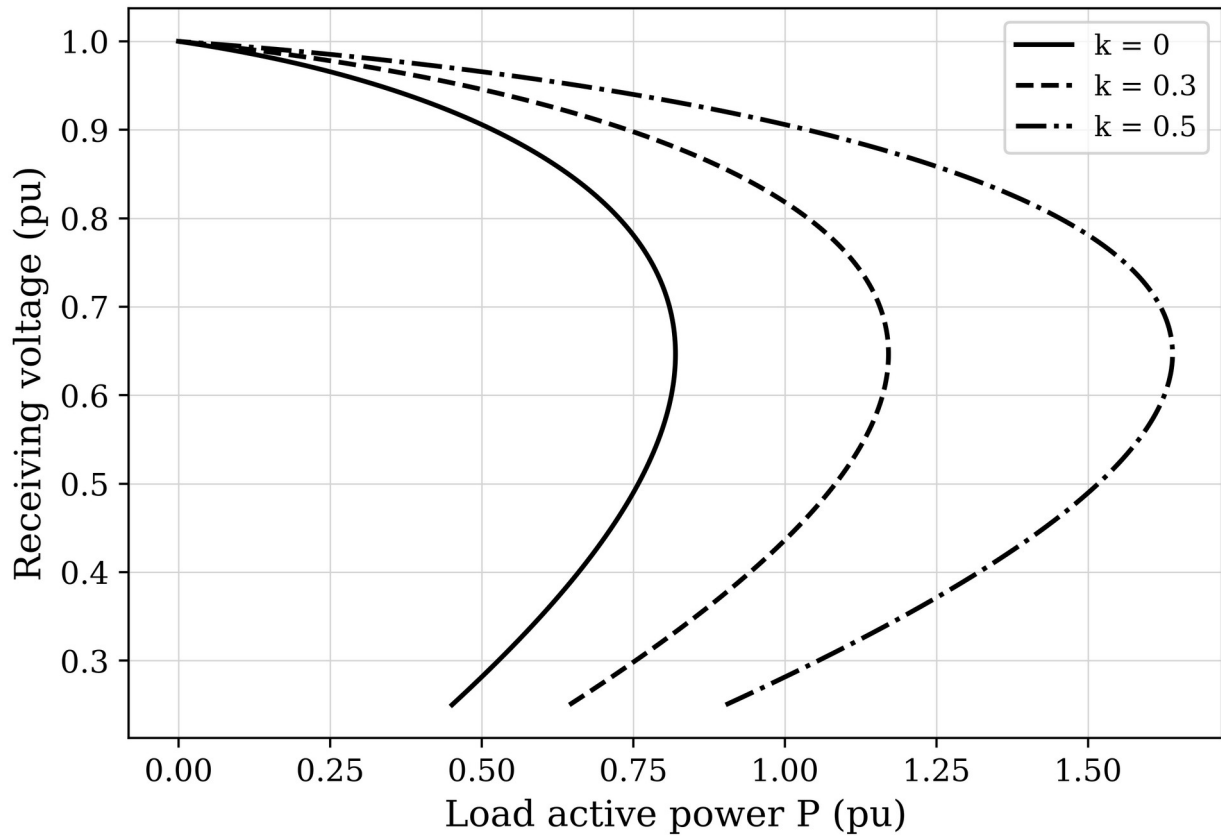


Figure 7.4. Static radial P–V curves from $E^2 V_r^2 = (V_r^2 + X_{eff} Q)^2 + (X_{eff} P)^2$, with $E = 1$ pu, uncompensated $X = 0.5$ pu and $Q/P = 0.2$. Both solution branches are shown; this is a static model.

Example 7.1. Effect of series compensation on steady-state transfer capability

A lossless 220 kV corridor is represented by a series reactance of $40 \, \Omega$ per phase. Both end voltages are 220 kV line-to-line and the transmission angle is 30° . Determine the transmitted power before compensation and with 40% series compensation.

Solution. With line-to-line voltage, the three-phase power of a lossless line is $P = V_{LL}^2 \sin \delta / X$. Without compensation, $P = 220^2 \times 0.5 / 40 = 605$ MW. With $k = 0.40$, $X_{eff} = 40(1 - 0.40) = 24 \, \Omega$ and $P = 220^2 \times 0.5 / 24 = 1008.3$ MW. The same angle now carries 66.7% more power, exactly the factor $1/(1 - k)$.

Note. The result shows the leverage of series compensation, not a permissible transfer: thermal limits, transient stability, voltage profile, protection duty and subsynchronous interaction must still be checked.

7.1.4 Transient stability and power oscillation damping

Raising the power-angle curve does two things in a transient. It lowers the pre-fault angle for the same mechanical power, and it enlarges the decelerating area available after clearing. Figure 7.5 illustrates the equal-area construction for an uncompensated line and for the same line with $k = 1/3$. For a fault that reduces the transfer capability to 0.4 pu, the critical clearing angle rises from about 91° to about 109° (Example 7.18). Controlled series compensation goes further. By switching to maximum compensation immediately after the fault, the controller raises the post-fault curve exactly when it matters, and then returns to the steady-state level once the first swing is over.

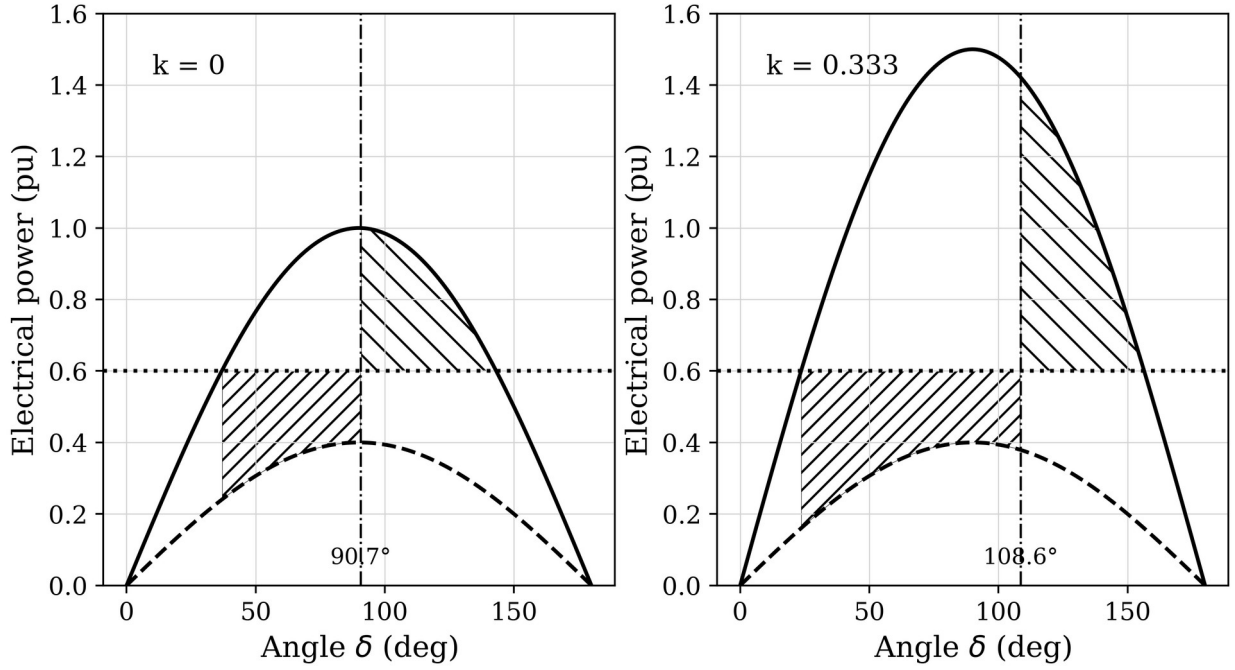


Figure 7.5. Equal-area constructions for Example 7.18: $P_m = 0.6$ pu, fault maximum 0.4 pu and post-fault maxima 1.0 and 1.5 pu. Forward and backward hatching distinguish acceleration and deceleration areas.

Power-oscillation damping modulates electrical power with the proper modal phase. In $M d \Delta \omega / dt = \Delta P_m - \Delta P_e$, a positive component $\Delta P_e = K_D \Delta \omega$, $K_D > 0$, removes oscillatory kinetic energy. Positive speed deviation means angle is increasing relative to the reference, not necessarily that the machine is accelerating. A component proportional only to angle changes synchronizing stiffness. The controller therefore includes washout, phase compensation and limits; the sign depends on the measured signal and network sensitivity. Section 7.4.8 gives a reproducible ideal SMIB illustration.

7.1.5 Subsynchronous resonance

A series capacitor forms a resonant circuit with the total inductance of the network it is connected to, which includes generators, transformers and lines. If that total reactance at system frequency f_0 is X , the electrical natural frequency is

$$f_e = \frac{1}{2\pi\sqrt{LC}} = f_0 \sqrt{\frac{X_C}{X}} = f_0 \sqrt{k} \quad (7.5)$$

Here f_e is the natural frequency of the ideal series LC circuit, using the **total** inductance of the resonant path. A stator component at $f_e < f_0$ produces a rotor-frame component near $f_0 - f_e$ for a rotor near synchronous speed. Coincidence with a turbine-generator torsional mode identifies a possible interaction, not proof of instability: modal coupling and net damping determine growth. The Mohave shaft failures in 1970 and 1971 brought this mechanism to engineering attention [10], [11]. The induction-generator effect and transient torque amplification are related risks that can occur without the same torsional coincidence.

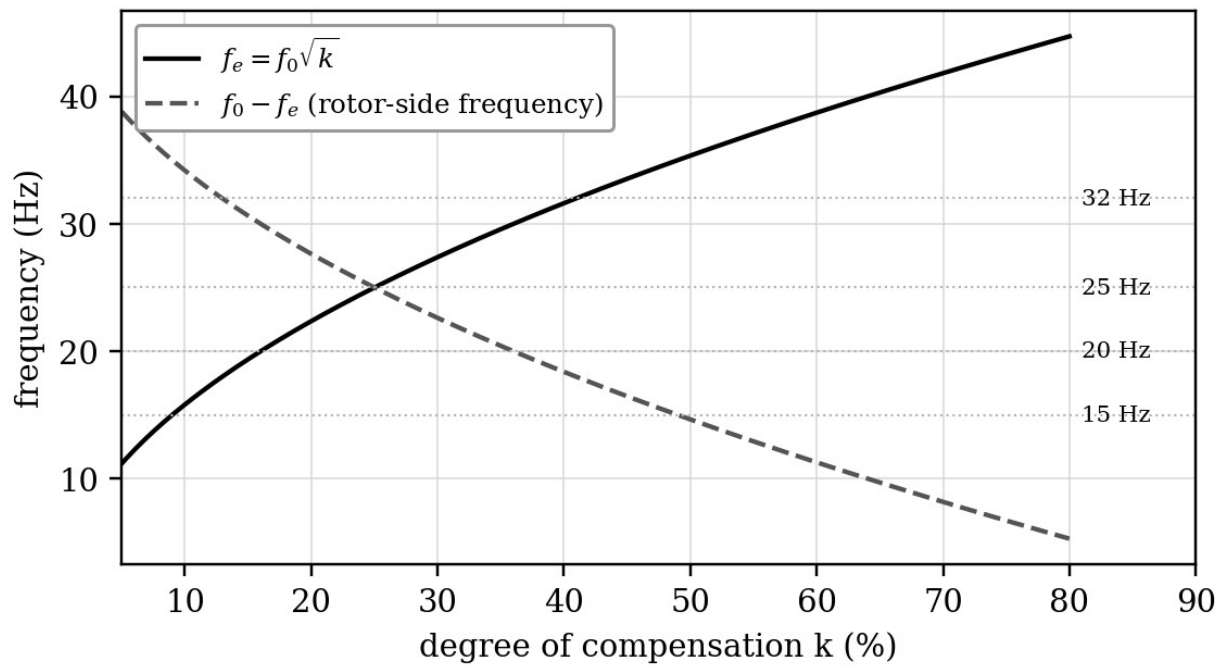


Figure 7.6. Electrical resonant frequency and the complementary rotor frequency versus degree of compensation for a 50 Hz system, with typical torsional mode frequencies.

Fixed compensation near turbine-generators is chosen from frequency scans, coupled torsional studies and contingency cases rather than from a fixed percentage. A TCSC can reshape its subsynchronous impedance through suitable firing control, and an SSSC produces its fundamental injection without a physical series capacitor, but both still require SSR, SSCI and control-interaction studies. Examples 7.17 and 7.30 show the frequency screening that starts such a study.

7.1.6 Practical limits and protection of series capacitors

An inserted capacitor carries line current and can experience severe fault overvoltage. Figure 7.7 shows a typical coordinated arrangement: a zinc-oxide MOV clamps voltage and absorbs energy; a triggered gap with a damping reactor, when provided, protects the MOV against excessive duty; and a bypass breaker provides sustained bypass. The protective level and energy limits are project-specific. The platform is insulated for its line-to-ground potential, while each segment has its own differential-voltage duty. Control and measurement signals use optical isolation; auxiliary supplies, valve cooling and insulation require their own engineered isolation arrangements.

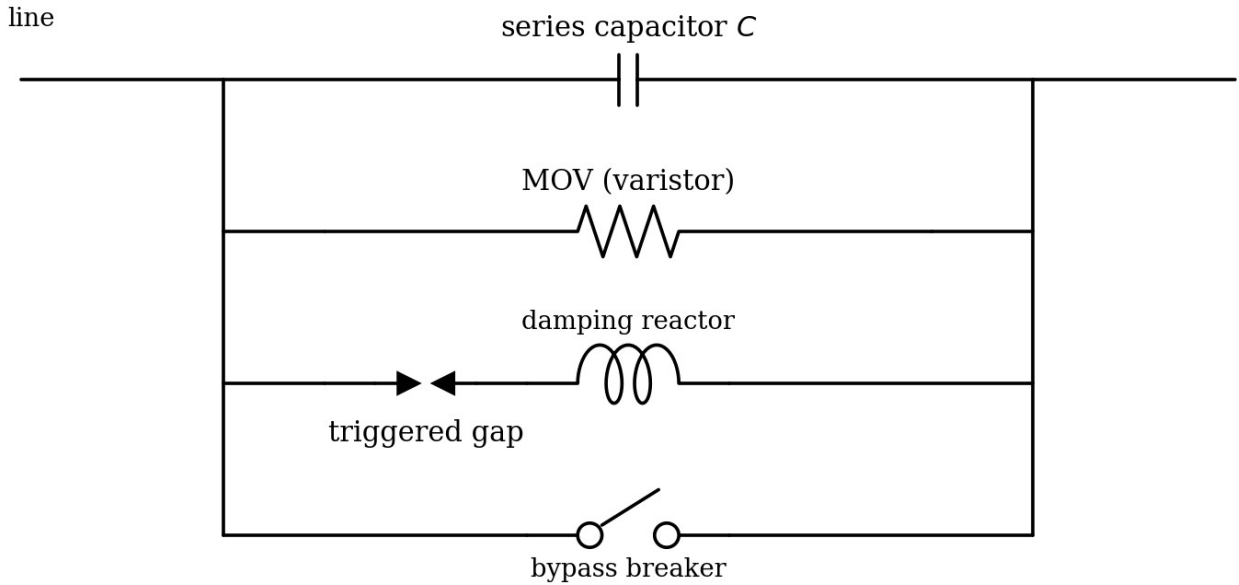


Figure 7.7. Protection of a series-capacitor segment: MOV, triggered gap with damping reactor, and bypass breaker.

Higher compensation reduces the net reactance and can raise the current, the local capacitor-terminal voltage and the sensitivity to outages. It also affects distance protection and subsynchronous interactions. Installations at 70% to 75% are demanding designs rather than general limits; the permissible range follows from the complete network, equipment and protection study.

7.1.7 Development of controlled series compensation: a short literature survey

Controlled series compensation developed through the FACTS work of the late 1980s. Vithayathil and co-workers reported conventional and novel reactive-control case studies in 1988 [1]. Kayenta (230 kV, 1992) and Slatt (500 kV, 1993) were early US applications [5], [9]. Later projects addressed SSR mitigation, inter-area damping and interconnection transfer; Table 7.3 provides traceable historical examples. Hingorani's NGH damper [2] illustrates how controlled dissipation can suppress subsynchronous components. Karady and co-workers analyzed the GCSC [3], and Gyugyi, Schauder and Sen developed the voltage-source SSSC treatment [4]. The texts [5]–[8] and TCSC studies [9] support the analytical progression of this chapter. Table 7.2 collects the angle reference used for each device in the rest of the chapter.

Table 7.2 Angle references used in this chapter (radians in derivations, degrees in numerical results).

Device/model	Reference	Controlled interval
GCSC	γ after line-current crest	Capacitor inserted γ to $\pi - \gamma$
TSSC	Natural valve-current zero	Whole stages; no continuous firing law
Approximate/exact TCSC	β half conduction; $\alpha = \pi/2 - \beta$	Pulse centered on voltage-zero/line-current-crest reference
Chapter 6 TCR	Delay usually from voltage zero	Convert reference before reusing a law

7.2 The GTO-Thyristor Controlled Series Capacitor (GCSC)

7.2.1 Circuit and principle

The GCSC (Figure 7.8) places a bidirectional, turn-off-capable valve across a series capacitor. The drawing represents two ideal **reverse-blocking turn-off valves**; it is not a direct substitution diagram for every GTO, IGCT or IGBT. An ordinary IGBT with an antiparallel diode does not block reverse voltage, so an IGBT realization needs a bidirectional blocking arrangement, for example anti-series IGBT devices with their diode

paths and appropriate voltage ratings. When the valve conducts it bypasses the capacitor; when it opens the line current charges the capacitor. Control adjusts its fundamental voltage and hence its effective capacitive reactance.

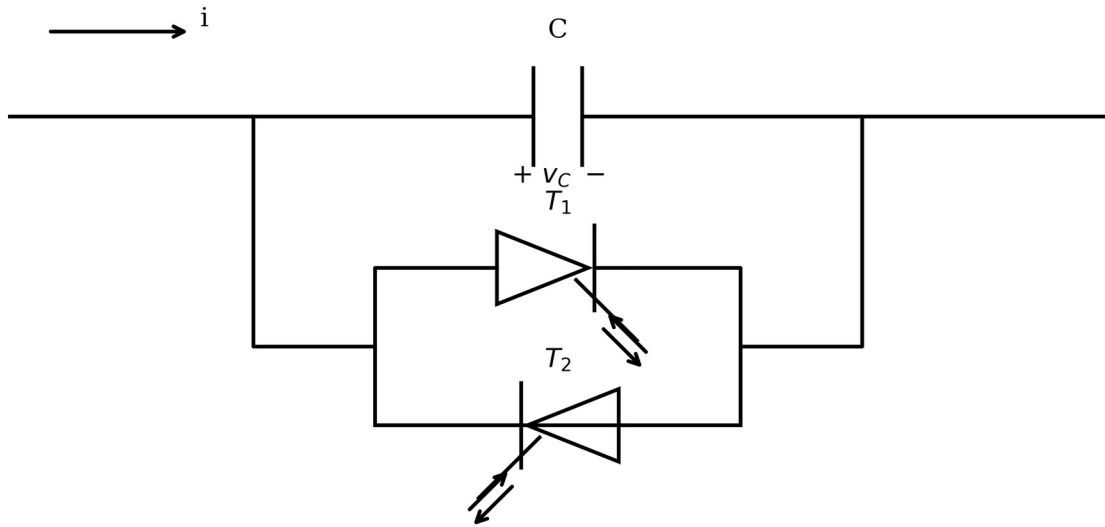


Figure 7.8. GCSC module with two ideal reverse-blocking turn-off valves. Gate arrows indicate commanded turn-on and turn-off. Ordinary diode-equipped IGBTs require a suitable bidirectional blocking circuit.

Let the line current be $i = I \cos \omega t$, so that its crests occur at $\omega t = 0, \pi, 2\pi, \dots$. The valve is opened (turned off) at a delay angle γ after a current crest. From that instant the capacitor integrates the current. The voltage rises to a maximum at the next current zero and returns to zero at $\omega t = \pi - \gamma$, symmetrically placed about the current zero. At that instant the valve is forward biased with zero voltage across it, and it turns on again to keep the capacitor bypassed until γ after the next crest. The cycle repeats with opposite polarity. Figure 7.9 shows the result for three delay angles.

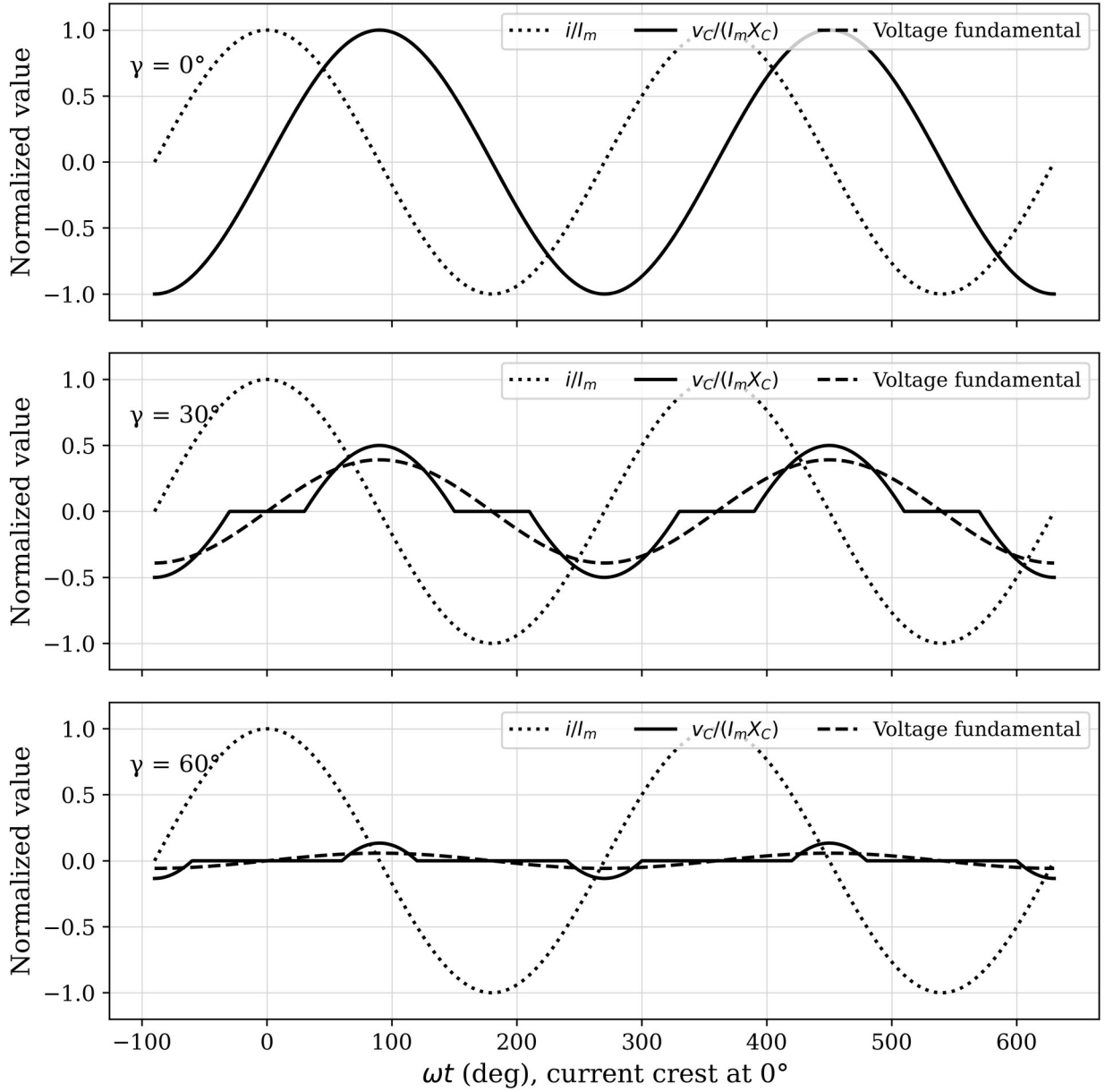


Figure 7.9. Ideal GCSC waveforms at $\gamma = 0^\circ$, 30° and 60° , measured after a line-current crest. Bases are I_m and $I_m X_C$; the dashed voltage is its Fourier fundamental.

7.2.2 Capacitor voltage

During the insertion interval of the positive half-cycle,

$$v_C(t) = \frac{1}{C} \int_{\gamma/\omega}^t I \cos \omega \tau d\tau = \frac{I}{\omega C} (\sin \omega t - \sin \gamma), \gamma \leq \omega t \leq \pi - \gamma \quad (7.6)$$

where v_C is the capacitor voltage, I the peak line current, C the capacitance and γ the turn-off delay measured from the current crest. Outside this interval $v_C = 0$, and the negative half-cycle is the mirror image. The term $(I/\omega C) \sin \gamma$ is a γ -dependent offset that shifts the sinusoid obtained at $\gamma = 0$ downward, so that the voltage starts and ends at zero. The peak capacitor voltage, at the current zero, is $(I/\omega C)(1 - \sin \gamma)$.

7.2.3 Fundamental reactance and the duality with the TCR

The ideal GCSC and TCR are duals: a current-driven capacitor controlled by valve opening corresponds to a voltage-driven reactor controlled by valve closing. Natural extinction occurs at capacitor-voltage zero in the GCSC bypass transition and at branch-current zero in the TCR. The TCR delay may be measured from voltage zero or crest; this chapter measures GCSC γ from a **line-current crest**, so numerical angle labels

cannot simply be transferred. For the fundamental, $V_{CF} = (2/\pi) \int_{\gamma}^{\pi-\gamma} v_C(\theta) \sin \theta d\theta$. Substituting Equation (7.6) gives

$$V_{CF}(\gamma) = \frac{I}{\omega C} \left(1 - \frac{2\gamma}{\pi} - \frac{\sin 2\gamma}{\pi} \right) \quad (7.7)$$

where V_{CF} is the peak fundamental capacitor voltage. Its ratio to the current defines the effective capacitive reactance

$$X_C(\gamma) = \frac{V_{CF}}{I} = X_C \left(1 - \frac{2\gamma}{\pi} - \frac{\sin 2\gamma}{\pi} \right), X_C = \frac{1}{\omega C} \quad (7.8)$$

where $X_C(\gamma)$ is the fundamental series reactance and X_C the reactance of the bare capacitor. The capacitor is fully inserted at $\gamma=0$ and fully bypassed at $\gamma=\pi/2$. The angle during which the capacitor is in circuit in each half-cycle, the *blocking angle*, is $\xi = \pi - 2\gamma$. Figure 7.10 plots Equation (7.8). As with the TCR, the law is nonlinear and steep near full insertion, so the controller uses an inverse look-up table (Example 7.22).

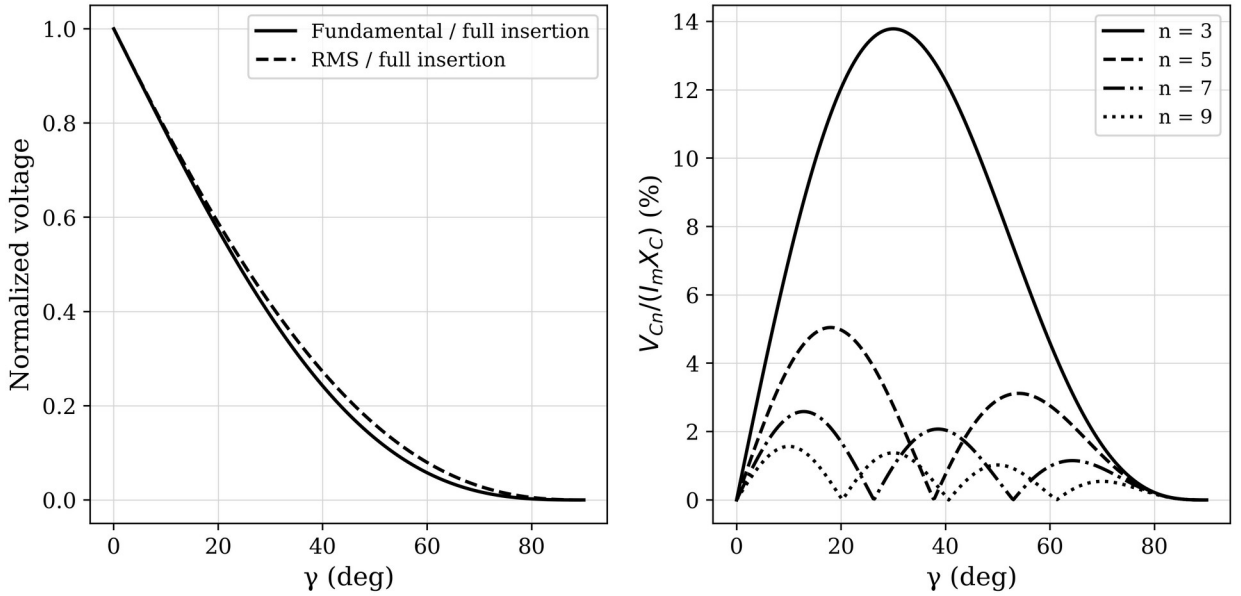


Figure 7.10. GCSC fundamental and total RMS voltage normalized to their full-insertion values (left), and peak harmonic magnitudes on $I_m X_C$ (right). Harmonic percentages and THD use different bases.

The total RMS capacitor voltage and the individual harmonics follow by the same duality. With γ playing the role of the TCR's peak-referenced delay,

$$V_{C,rms} = \frac{I}{\omega C} \sqrt{\frac{(\pi - 2\gamma) \left(\frac{1}{2} + \sin^2 \gamma \right) - \frac{3}{2} \sin 2\gamma}{\pi}} \quad (7.9)$$

$$V_{Cn} = \frac{I}{\omega C} \cdot \frac{4}{\pi} \cdot \frac{\sin \gamma \cos n\gamma - n \cos \gamma \sin n\gamma}{n(n^2 - 1)}, n=3, 5, 7, \dots \quad (7.10)$$

Here $V_{C,rms}$ is the RMS of the complete chopped voltage, and V_{Cn} the peak magnitude of harmonic n .

Equation (7.9) follows from $V_{C,rms}^2 = (I X_C)^2 \pi^{-1} \int_{\gamma}^{\pi-\gamma} (\sin \theta - \sin \gamma)^2 d\theta$. Expanding the square and integrating gives the displayed radicand. The third harmonic reaches about 13.8% of $I X_C$ at $\gamma=30^\circ$. The term is $\sin^2 \gamma$, not $\sin^2(2\gamma)$. The incorrect alternative agrees at the endpoints 0° and 90° and at the only interior coincidence, 60° ; it must not be used as a design equation. Harmonic voltage ratings use peak or RMS magnitudes consistently, and THD uses the actual fundamental, not the fixed base $I X_C$.

7.2.4 Operating modes and characteristics

A GCSC is operated in one of two modes. In the *voltage compensating mode*, the device maintains a rated compensating voltage V_{Cmax} over a range of line currents $I_{min} \leq I \leq I_{max}$. The capacitor is chosen so that the full capacitor produces V_{Cmax} at the minimum current, $X_C = V_{Cmax}/I_{min}$; at higher currents γ is increased to keep the fundamental voltage constant. In the *impedance compensating mode*, the device maintains a chosen reactance at any current up to I_{max} ; the capacitor is then sized for $X_C = V_{Cmax}/I_{max}$, and any reactance between zero and X_C is available. Figure 7.11 shows the two V-I characteristics. Losses are mainly conduction losses of the valve, which carries the line current while the capacitor is bypassed. They are therefore largest at small reactance and high current, which is the opposite of the TCSC (Section 7.4.7).

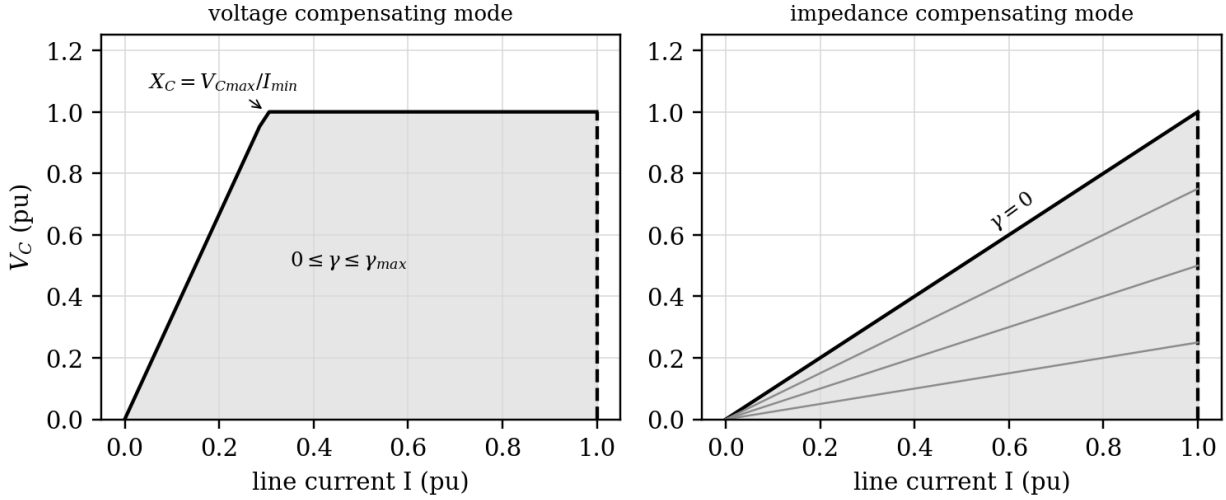


Figure 7.11. GCSC V-I operating areas in voltage compensating mode (left) and impedance compensating mode (right).

Example 7.2. Effective reactance of a GCSC at a given turn-off delay

A GCSC has a fully inserted capacitive reactance of $8 \, \Omega$. Find the effective fundamental reactance at $\gamma = 30^\circ$, measured from the current crest.

Solution. From Equation (7.8) with $\gamma = \pi/6$, the bracket is $1 - 1/3 - \sin(\pi/3)/\pi = 0.3910$, so $X_C(\gamma) = 8 \times 0.3910 = 3.13 \, \Omega$.

Note. The angle reference is part of the equation. A delay measured from a current zero must be converted ($\gamma = \gamma_0 - 90^\circ$) before Equation (7.8) is used.

7.2.5 Harmonics, sequential control and practical aspects

Because a GCSC is inserted directly in the line without a transformer, the multi-pulse cancellation used in converters is not readily available. A small number of publications propose six- and twelve-pulse arrangements through series transformers, but the standard remedy is *sequential control*. The total capacitor is divided into m series modules, each rated $1/m$ of the total voltage. All but one module are either fully inserted or fully bypassed, and only one module is delay-angle controlled. The harmonic voltages, expressed on the total rating, then fall by roughly the factor m , exactly as for the segmented TCR (Example 7.25). Splitting the high-voltage valve into four or more modules is also attractive on its own, since each valve then sees a fraction of the total voltage.

The GCSC is principally encountered in the literature as a conceptual and prototype-level controlled-series-capacitor topology; transmission practice has been dominated by fixed series capacitors, TCSC and, increasingly, converter-based series compensation. Turn-off semiconductors have lower surge capability than thyristors, and the valve must interrupt full line current twice per cycle. A short-circuit or fault current through the valve requires a rapid thyristor or mechanical bypass. The concept remains valuable both for its clean theory, as the dual of the TCR, and in distribution-level series compensators where IGBTs are adequate.

7.2.6 Gate timing, valve stress and practical control

The GCSC control variable is not a conventional thyristor firing delay. The bidirectional turn-off valve conducts while the capacitor is bypassed and is commanded off after the selected current crest. From that instant until the valve turns on again at zero capacitor voltage, the line current charges the capacitor. A practical controller therefore begins with current synchronization, calculates the requested fundamental reactance, converts it into a turn-off delay through the inverse of Equation (7.8), and issues turn-off commands under protection supervision. The current reference must be filtered enough to reject harmonics without introducing a phase error large enough to shift the intended insertion interval.

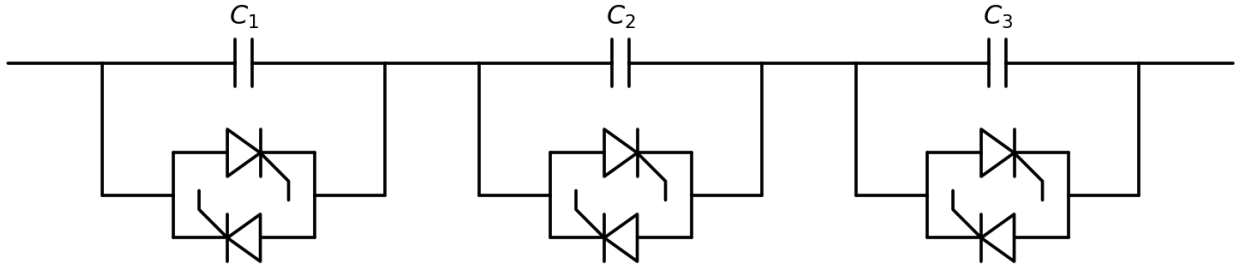
The valve has two complementary duties. During bypass it carries the line current and produces conduction loss. During the open interval it must withstand the capacitor voltage. The blocking-voltage rating is therefore linked to maximum inserted capacitor voltage, whereas current rating is linked to corridor current and fault-bypass duty. Modularization reduces the blocking voltage per valve and enables sequential control in which most modules are either fully inserted or fully bypassed and only one module is partially controlled.

The waveform should be interpreted physically. At full insertion the device behaves as an ordinary series capacitor. As the bypass interval is enlarged, capacitor voltage is clamped to zero for a larger fraction of each half-cycle. The fundamental component decreases continuously, but odd-harmonic voltage changes nonlinearly with delay. Fundamental-reactance control and harmonic-voltage duty must therefore be checked together.

7.3 The Thyristor-Switched Series Capacitor (TSSC)

7.3.1 Circuit and principle

A TSSC (Figure 7.12) consists of several capacitors in series, each shunted by a bypass valve made of antiparallel thyristor strings. The structure resembles the sequentially controlled GCSC, but the valves are ordinary thyristors, which cannot be turned off by the gate. A capacitor is *inserted* by withdrawing the gate pulses of its valve, and the valve then stops conducting at the next current zero. It is *bypassed* by firing the valve. The degree of compensation therefore changes in steps, set by the number of stages in service, and each stage is either fully in or fully out. No partial conduction occurs, so a TSSC generates no harmonics in steady state.



each stage: capacitor + antiparallel thyristor bypass valve

Figure 7.12. TSSC with three stages, each a capacitor with an antiparallel thyristor bypass valve.

7.3.2 Insertion at current zero and the DC offset

Since a thyristor can only commute naturally, a stage is necessarily inserted at a zero crossing of the line current. Taking $i = I_m \sin \omega t$ with a discharged capacitor inserted at $\omega t = 0$,

$$v_C(t) = \frac{1}{C} \int_0^t I_m \sin \omega \tau d\tau = I_m X_C (1 - \cos \omega t) \quad (7.11)$$

Here I_m is peak line current and X_C the stage reactance magnitude. Under the imposed-sinusoidal-current approximation, the discharged capacitor swings from zero to $2I_m X_C$. Its DC offset persists in that ideal model. Leakage can relax trapped charge, while a real network changes the current and creates a switching transient; series resistance alone cannot remove the offset if current is prescribed independently. A capacitor initially at $-I_m X_C$ follows the offset-free waveform in Figure 7.13(b). Example 7.28 calculates the insertion and bypass stresses.

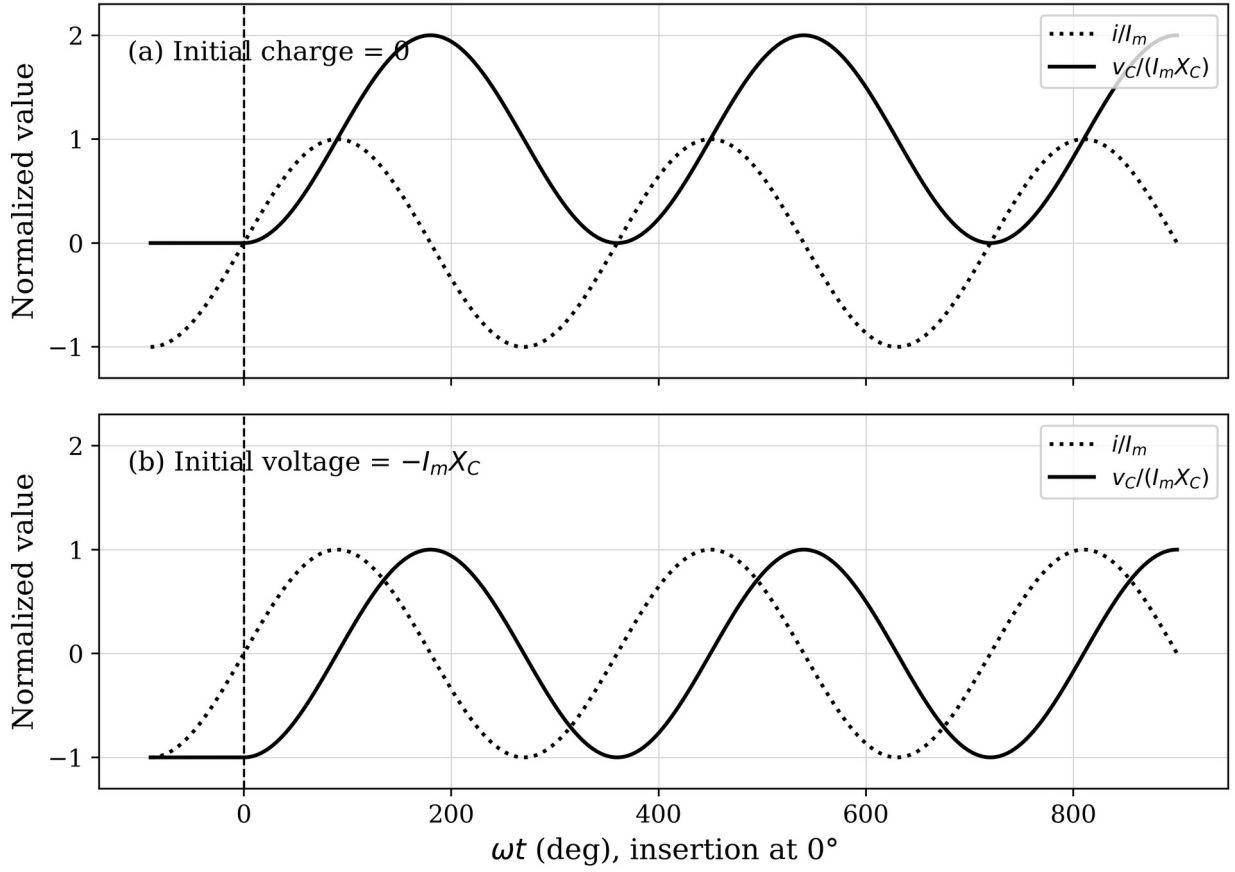


Figure 7.13. TSSC insertion at a current zero under an imposed sinusoidal current: (a) initially discharged; (b) initially at $-I_m X_C$. Prior to insertion the capacitor holds its initial charge; the line-current trace is shown for reference.

The bypass rule is the counterpart for the valve. Firing the valve when the capacitor holds a voltage V_{C0} discharges the capacitor through the small inductance L_v of the valve circuit, giving a surge current of peak value

$$I_{surge} \approx |V_{C0}|/Z_0, Z_0 = \sqrt{L_v/C} \quad (7.12)$$

The estimate neglects damping and the additional line-current component in the bypass valve. Its initial discharge slope is $|V_{C0}|/L_v$. Low-stress bypass is therefore coordinated with capacitor-voltage zero. In the ideal offset case this occurs once per cycle; without offset it occurs at successive line-current crests. During faults, protection may require earlier bypass, so the valve, reactor and MOV must withstand the resulting surge. Example 7.28 is a component-stress calculation, not a complete fault simulation.

7.3.3 Operating modes and characteristics

In the *voltage compensating mode*, the TSSC is asked to approximate a rated compensating voltage over a range of line current. If m equal stages of reactance X_C are provided, the rated voltage at the minimum current is obtained with all stages in service, $V_{Cmax} = m X_C I_{min}$; as current rises, stages are bypassed to keep the voltage at or below V_{Cmax} . The frequently quoted relation $V_{Cmax} = 4 X_C I_{min}$ is simply the case $m=4$. In the *impedance compensating mode*, the number of stages in service is held constant, giving a fixed series reactance at any current up to I_{max} . Figure 7.14 shows the two stepped characteristics, which are the discrete counterparts of Figure 7.11.

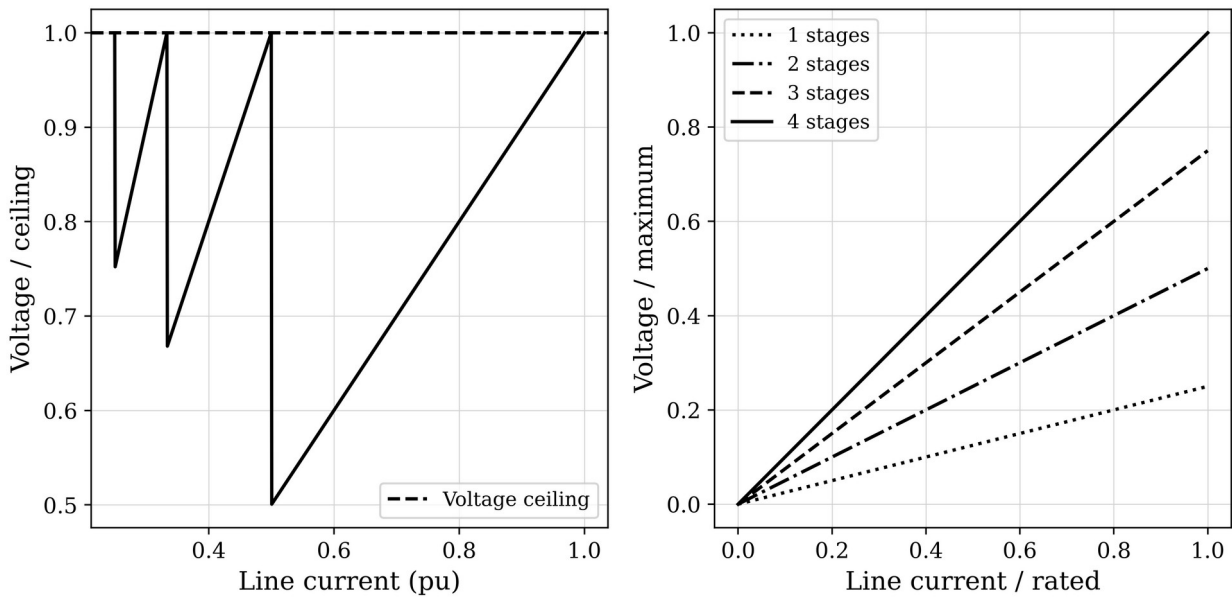


Figure 7.14. Illustrative TSSC voltage-ceiling control with four equal stages (left), and fixed-stage reactance characteristics (right). The left model has per-stage $X = 1$ normalized unit and a ceiling of 1; actual switching is naturally commutated.

Bypassed-stage valves carry line current and contribute conduction loss; inserted capacitors, reactors, auxiliary equipment and cooling also have losses. Each valve must block the capacitor peak voltage, including offsets and protective clamping, and meet continuous and fault-bypass current duties. Total branch RMS and individual-thyristor RMS are different ratings.

Example 7.3. Distinct compensation levels of an equal-stage TSSC

A TSSC has four equal stages of $3\ \Omega$ each and is installed in a line of $30\ \Omega$. Find the available degrees of compensation.

Solution. The inserted reactance can be 0, 3, 6, 9 or $12\ \Omega$, giving 0%, 10%, 20%, 30% and 40%. Although four bypass valves allow $2^4 = 16$ switching combinations, equal stages produce only five distinct levels.

Note. Binary-weighted stages of 1.5, 3, 6 and $12\ \Omega$ would give sixteen levels in 5% steps from 0% to 75% with the same number of valves; their capacitances are weighted in the inverse ratio.

7.3.4 Response and SSR

After gate withdrawal, insertion waits for the next valve-current zero, within about half a cycle for an ideal sinusoid. Low-stress bypass waits for a capacitor-voltage zero: up to half a cycle in offset-free operation and up to one cycle with the ideal insertion offset. Supervisory delays and fault priorities add to these bounds. A TSSC can schedule stepped flow and can contribute damping with suitable stage logic, but an inserted stage retains fixed-capacitor SSR behavior. Its quantization and commutation constraints must be included in the interaction study; a nearby torsional mode does not alone prove that every TSSC arrangement is unsuitable.

7.3.5 Stage coordination, binary weighting and switching logic

Equal TSSC stages are easy to manufacture and share voltage uniformly, but n equal stages provide only $n+1$ distinct reactance levels. Binary-weighted reactances X , $2X$, $4X$ and so on provide 2^n levels with the same number of valves. Because capacitor reactance is inversely proportional to capacitance, the capacitances are weighted in the opposite order, C , $C/2$, $C/4$ and so on. This is the reverse of a binary shunt capacitor bank, in which the capacitance (susceptance) itself is weighted.

A supervisory controller chooses the combination of inserted stages from a power-flow, current, voltage or impedance reference. The requested change cannot be imposed at an arbitrary instant. Insertion waits for natural current extinction of the bypass valve, while low-stress bypass is coordinated with capacitor voltage. When several

stages change state, their operations can be staggered to reduce simultaneous voltage steps and platform stress. The response is therefore discrete in magnitude and constrained by natural commutation in time.

7.3.6 Protection, ratings and practical limitations

Each inserted capacitor carries full line current and must be rated for steady-state voltage, insertion offset and temporary overload. Each bypass valve carries full line current while its stage is out of service and must withstand the capacitor voltage while the stage is inserted. A small reactor in the valve branch limits the discharge current and di/dt when a charged stage is bypassed (Equation (7.12)). MOV protection and a mechanical bypass path are coordinated with the series-capacitor protection described in Section 7.1.6.

The attraction of the TSSC is its simplicity. Once a stage is fully inserted, it is a linear capacitor and does not create phase-control harmonics. Its limitations are equally important: compensation is quantized, switching is constrained by natural commutation, and an inserted stage retains the frequency-dependent behaviour of a physical series capacitor. TSSC is therefore appropriate for stepped scheduling and coarse power-flow redistribution, while TCSC or SSSC is preferable when continuously variable damping or wide-range modulation is required.

7.4 The Thyristor-Controlled Series Capacitor (TCSC)

7.4.1 Circuit, purpose and installations

The TCSC places a controlled reactor in parallel with a series capacitor (Figure 7.15). Its distinct modes are blocked capacitor, capacitive vernier, inductive vernier and low-impedance valve/reactor bypass. A fixed bank often supplies most of the corridor compensation, with the TCSC providing modulation. MOV, valve and breaker protection are coordinated; whether a separate triggered gap can be omitted depends on the design.

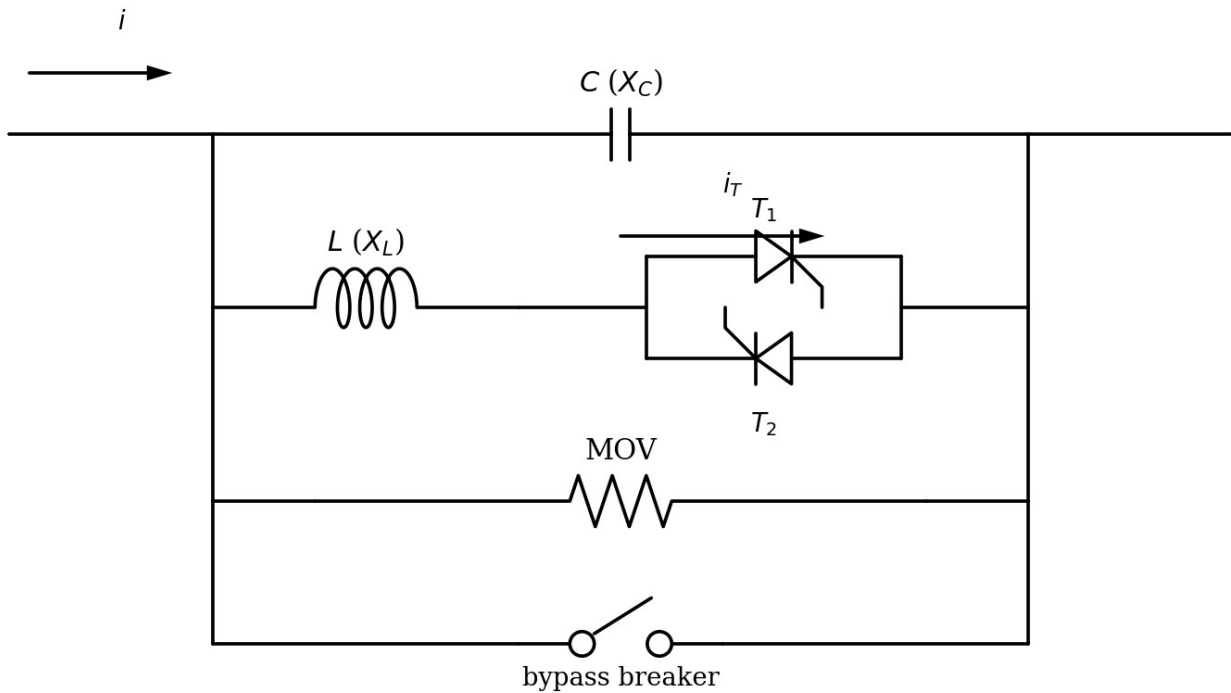


Figure 7.15. TCSC module: capacitor in parallel with a reactor and antiparallel thyristor valve, with MOV and bypass breaker.

Table 7.3 gives selected historical installations, each with its own source. Dates are reported commissioning or test-commissioning dates, and ratings are omitted where the cited record does not state them. The table illustrates applications; it is not an equipment specification or a claim of precedence.

Table 7.3 Representative TCSC installations.

Installation	Year	Voltage	Purpose and evidence
Kayenta, WAPA, USA	1992	230 kV	Early controllable section within a fixed bank; power-flow control [5]

Installation	Year	Voltage	Purpose and evidence
Slatt, BPA, USA	1993	500 kV	Six-module prototype; control and SSR studies [9]
Stöde, Sweden	1998	400 kV	SSR mitigation; operator report records operation in 1998 [24]
Imperatriz, North–South interconnection, Brazil	1999	500 kV	Inter-area damping; supplier records TCSC commissioning in 1999 [25]
Kanpur-Ballabgarh, India	Test commissioning reported 2005–2006	400 kV	Indigenous BHEL/PGCIL/DIT project; power flow and damping [23]
Rourkela-Raipur, India	2004	400 kV, double circuit	First TCSC in India; two banks at the Raipur end for East–West transfer and inter-area damping [26]

7.4.2 The simplified (voltage-source) model

The approximate voltage-source model treats the capacitor voltage as sinusoidal. Let α be the peak-referenced delay and $2\beta = \pi - 2\alpha$ the reactor conduction interval, with $\beta = \pi/2 - \alpha$. This differs by 90° from a zero-referenced TCR delay in Chapter 6. The resulting reactor reactance is

$$X_L(\alpha) = X_L \frac{\pi}{\pi - 2\alpha - \sin 2\alpha} = X_L \frac{\pi}{2\beta - \sin 2\beta}, X_L = \omega L \quad (7.13)$$

where $X_L(\alpha)$ is the equivalent reactance of the controlled reactor and X_L its full-conduction value. In parallel with the capacitor, and using the signed convention in which positive reactance is inductive,

$$X_{TCSC}(\alpha) = \frac{X_C X_L(\alpha)}{X_C - X_L(\alpha)} \quad (7.14)$$

where X_{TCSC} is the signed reactance of the module. With $X_L < X_C$, the module is inductive for small α , passes through a parallel resonance where $X_L(\alpha) = X_C$, and becomes capacitive with a reactance magnitude larger than X_C beyond it, falling back to $-X_C$ as $\alpha \rightarrow 90^\circ$. The ratio of reactor to line current follows from current division, $I_{TCR}/I = X_C/(X_C - X_L(\alpha))$. Close to resonance this ratio is large, which is why the capacitive boost must be limited (Example 7.32).

7.4.3 Why the simplified model is not accurate

The simplified model assumes that the reactor sees a sinusoidal voltage. In a TCSC it does not. The module is driven by the *line current*, which the rest of the network imposes almost independently of what the TCSC does. The capacitor voltage is shaped by the reactor pulses themselves. When a thyristor fires, the charged capacitor discharges through the reactor in a resonant half-period of the LC loop, reversing its charge, while the line current continues to flow. This *charge reversal* is the central mechanism of the TCSC, and it is absent from the voltage-source picture. The consequences are not small. For the module of Example 7.31 ($C = 1000 \mu\text{F}$, $L = 1.5 \text{ mH}$), the simplified model places the resonance at $\alpha = 48.2^\circ$, whereas the exact analysis places it at 55.4° . At $\alpha = 50^\circ$ the simplified model predicts a capacitive reactance of 27.6Ω , while the actual module is *inductive*, $+5.13 \Omega$. A design based on the simplified model can therefore select an operating point on the wrong side of resonance.

Example 7.4. Locating the simplified TCSC parallel-resonance point

A TCSC has $X_C = 20 \Omega$ and a reactor of $X_L = 4 \Omega$. Use the simplified model, Equation (7.13), to find the firing angle at which parallel resonance occurs, and compare it with the exact current-driven model.

Solution. In the simplified model resonance occurs when $X_L(\alpha) = X_C$, that is when $\pi - 2\alpha - \sin 2\alpha = \pi X_L/X_C = \pi/5$. Solving numerically gives $\alpha = 43.40^\circ$ ($\beta = 46.60^\circ$). The exact model has its first resonance at $\beta_r = \pi/(2\varpi)$ with $\varpi = \sqrt{X_C/X_L} = 2.236$, so $\beta_r = 40.25^\circ$ and $\alpha_r = 49.75^\circ$. The two models place the resonance 6.4° apart.

Note. Between $\alpha = 43.4^\circ$ and 49.8° the two models disagree on whether the module is inductive or capacitive, so the permitted firing band and its resonance margin must be set from the exact characteristic.

7.4.4 Exact analysis of the line-current-driven TCSC

Let the line current be $i = I_m \sin \theta$, $\theta = \omega t$, and let v_C be the capacitor voltage and i_T the thyristor current. While both thyristors block, $C dv_C/dt = i$. While a thyristor conducts,

$$C \frac{dv_C}{dt} = i - i_T, L \frac{di_T}{dt} = v_C \Rightarrow \frac{d^2 v_C}{dt^2} + \omega_0^2 v_C = \frac{1}{C} \frac{di}{dt}, \omega_0 = \frac{1}{\sqrt{LC}} \quad (7.15)$$

Equation (7.15) is valid only while a valve conducts; otherwise $i_T = 0$ and $C dv_C/dt = i$. The solution has a forced term at ω and a free term at ω_0 . For the symmetric periodic pulse, define $x = \theta - \pi/2$, measured from the line-current crest and the central capacitor-voltage zero. Conduction occupies $-\beta \leq x \leq \beta$. Boundary matching imposes $i_T(-\beta) = i_T(\beta) = 0$. Fourier extraction of the resulting capacitor voltage yields [9]

$$X_{TCSC}(\beta) = -X_C + C_1(2\beta + \sin 2\beta) - C_2 \cos^2 \beta (\varpi \tan \varpi \beta - \tan \beta) \quad (7.16)$$

with

$$\varpi = \frac{\omega_0}{\omega} = \sqrt{\frac{X_C}{X_L}}, X_{LC} = \frac{X_C X_L}{X_C - X_L}, C_1 = \frac{X_C + X_{LC}}{\pi}, C_2 = \frac{4 X_{LC}^2}{\pi X_L} \quad (7.17)$$

where ϖ is the tuning ratio of the module, X_{LC} the parallel reactance of capacitor and full reactor, and C_1, C_2 are constants of the module. Equation (7.16) gives $-X_C$ at $\beta = 0$ (blocked) and the full-conduction parallel reactance X_{LC} at $\beta = \pi/2$ (bypass). It becomes infinite where $\tan \varpi \beta$ does, so the resonances lie at

$$\beta_r = \frac{(2m-1)\pi}{2\varpi}, m = 1, 2, \dots \quad (7.18)$$

For $1 < \varpi < 3$ there is one formal pole in $0 < \beta < \pi/2$; for $\varpi > 3$ additional poles enter the range. A practical operating band must also satisfy natural commutation, valve polarity, current, voltage and thermal constraints. A formal continuation beyond a pole is not automatically realizable by thyristors: check whether a pulse reaches zero before its assumed extinction. Typical design ratios near $X_L/X_C = 0.11$ – 0.25 illustrate the trade-off, not a mandatory interval. Figure 7.16 compares the exact and simplified formulas for $X_L/X_C = 0.133$. Example 7.31 gives numerical values for this derivation, and Example 7.33 connects angle, waveform stress and reactor selection.

Waveform bridge and reproducibility. Set $a = \varpi^2$, $c = \cos \beta / \cos(\varpi \beta)$ and normalize $y = v_C/(I_m X_C)$, $t = i_T/I_m$. During conduction, $y = [-\sin x + \varpi c \sin(\varpi x)]/(a-1)$ and $t = a[\cos x - c \cos(\varpi x)]/(a-1)$. For $\beta < x < \pi/2$, $y = \sin x + D$, where $D = y(\beta) - \sin \beta$; use odd symmetry about $x = 0$ and half-wave antisymmetry to complete the cycle. Thus $X_{TCSC}/X_C = -(2/\pi) \int_{-\pi/2}^{\pi/2} y \sin x dx$, and branch RMS follows from integrating t^2 . These expressions reproduce Equation (7.16) without assuming sinusoidal capacitor voltage.

As a numerical check, the waveform integrals were evaluated separately on each switching interval, with tolerances of 10^{-10} , at 180 values of β for $X_L/X_C = 0.133, 0.15$ and 0.20 , excluding the immediate neighbourhood of the poles. Equation (7.16) agreed with the integrated waveforms to within 10^{-10} . This confirms the analytical result for the ideal periodic waveform; it is not a substitute for EMT or hardware validation.

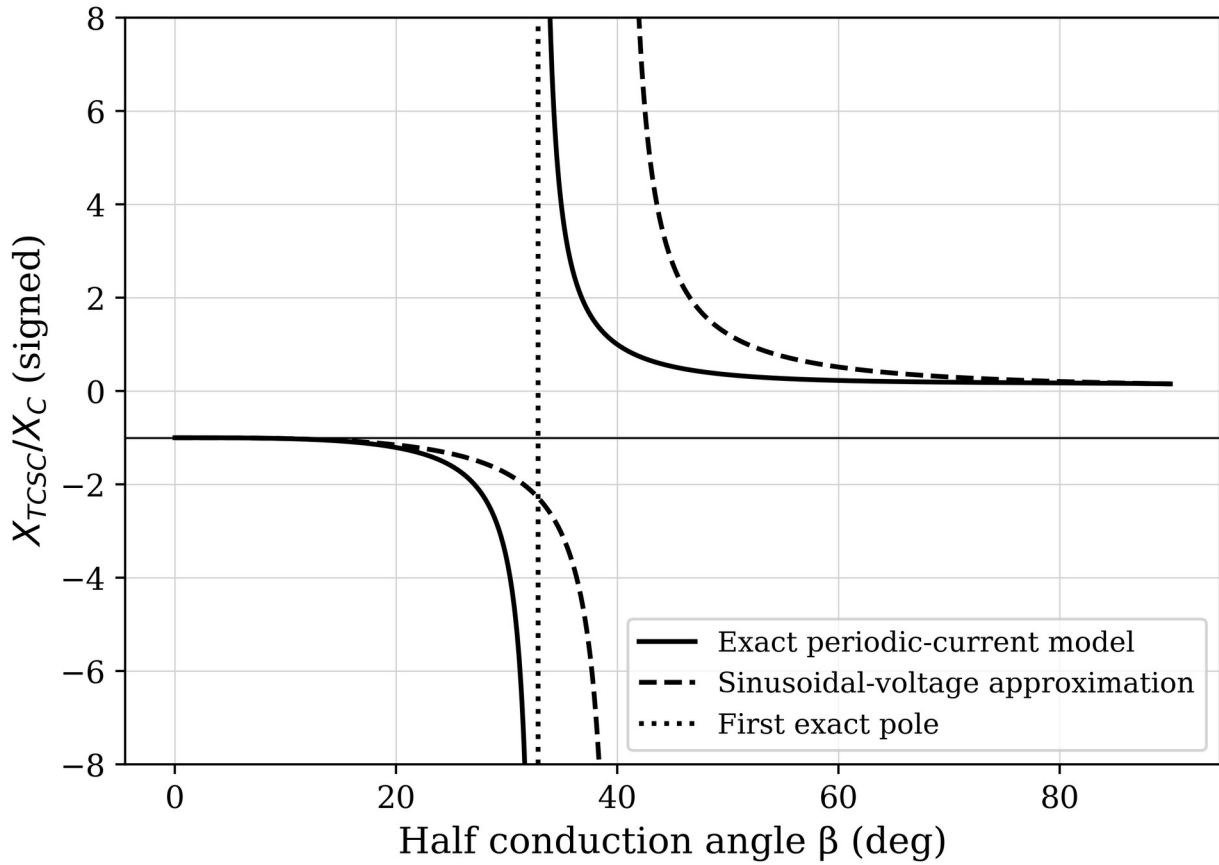


Figure 7.16. Signed exact and approximate TCSC reactance for $X_L/X_C = 0.133$, versus half conduction angle. Curves are split near poles; formal branches still require natural-commutation and component-limit checks.

7.4.5 Waveforms and the charge-reversal mechanism

Figures 7.17 and 7.18 show steady-state waveforms obtained from the exact model for a module with $X_L/X_C = 0.133$. In the capacitive vernier ($\beta = 25^\circ$), each thyristor fires shortly before the capacitor voltage crosses zero. The reactor current pulse reaches about 1.7 times the peak line current and flows against it, so during the reversal the capacitor current is boosted to about 2.7 times the peak line current, and the capacitor voltage is pushed further than the line current alone would push it. The fundamental voltage is 1.6 times that of the bare capacitor. In the inductive vernier ($\beta = 55^\circ$), conduction is long, the reactor carries more than the line current, and the capacitor voltage fundamental leads the current: the module is net inductive at $0.27 X_C$.

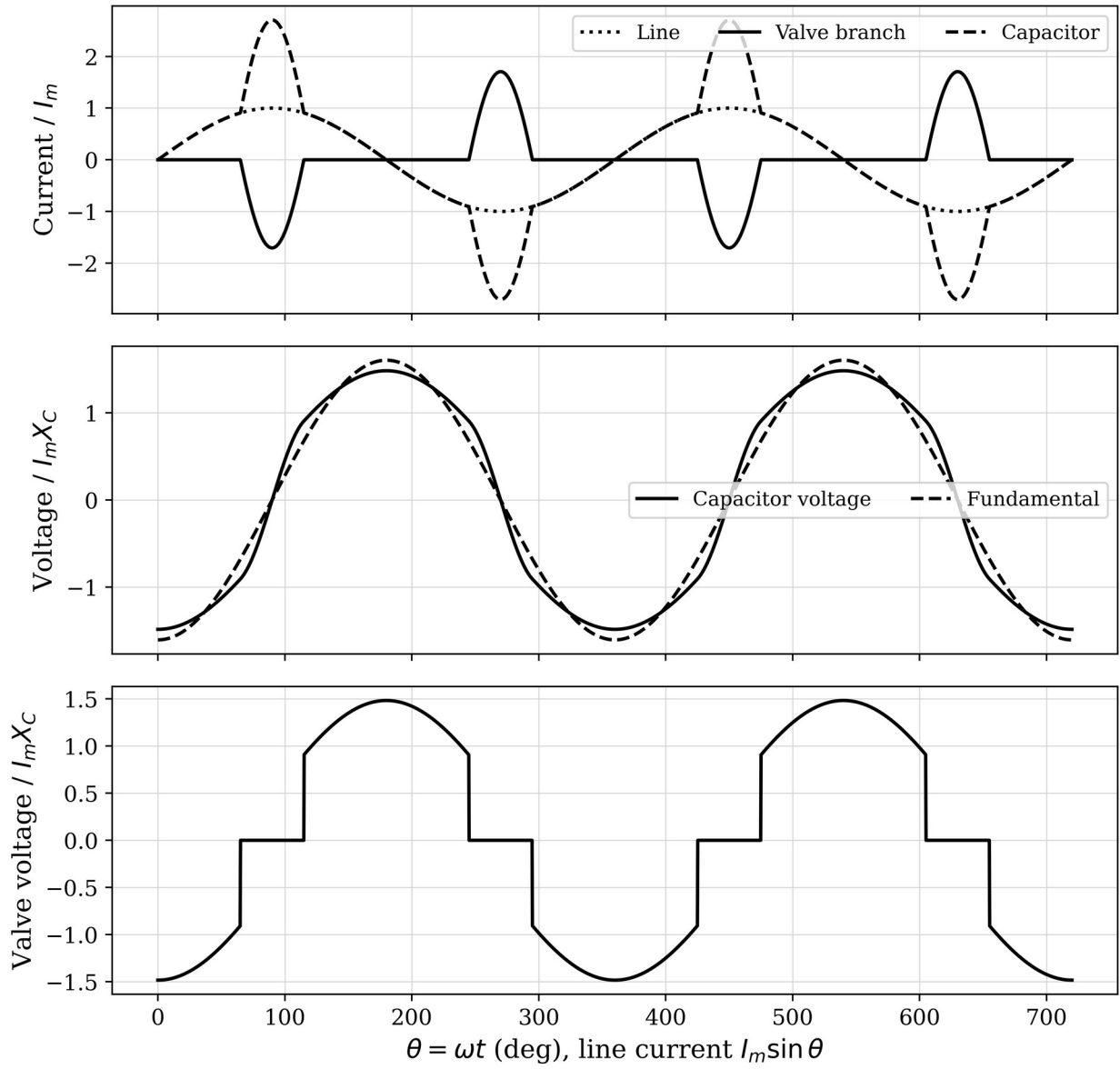


Figure 7.17. Exact ideal TCSC capacitive-vernier waveforms for $X_L/X_C = 0.133$, $\beta = 25^\circ$. Current base is I_m ; voltage base $I_m X_C$. The valve voltage is zero during conduction; θ is measured from line-current zero.

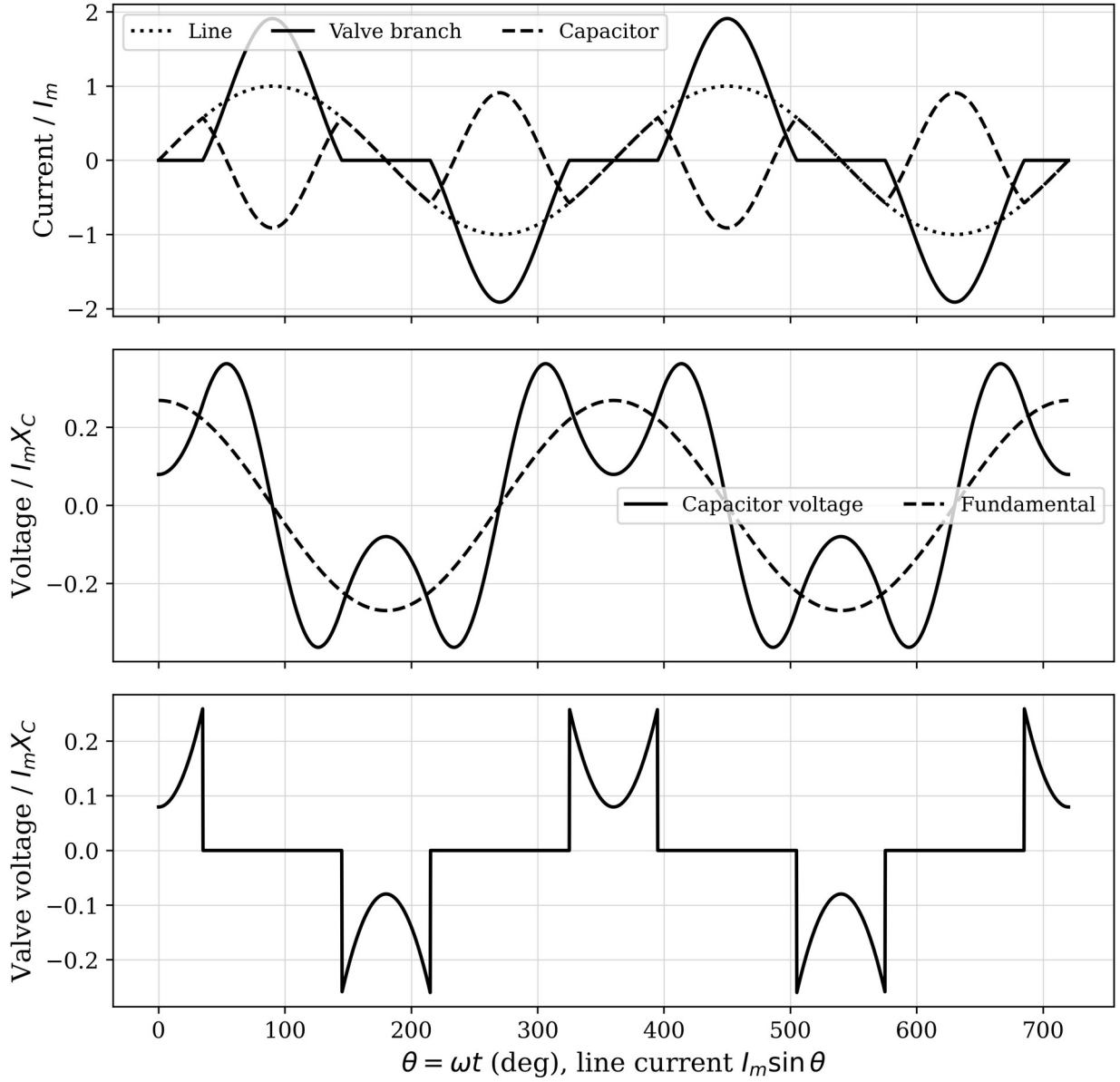


Figure 7.18. Exact ideal TCSC inductive-vernier waveforms for $X_L/X_C = 0.133$, $\beta = 55^\circ$, with the same reference and bases as Figure 7.17.

The charge-reversal picture explains voltage boost: reactor pulses redistribute capacitor charge while the line supplies current. An idealized instantaneous reversal separates an ordinary current-proportional capacitor voltage from an additional waveform caused by changed charge. It is useful intuition, but a finite-reactor TCSC in periodic steady state still has $V_{C1} = I_1 |X_{TCSC}(\beta)|$. As current tends to zero, a bounded boost cannot maintain arbitrary rated voltage. Stored initial charge can affect a transient, not sustain current-independent voltage indefinitely. Low-current control must account for synchronization, valve holding current and permitted boost.

7.4.6 Choice of the reactor and harmonics

At a **fixed required capacitive boost**, a larger reactor can widen the angular margin to the first pole and reduce peak and RMS valve-branch current and capacitor-voltage THD, while slowing charge reversal. Example 7.33 quantifies the trend for ratios 0.10, 0.15 and 0.20. This comparison must not be confused with one at fixed firing angle. A smaller reactor gives a lower-impedance protective bypass but higher reversal-current stress. For $X_L/X_C < 1/9$ a second formal pole appears below full conduction. Transitions between vernier and bypass require protected switching logic validated in EMT; commanding a rapid continuous sweep across a pole is not a safe design rule.

The imposed-current model predicts capacitor-voltage harmonics and internal circulating-current pulses. In a real network these voltage harmonics drive line-current harmonics through the frequency-dependent network impedance, so the line current is not perfectly sinusoidal. Figure 7.19 plots peak harmonic voltage on the fixed base $I_m X_C$, with the region around the first pole excluded. At $\beta = 25^\circ$ and $X_L/X_C = 0.133$ the third harmonic is 17.3% of that base, or 10.8% of the actual fundamental. Harmonic duty grows rapidly towards resonance, so the capacitive vernier cannot be assumed to produce only a few percent of harmonic voltage.

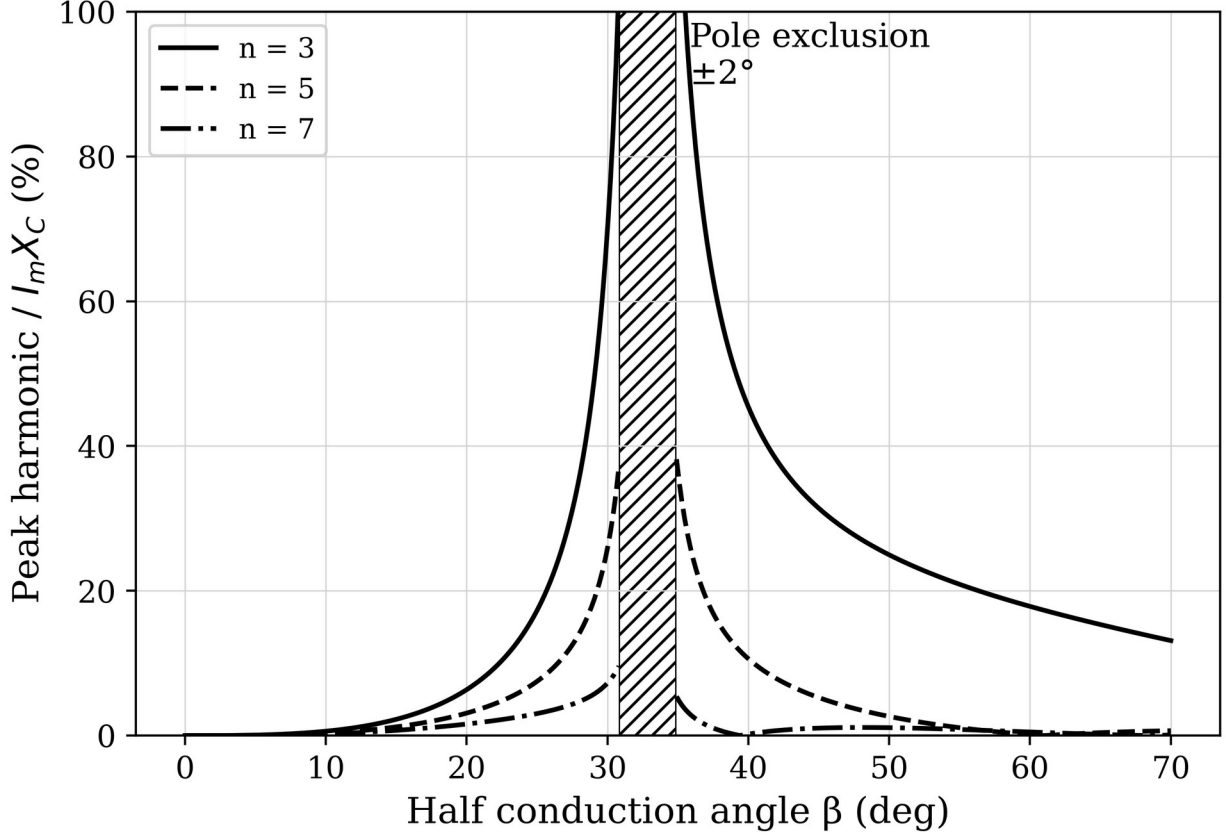


Figure 7.19. Peak TCSC harmonic voltages from Fourier integration of the stated periodic waveforms, $X_L/X_C = 0.133$. A $\pm 2^\circ$ band around the first pole is excluded. Very large near-pole values outside the ordinate are not admissible operating points.

7.4.7 Operating modes, characteristics and losses

The blocking mode leaves the bare capacitor, $X_{TCSC} = -X_C$. Continuous bypass gives the small positive parallel reactance X_{LC} in the ideal model. In the capacitive vernier define the **positive** boost $k_B = |X_{TCSC}|/X_C = -X_{TCSC}/X_C$, often limited to an application-dependent value around 2–3. The inductive vernier can reduce flow, but severe-fault protection generally prioritizes robust bypass over normal vernier regulation. Signed X_{TCSC} remains the quantity used in network equations.

The attainable band depends on line current, waveform voltage, valve/reactor current, thermal duty and angle margin (Figure 7.20). At low current a finite maximum boost still gives voltage proportional to current; at higher current the voltage ceiling reduces the allowable boost. RMS fundamental voltage alone does not describe the capacitor's peak or total RMS stress. The gap between the blocked capacitor and inductive vernier prevents a single module from commanding every intermediate reactance. Series combinations of fixed, bypassable and controlled modules can adjust the aggregate range.

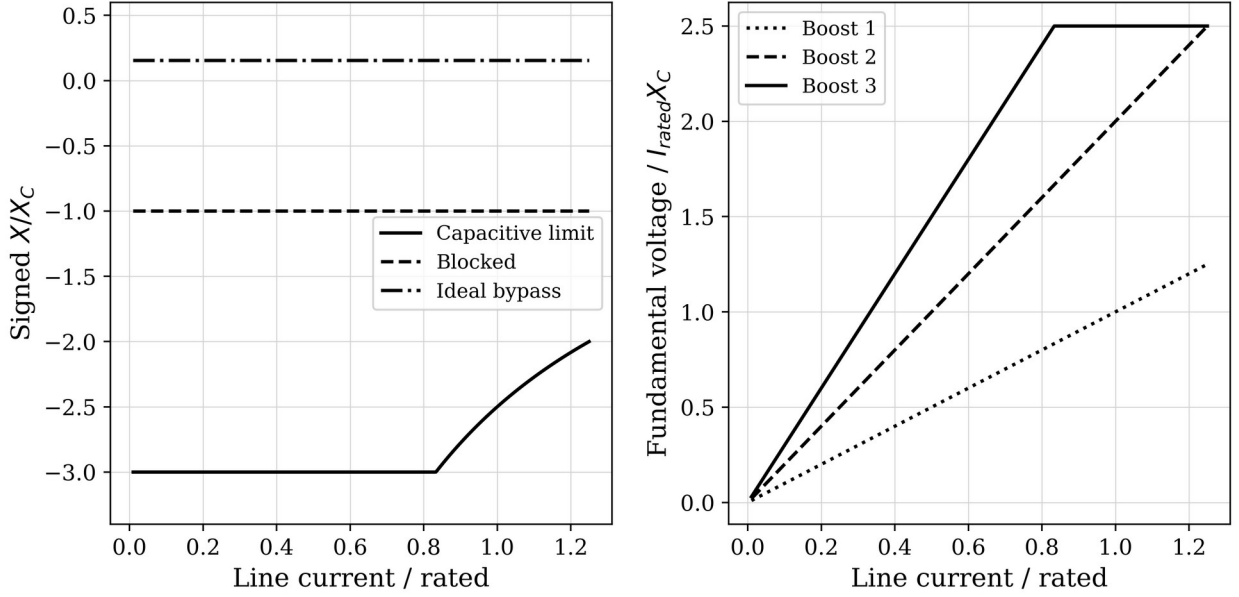


Figure 7.20. Illustrative TCSC limits: signed reactance with finite capacitive boost and ideal bypass (left), and positive capacitive-voltage magnitudes (right). Maximum boost 3 and normalized fundamental-voltage ceiling 2.5 do not replace peak/RMS/thermal ratings.

7.4.8 Control

The internal controller estimates the fundamental current and capacitor voltage, forms a signed reactance estimate and maps a limited order to synchronized valve pulses. Current synchronization needs filtering that preserves the intended firing phase, and capacitor-voltage prediction uses both measured voltage and current. Suitable firing can improve subsynchronous behaviour, although equally spaced reversals do not by themselves ensure positive damping. The reactance estimator must use phase as well as magnitude, for example $Im(\dot{V}_{C1}/\dot{I}_1)$ with consistent terminal polarity, and must be disabled or conditioned near zero current.

External control. Figure 7.21 shows the external structure. Depending on the application, the regulated quantity is line power or current (power-flow control), the transmission angle, or a reactance schedule. Supplementary loops add power oscillation damping and, if required, SSR damping, and a transient-stability function drives the module to maximum boost immediately after a fault.

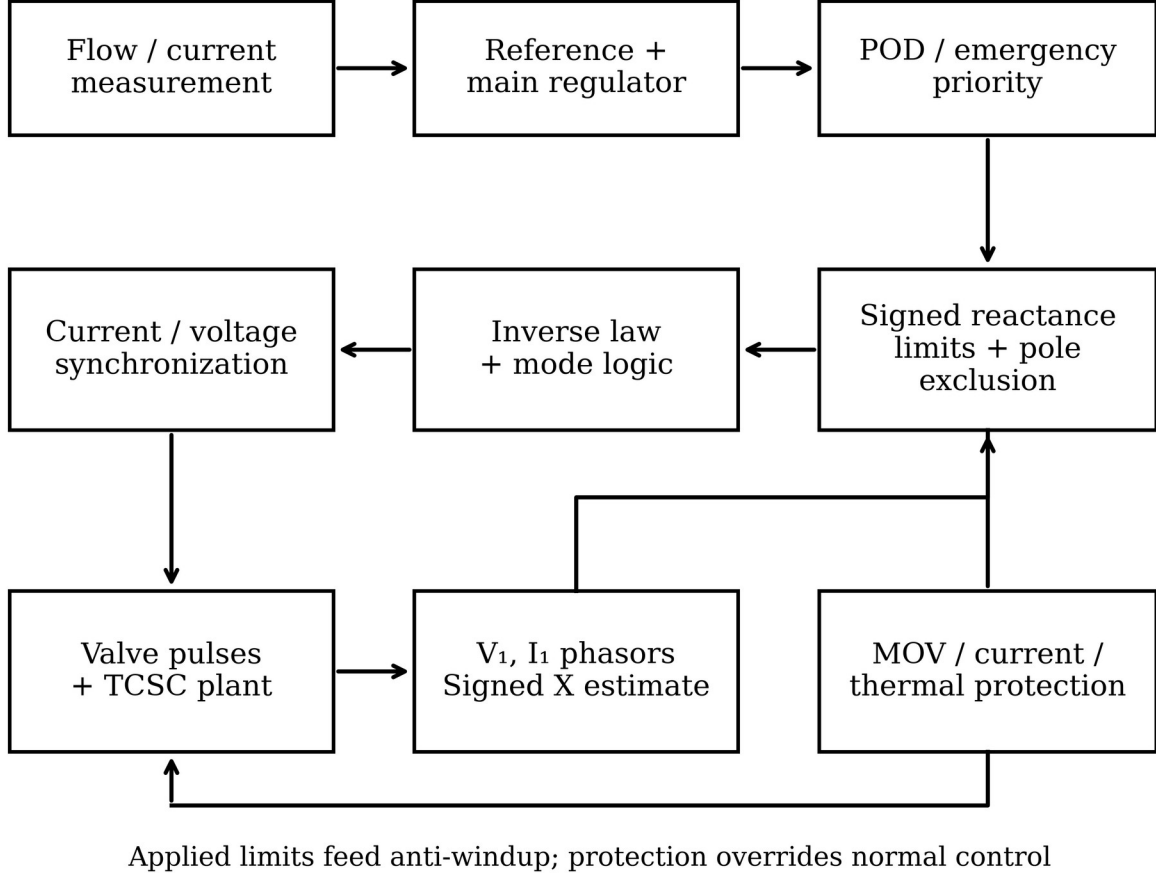


Figure 7.21. TCSC control path with signed reactance estimation, synchronization, pole exclusion, limits and protective override. POD priority and applied-limit anti-windup must be designed for the actual application.

The complete TCSC controller solves two problems, an external one and an internal one. The external controller decides what the corridor requires, such as a line-power order, scheduled reactance, transient-stability boost or damping modulation. The internal controller realizes that request without crossing the resonance exclusion zone or violating capacitor-voltage, reactor-current, valve-current and thermal limits. The reactance order therefore passes through limiters and an inverse characteristic before becoming synchronized firing pulses.

Figure 7.22 compares two ideal SMIB simulations that the reader can reproduce. The model is $\dot{\delta} = \omega_b \Delta \omega$, $2H \Delta \dot{\omega} = P_m - P_e - D \Delta \omega$, with $H = 4$ s, $f_0 = 50$ Hz, $P_m = 0.6$ pu, $D = 0$, $V = 1$ pu and total uncompensated $X = 1$ pu. Initially $x_C = 0.25$ pu and $\delta_0 = \sin^{-1}[P_m(1 - x_C)]$. During an 80 ms fault $P_e = 0$ and the device is bypassed; afterwards $P_e = \sin \delta / (1 - x_C)$. The damping case uses $\dot{x}_C = 0.25 + 8 \Delta \omega$, limited to 0.10 to 0.40 pu, through a 20 ms lag. The damping shown follows from these assumptions and is not a prediction of field performance.

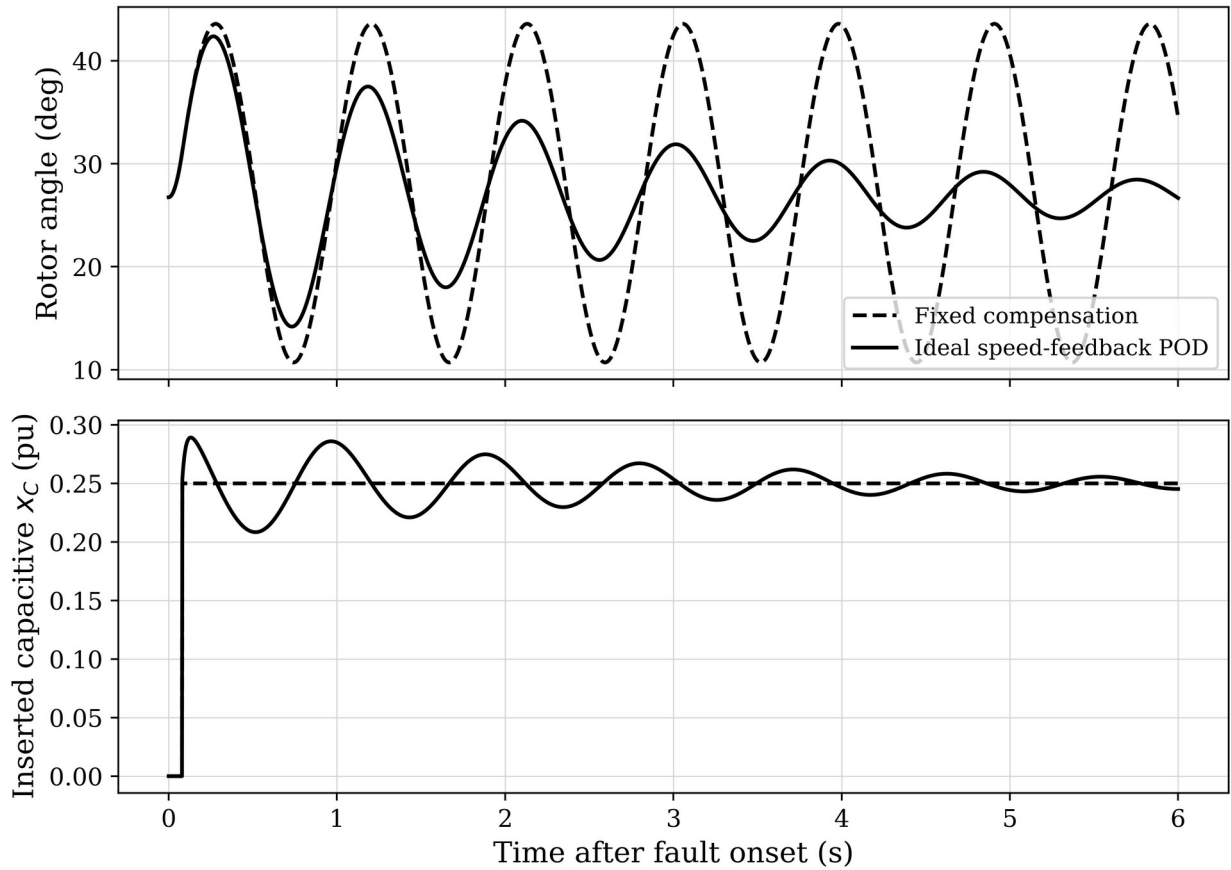


Figure 7.22. Reproducible ideal SMIB comparison with an 80 ms zero-transfer fault and fixed or speed-feedback series compensation. $H = 4$ s, $P_m = 0.6$ pu, $X = 1$ pu, $D = 0$; the POD order is $0.25 + 8\Delta\omega$ with 0.10–0.40 pu bounds and a 20 ms lag. Device firing and detailed protection are omitted.

The supplementary POD channel filters its selected measurement, applies washout and lead-lag compensation, limits the modulation, and adds it to the scheduled reactance. With positive $\Delta\omega$, a positive damping component of **electrical power** is required in the swing equation; its torque opposes the speed deviation. The sign of a reactance or voltage command follows from the local power sensitivity. Filters, communication, firing dynamics and operating-point changes contribute phase, so increasing gain alone does not establish positive damping.

Example 7.5. Supplementary damping command superimposed on TCSC reactance control

A TCSC operates at a steady capacitive reactance order of 12Ω . A power-oscillation damping controller adds a sinusoidal modulation of $\pm 1.5 \Omega$. State the commanded range and the checks required before the modulation is applied.

Solution. The order varies between 10.5 and 13.5 Ω . The internal controller must map this range into admissible firing instants while respecting the capacitive operating band, the resonance exclusion margin, and the capacitor-voltage, reactor-current and valve thermal limits. The damping channel must also keep its intended phase compensation over the range of modal frequencies.

Note. A damping controller is not validated by the amplitude of its reactance order. Positive damping must be demonstrated for the actual network, measurement filters, firing logic and operating points.

7.4.9 Subsynchronous behaviour and the NGH principle

The NGH principle uses a controlled resistor across a series capacitor to dissipate subsynchronous voltage components [2]. TCSC charge reversals provide a related opportunity to shape the subsynchronous terminal response. Published studies and project tests demonstrate favorable behavior for particular control designs [9], [24]. Whether the effective response is inductive, resistive and positively damped depends on firing logic, synchronization, operating point and network frequency coupling. A lossless steady-periodic reactance law at

50 Hz cannot establish that property. Assess the closed-loop small-signal/EMT response and coupled torsional modes for each relevant contingency.

7.4.10 Representation in power-flow and stability studies

For steady-state studies the TCSC is represented as a variable series reactance X_{TCSC} in the branch admittance matrix. With the line reactance X , the branch admittance becomes $1/[j(X + X_{TCSC})]$, and X_{TCSC} (or β) is added as a state variable regulated to a power-flow target. An equivalent *power-injection model* replaces the TCSC by the active and reactive powers it injects at the two end buses, which leaves the admittance matrix unchanged. For stability studies a first-order lag between the reactance order and the actual reactance, with a time constant of about one cycle, is usual. Example 7.37 works through the admittance form.

No single TCSC model is appropriate for every study. A variable-reactance model is suitable for load-flow scheduling and many electromechanical studies. A first-order dynamic reactance model can represent dominant command response in transient-stability simulations. SSR, harmonic, protection and switching studies require a waveform or EMT representation containing the capacitor, reactor, thyristor firing logic, MOV and bypass path. Dynamic-phasor and frequency-dependent models provide an intermediate level when subsynchronous behaviour must be retained without the full cost of switching simulation.

Model fidelity should be chosen from the phenomenon being studied. A damping controller that appears satisfactory with a first-order reactance model can change after current synchronization, resonance-exclusion logic and firing dynamics are included. Conversely, a detailed EMT model is unnecessary for an optimization that only requires the steady-state branch reactance.

7.4.11 Ratings and protection

Equipment selection uses maximum capacitor peak and total RMS voltage, reactor/valve-branch RMS current, individual-thyristor RMS and average current, peak pulse current, fault duty and the specified thermal overload profile. For symmetric antiparallel pulses, $I_{T,device,rms} = I_{T,branch,rms}/\sqrt{2}$, while the pulse peak is unchanged. Reactor copper loss uses branch RMS. Example 7.36 makes these definitions explicit. MOV energy, protective bypass and recovery must be coordinated; temporary overload allowances are specification-specific and cannot be inferred from a single steady rating.

During severe faults, normal vernier control is subordinated to protection. Depending on the design, the thyristor valve is driven to bypass, the bypass breaker may close, and the MOV absorbs part of the capacitor energy. Recovery after fault clearing must avoid an uncontrolled transition through the resonant region. Modern TCSC specifications therefore treat operating-mode transitions, protection coordination and real-time control-system validation as part of the controller design.

7.5 The Static Synchronous Series Compensator (SSSC)

7.5.1 Definition and circuit

IEEE and CIGRE define the SSSC as a static synchronous generator operated without an external energy source as a series compensator, whose output voltage is in quadrature with, and controllable independently of, the line current, so as to increase or decrease the overall reactive voltage drop across the line and thereby control the transmitted power. The SSSC may include energy storage or a power supply on its DC side, in which case it can also exchange active power with the line.

Physically, an SSSC is a voltage-source converter, multi-pulse or multilevel, connected to the line through a series coupling transformer (Figure 7.23). The converter is a synchronous voltage source like the STATCOM of Chapter 6, but it is synchronized to the *line current* rather than to a bus voltage. Proposed by Gyugyi in 1989 [4], it has been realized in the converter-based installations of Inez and Marcy in the United States, where the series converter of a UPFC or CSC can be operated as an SSSC, and more recently in transformerless *modular* SSSCs that hang directly on transmission conductors.

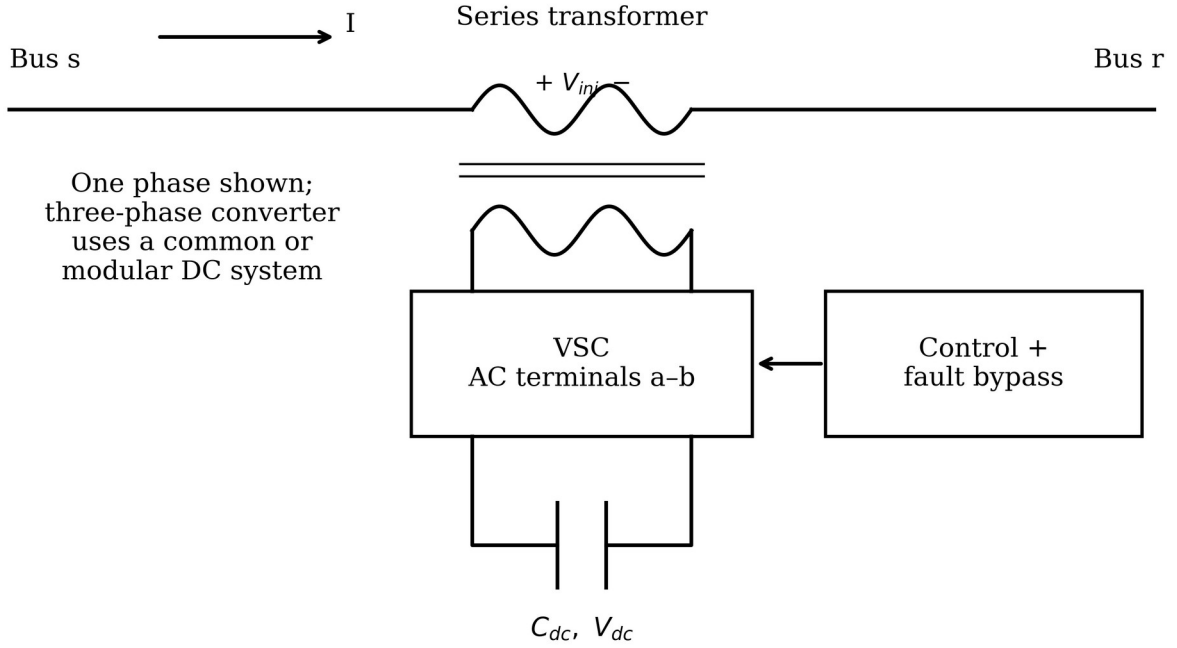


Figure 7.23. One-phase equivalent of a transformer-coupled SSSC. The primary carries line current, the secondary connects to converter AC terminals, and the DC capacitor connects across its DC terminals. Balanced three-phase and modular implementations require the corresponding converter and protection circuits.

7.5.2 Power-angle characteristic

Use the symmetric reference of Section 7.1.2 and define a **signed voltage drop** across the SSSC in the forward line direction as $\underline{V}_{inj} = -jV_q$. Thus $V_q > 0$ is capacitive for positive forward current, and $V_q < 0$ is inductive before reversal. KVL gives $\underline{V}_s - \underline{V}_r = jXI - jV_q$, hence the signed current on the fixed real reference axis is

$$I = \frac{2V \sin(\delta/2) + V_q}{X} \quad (7.19)$$

Here I is a signed scalar; its physical RMS magnitude is $|I|$. For $I > 0$ the injected drop lags current by 90° when $V_q > 0$. Since the receiving-end real part is $V \cos(\delta/2)$, the per-phase receiving power is

$$P_q = \frac{V^2}{X} \sin \delta + \frac{V V_q}{X} \cos \frac{\delta}{2} \quad (7.20)$$

where P_q is the power transmitted with the SSSC. Compare this with the series capacitor, $P_c = V^2 \sin \delta / [(1-k)X]$. The capacitor raises power by a fixed *fraction of the power at the same angle*, so it adds nothing at $\delta = 0$ and contributes most near 90° . The SSSC adds a term proportional to $\cos(\delta/2)$, a fixed *fraction of the maximum power* V^2/X . This term is largest at small angles and does not vanish at $\delta = 0$. Figure 7.24 shows both families. Two features follow directly from Equation (7.20).

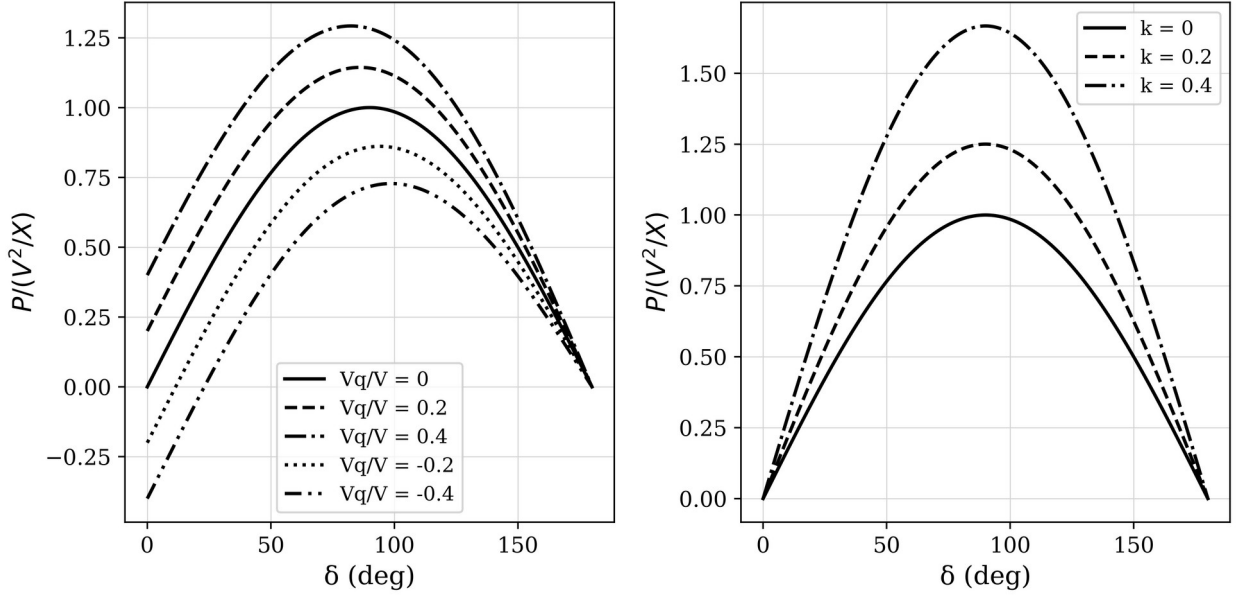


Figure 7.24. Ideal SSSC fixed-axis signed-voltage power-angle curves (left) and fixed-capacitor curves (right), normalized on V^2/X . Negative power denotes reversed current; zero-current controller transition is outside this phasor plot.

Symmetric voltage range. The ideal converter can command either voltage polarity within its voltage, current and MVA envelope, so it can increase or decrease corridor flow. The resulting flow-control range depends on the transfer sensitivity, angle, current and limits; it is not simply twice the converter rating.

Power reversal on a fixed reference. Equation (7.19) becomes negative when $V_q < -2V \sin(\delta/2)$, equivalently $|V_q| > 2V \sin(\delta/2)$ with $V_q < 0$. The current phasor then rotates by 180° . The same fixed-axis voltage now has a different capacitive/inductive relation to the reversed current, so “inductive operation” must not be extrapolated unchanged through reversal. A current-synchronized controller needs a defined zero-current transition and phase/polarity strategy. Example 7.39 gives the ideal reversal boundary, subject to converter voltage, DC-energy and current limits.

7.5.3 Operating modes and characteristics

In constant-voltage mode the SSSC commands a quadrature voltage up to its voltage ceiling; in reactance mode its magnitude is $V_q \leq X_q I \leq V$ until that ceiling is reached. Unlike a capacitor of finite reactance, it can hold a finite voltage at light load, provided the current stays above the synchronization threshold. At zero current the current angle is undefined, and charging, losses and auxiliary supply then need a defined operating strategy. Converter and transformer losses depend on topology, switching and loading, so their light-load behaviour has to be taken from the actual design.

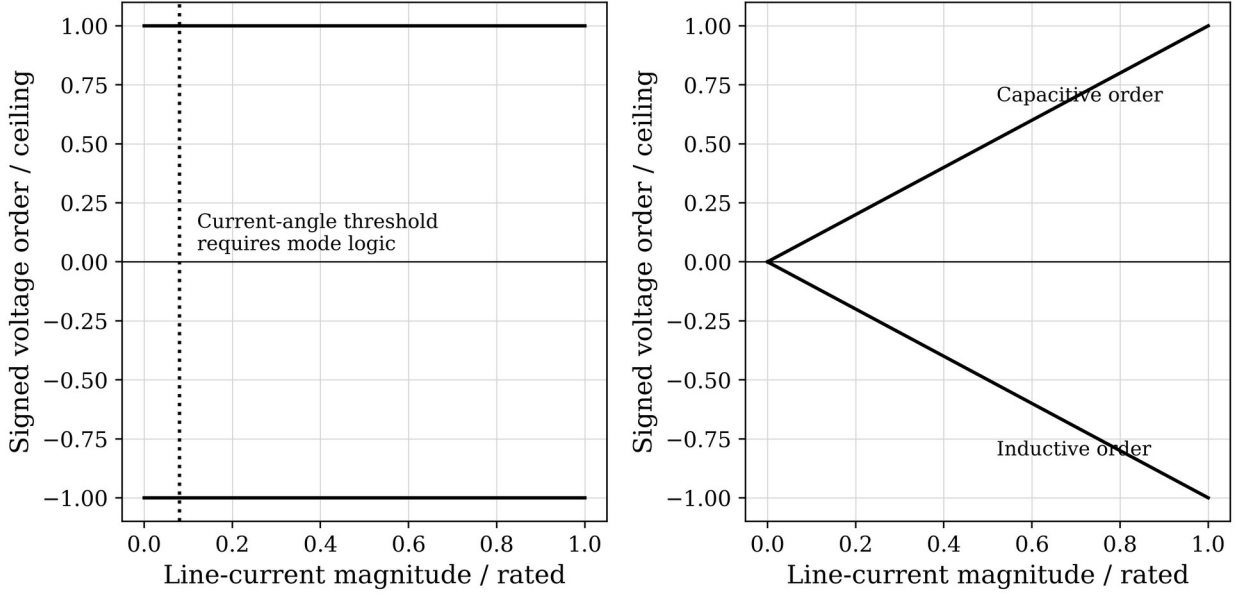


Figure 7.25. SSSC constant-voltage and constant-reactance envelopes. Capacitive and inductive labels refer to forward current; the minimum-current threshold shown is an example value.

Example 7.6. SSSC constant-voltage versus constant-reactance operation

An SSSC can inject up to 20 kV per phase in quadrature with the line current. Compare the injected voltage at line currents of 0.5 kA and 1.0 kA when it operates in constant-voltage mode and when it emulates a series reactance of 20 Ω .

Solution. In constant-voltage mode the order is 20 kV at both currents. In constant-reactance mode $V_q = I X_q$, giving 10 kV at 0.5 kA and 20 kV at 1.0 kA, where the converter reaches its voltage limit. Above 1.0 kA the emulated reactance can no longer be held.

Note. At very low current a constant-voltage order corresponds to a very large equivalent reactance V_q/I , so practical controllers use current thresholds and mode logic rather than extrapolating through zero current.

For balanced operation the converter rating is $S_{SSSC} = \sqrt{3} V_{q,LL} I$, where $V_{q,LL}$ is the injected line-to-line RMS voltage and I the line RMS current. This is a converter MVA rating, not a capacitor kvar rating. In constant-voltage mode the MVA demand grows in proportion to the line current. In constant-reactance mode $V_q = X_q I$, so the demand grows with I^2 until the voltage ceiling is reached; beyond that point the controller leaves the ideal constant-reactance characteristic and operates on its voltage limit. The same converter therefore follows different V-I trajectories under the two commands, a distinction that matters in both rating and power-flow studies.

7.5.4 Real power exchange and compensation of line resistance

With resistance R , the power delivered to the receiving end of an uncompensated line is

$$P = \frac{V^2}{X^2 + R^2} [X \sin \delta - R(1 - \cos \delta)] \quad (7.21)$$

Equation (7.21) is receiving-end power for equal terminal-voltage magnitudes with a series $R + jX$. An energy-supported SSSC can provide a series **rise** $\underline{V}_{src} = (V_p + jV_q)I/|I|$ for a positive compensating V_p and the appropriate quadrature polarity. With the drop convention used in the network, $\underline{V}_{inj} = -\underline{V}_{src}$; the effective impedance is $R - V_p/|I| + j[X - V_q/|I|]$. The physical line still dissipates $3I^2R$, while the source supplies $3V_p|I|$. Reducing reactance alone lowers effective X/R but need not reduce absolute received power; compute the operating point. Active and reactive damping contributions must be phase-coordinated and validated, rather than prescribed by angle direction alone.

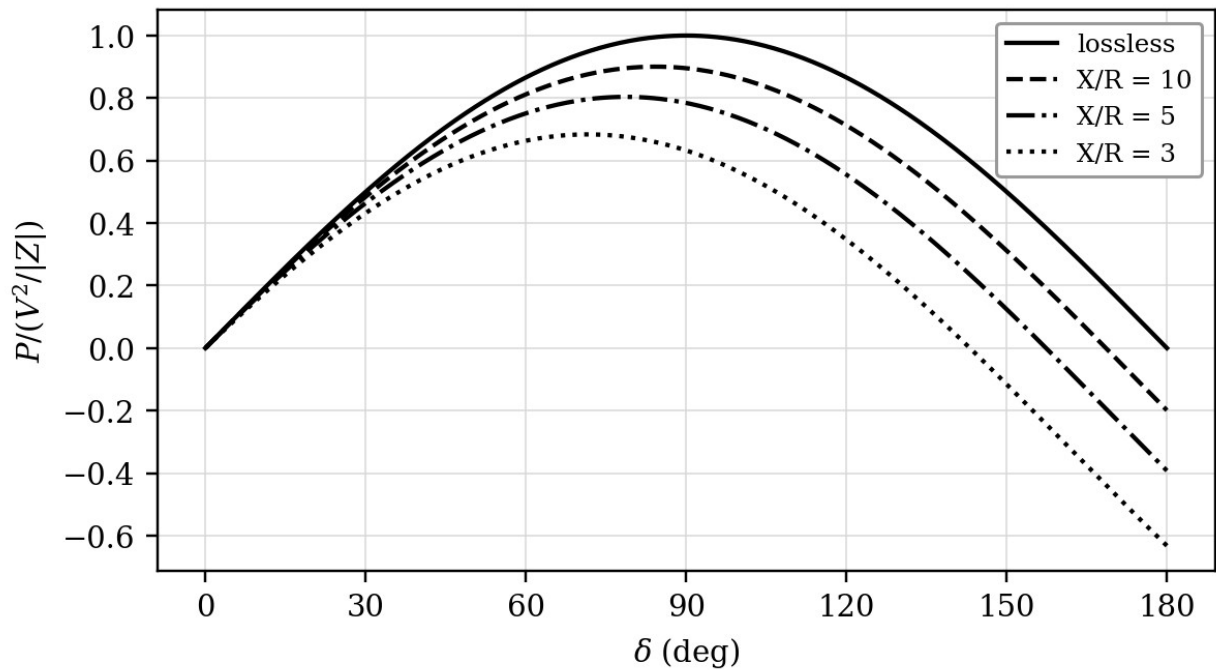


Figure 7.26. Normalized power-angle curves of a line with resistance for several X/R ratios.

Example 7.7. Active-power requirement for series resistance compensation

A line carries 800 A and has a series resistance of $1.5 \, \Omega$ per phase. Find the real power an energy-supported SSSC must supply to cancel the resistive voltage drop completely.

Solution. The in-phase voltage needed is $V_p = IR = 1.2 \, \text{kV}$ per phase, and the power supplied is $3 V_p I = 3 I^2 R = 3 \times 800^2 \times 1.5 = 2.88 \, \text{MW}$.

Note. This exchange is continuous, so it requires an external energy source or a power path to the DC side; a DC capacitor alone cannot sustain it.

7.5.5 Converter, transformer and protection

The converter is built like a STATCOM converter: early designs used multi-pulse two-level or three-level bridges with harmonic-neutralizing series transformers, and new designs use cascaded H-bridge or MMC topologies. The series transformer must carry full line current, including fault current, and its secondary is protected by a thyristor bypass switch and a breaker that short the winding during faults. The converter is thereby isolated from fault currents it could not interrupt. A transformerless variant floats the converter at line potential, as in the modular SSSC units now installed in several networks.

The converter cannot carry the prospective line fault current, so the fast thyristor bypass and the mechanical bypass breaker are an essential part of its protection, and the series transformer must survive the fault current until they operate. Transformerless modular designs remove the series transformer but not the need for fault bypass, insulation coordination and protection-interaction studies (Section 7.7.2).

Multilevel converters synthesize the injected voltage from many smaller voltage steps, reducing harmonic content and semiconductor voltage stress. Cascaded and modular structures can also provide redundancy because a failed module may be bypassed while the remaining modules continue at reduced capability.

A modular SSSC distributes the required series voltage among several converter modules. Instead of one large substation converter and series transformer, smaller controllable voltage sources can be deployed at selected locations and expanded or relocated as the network changes. The control problem then becomes coordinated: modules must share voltage orders, respect local current and thermal limits, and respond coherently to faults and protection commands.

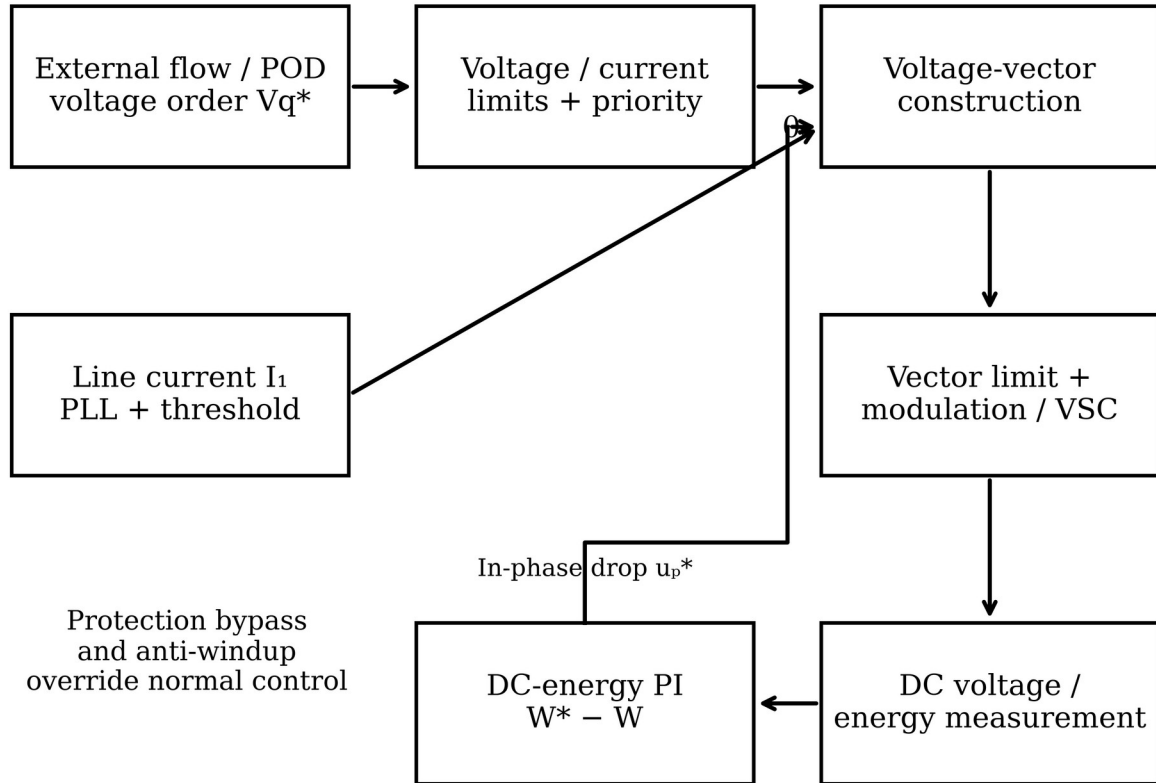
7.5.6 Series resonance and subsynchronous interactions

An ideal commanded series-voltage source does not introduce the literal $1/(sC)$ impedance of a fixed series capacitor. This removes that particular resonance mechanism, but its **closed-loop incremental impedance** is not necessarily only transformer leakage. DC-link dynamics, current synchronization, voltage modulation, control delays and any emulated reactance create frequency-dependent coupling. A subsynchronous AC component can exchange energy with the DC capacitor and create sidebands. Increasing capacitance may reduce ripple, but neither large capacitance nor fundamental quadrature injection proves SSR neutrality. Storage adds energy capability while introducing further controlled dynamics; assess the full terminal impedance and EMT response [12], [18], [20].

The absence of classical series resonance must not be read as immunity from every subsynchronous phenomenon. The converter, its synchronization loop, DC-link dynamics, modulation and limiters, together with the surrounding network, form a controlled dynamic system. That system can take part in subsynchronous or supersynchronous control interactions even though no physical capacitor reactance crosses the line inductive reactance. Section 7.7.3 develops the impedance-based analysis used to screen such interactions.

7.5.7 Control

The internal control of an SSSC (Figure 7.27) synchronizes to the line current through a PLL. It constructs the output voltage vector from a quadrature component V_q , demanded by the external control, and a small in-phase component V_p , demanded by the DC-voltage regulator to cover losses or by the external control when storage is available. With an *indirectly controlled* converter, the DC voltage is varied through a transient angle shift, as in the STATCOM of Chapter 6; with a *directly controlled* converter, the voltage magnitude is set by modulation at constant DC voltage. The external control regulates line power, current or effective impedance, and adds damping and transient-stability functions.



$$\text{Forward-current drop: } V_{inj}^* = (u_p^* - jVq^*) \exp(j\theta_i)$$

Figure 7.27. SSSC control with current-angle synchronization kept separate from voltage magnitude, connected DC-energy regulation and converter limits. A positive in-phase series drop absorbs charging power; a positive capacitive Vq gives a negative quadrature drop.

In both schemes the power-system controller should be kept separate from the fast converter controller. The former requests a series voltage, a power flow or an equivalent reactance; the latter produces a feasible voltage vector within modulation, current, DC-voltage and semiconductor limits. With indirect control, the switching pattern and the DC-link voltage set the synthesized AC voltage, and a small phase displacement from exact quadrature supplies the converter losses and regulates the DC-link energy. With direct control, PWM or multilevel modulation sets the output-voltage vector explicitly.

For a purely reactive SSSC without external storage, the steady-state injected voltage is maintained approximately in quadrature with fundamental line current. Average real power exchanged with the AC system is then only that required to cover losses. A small in-phase voltage component regulates the DC link. If an energy source is connected to the DC side, the in-phase component can become an intentional control variable and sustained active-power exchange becomes possible subject to converter MVA and stored-energy limits.

For steady power-flow control, the SSSC adjusts the sign and magnitude of quadrature voltage so that measured line power or current follows its reference. For damping, a supplementary controller modulates the voltage order with a small signal derived from local or remote measurements. Because the converter can reverse the sign of series voltage without crossing a capacitor resonance, it can move smoothly between capacitive and inductive action. During a large disturbance, a transient-stability controller may temporarily request maximum capacitive injection to raise electrical power during the first swing, provided current and voltage limits permit it.

These functions compete for the same converter capability. Practical priority logic allocates available series voltage among the steady power-flow order, damping signal, DC-voltage requirement and emergency support. If storage is present, active-power exchange introduces an additional energy constraint. A complete SSSC model therefore needs both an AC-side converter limit and a DC-side energy balance.

Energy and capability bridge. With the series-drop orientation used here, the AC power absorbed by a balanced converter is $P_{abs} = 3\Re(\underline{V}_{inj}\underline{I}^*)$. For a common DC link, $W = 1/2 C_{dc} V_{dc}^2$ and $\dot{W} = P_{abs} + P_{dc} - P_{loss}$, where P_{dc} is external power into the link. A positive in-phase drop absorbs charging power; a series rise exports stored power. Enforce $W_{min} \leq W \leq W_{max}$, charge/discharge power limits, voltage/current ceilings and $3|V_{inj}||I| \leq S_{rated}$. Floating-cell topologies additionally need individual-cell energy balancing.

7.5.8 Hybrid schemes

A fixed capacitor combined with an SSSC can provide bulk compensation and a controllable voltage reserve. The two must be rated on a common voltage, current and power base; a per-unit reactance and a per-unit voltage are not interchangeable. Targeted SSSC modulation can mitigate capacitor-related subsynchronous interactions, but this requires a validated damping design and protection study. Example 7.43 compares separate capacitor and SSSC solutions at one operating point; it does not itself size a hybrid.

7.6 Comparison and Selection

Figure 7.28 contrasts ideal limiting envelopes. A fixed capacitor or inserted TSSC stage has voltage proportional to current. A TCSC with finite boost also approaches zero voltage as current approaches zero; it is eventually voltage-limited at high current. An SSSC can command a voltage plateau above its minimum-current synchronization threshold. These are illustrative envelopes, not complete hardware V-I ratings. Table 7.4 summarizes the differences.

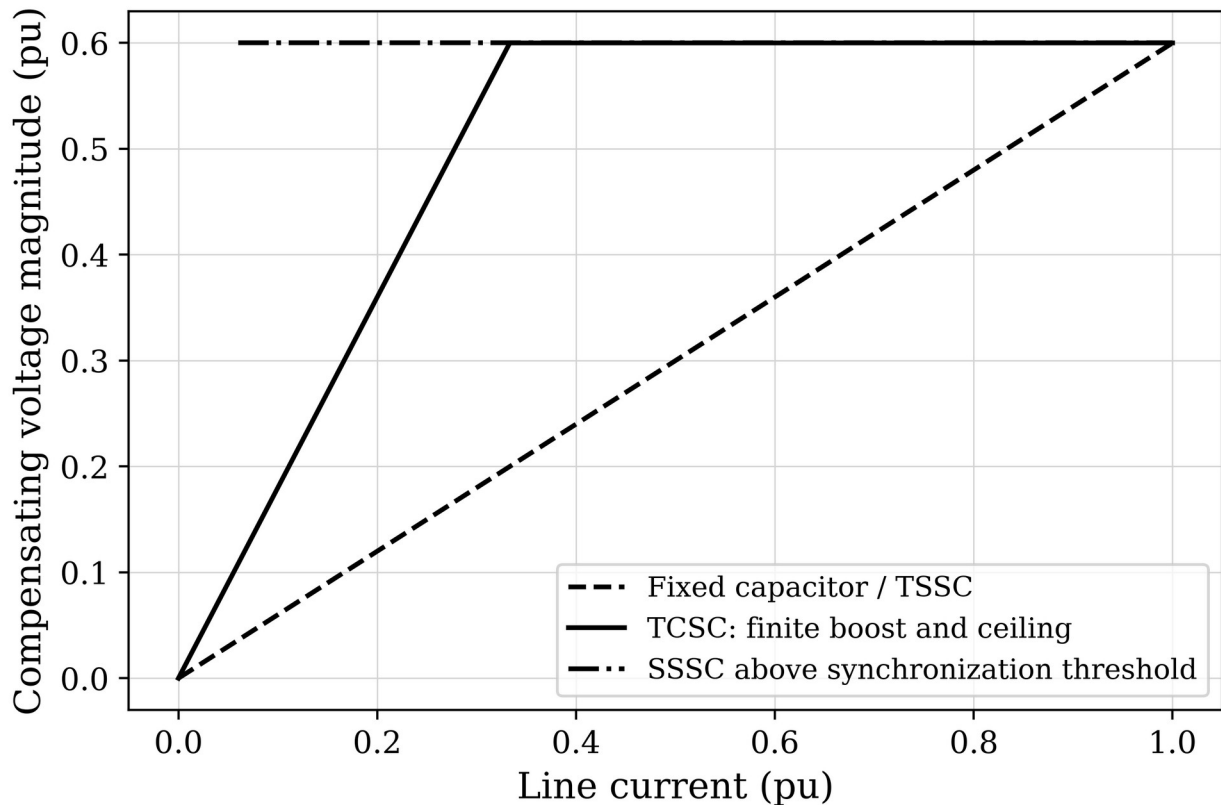


Figure 7.28. Illustrative compensating-voltage limits: finite-reactance capacitor/TSSC, finite-boost TCSC and a voltage-controlled SSSC above its synchronization threshold. No finite-boost TCSC sustains its voltage plateau at zero current.

Table 7.4 Comparison of series FACTS controllers.

Attribute	GCSC	TSSC	TCSC	SSSC
Control	Continuous, capacitive only	Stepped, capacitive	Continuous, capacitive boost and inductive	Continuous, capacitive and inductive, symmetric
Switching devices	GTO/IGCT (turn-off)	Thyristors	Thyristors	IGBT/IGCT converter

Attribute	GCSC	TSSC	TCSC	SSSC
Harmonics	Voltage harmonics (TCR dual)	None in steady state	Voltage harmonics, small at modest boost	Low with multilevel converter
Output at low current	Falls with current	Falls with current	Boost possible, falls eventually	Independent of current
SSR behaviour	Capacitive (control dependent)	Fixed-capacitor behavior; switching/control must be studied	Favorable behavior possible with validated control; not automatic	No literal fixed-capacitor resonance; controlled interactions remain
Power reversal	No	No	No (inductive vernier only reduces flow)	Yes
Active power	No	No	No	Yes, with DC source/sink, storage or linked converter
Response	Half cycle	Insertion: next current zero; low-stress bypass: next voltage zero	Firing and mode dependent; often approximated by a cycle-scale lag	Voltage/modulation, synchronization and bypass dependent
Installation	Platform at line potential	Platform	Platform	Series transformer and ground-level converter, or modular at line potential
Maturity	Concept and distribution use	Limited	Widely installed	Centralized and modular field applications; project-specific capability

The choice among them depends on what the compensation must achieve. For bulk transfer increase over long lines, fixed capacitors remain the cheapest per megavar, with a TCSC module added where damping or SSR control is needed. For power-flow control in meshed networks, where the flow may need to be reduced as well as increased, the SSSC's symmetric range and power-reversal ability are decisive. Where only coarse scheduling is required and SSR is not a concern, a TSSC or mechanically switched capacitor is adequate. Thyristor-controlled series reactors (TCSR) and thyristor-switched series reactors (TSSR), which control a series inductor instead of a capacitor, are mainly used for fault-current limiting and are treated with phase-angle regulators in Chapter 13.

Selection should be based on the required service rather than on a simple ranking, and Table 7.5 reorganizes the comparison in that way. The decisive questions are whether compensation must be continuous or stepped, whether the controller must both increase and decrease corridor flow, whether a classical series capacitor is acceptable from the SSR perspective, how much fault-current duty the series equipment must survive, and whether converter losses and protection complexity are justified by the additional controllability.

Table 7.5 Service-oriented comparison of the four series FACTS controller families.

Attribute	GCSC	TSSC	TCSC	SSSC	Engineering implication
Control resolution	Continuous capacitive	Discrete	Continuous with capacitive and inductive regions	Continuous voltage of either sign	Match resolution to flow-control and damping duty

Attribute	GCSC	TSSC	TCSC	SSSC	Engineering implication
Classical capacitor resonance	Present	Present	Present, with control-dependent behaviour	No literal fixed-capacitor element; emulated impedance remains dynamic	Still assess converter-control interactions
Low-current behaviour	Compensating voltage falls with current	Compensating voltage falls with current	Operating range depends on capacitor and reactor limits	Voltage-source command retained above minimum-current logic	Useful for bidirectional flow routing
Fault strategy	Fast bypass	Valve and breaker bypass	MOV, valve and bypass coordination	Converter bypass and relay coordination	Use EMT and protection studies
Key research issue	Stress and harmonics	Stage coordination	Dynamic model, SSR/SSCI and POD	Modularity, protection and converter-grid dynamics	Validate control and protection together

7.7 Recent Advances, Research Gaps and Emerging Directions

Sections 7.1 to 7.5 establish the circuit physics. Recent work extends them to modular power-flow routing, converter-related subsynchronous interactions, communication delay and protection-aware validation, issues that matter most where inverter-based generation and conventional machines share series-compensated networks. The models below are entry points for quantitative research, and their limitations matter as much as their numerical results. Table 7.6 summarizes the main gaps before they are examined in turn.

Table 7.6 Research gaps and possible contributions in series compensation.

Theme	Limitation of current practice	Potential contribution	Measurable impact
TCSC design models	Simplified LC model misplaces resonance	Exact or dynamic-phasor models in design tools	Correct mode and resonance margin
SSR and SSCI	Screening relies on case-by-case EMT studies	Impedance-based design of TCSC/SSSC damping	Positive net damping across 5 to 45 Hz
Modular SSSC fleets	Units dispatched individually	Optimal placement and coordinated dispatch with N-1 constraints	Congestion relief per installed MVA
Protection interaction	Converters often modelled as ideal shorts during faults	Bypass-aware relay studies and adaptive settings	Zone-1 security and dependability margins
Wide-area damping	Delay and loss of signals	Delay-robust, adaptive damping controllers	Damping ratio over the expected delay range

7.7.1 Modular and distributed series compensation for power-flow routing

A modular static synchronous series compensator (M-SSSC) distributes the series voltage of Section 7.5 among N small converter modules, each with its own bypass, mounted on the conductor or on a small platform. Field deployments now include a cross-border corridor of the Greek transmission system, where M-SSSC units are used to raise interconnection capacity and the integration of renewable generation [17], and several 220 kV circuits in Colombia, where operating experience with parallel lines has been reported [15]. Modularity changes the design question. The question is no longer only how far one line can be pushed, but how a fleet of voltage sources should be placed and dispatched in a meshed network.

If the active modules inject fundamental voltages aligned with a common reference, their contributions add:

$$\underline{V}_\Sigma = \sum_{k \in A} \underline{V}_k, |\underline{V}_k| \leq V_{k,max} \quad (7.22)$$

where $\underline{V}_\Sigma$ is the aggregate series voltage, A the set of available (non-bypassed) modules and $V_{k,max}$ the voltage rating of module k . A useful fleet metric is the headroom

$$h = 1 - \frac{|\underline{V}_\Sigma^i|}{\sum_{k \in A} V_{k,max}} \quad (7.23)$$

For equal co-phasal sharing, each active module supplies $\underline{V}_\Sigma^i/|A|$ while $|A| > 0$. The reserve h is zero at the aggregate voltage ceiling and negative for an infeasible order. If no modules are available, only zero injection is possible and the normalized headroom is undefined. Figure 7.29(b) uses six 8 kV units and a 24 kV order; three outages consume all voltage headroom. Current, thermal and fault limits can constrain sharing before this voltage ceiling is reached.

For flow routing use the signed **drop reactance** $X_{inj} = \Im(\underline{V}_{inj}/I)$, positive inductive. On the forward-current reference of Equation (7.19), $X_{inj} = -V_q/I$. This sign differs from the positive-capacitive voltage order. For two parallel paths with positive net reactances, fixed equal-end voltages and a common small angle, the DC power-flow approximation gives

$$P_1 = P_t \frac{X_2}{X_1 + X_{inj} + X_2}, \frac{\partial P_1}{\partial X_{inj}} = - \frac{P_t X_2}{(X_1 + X_{inj} + X_2)^2} \quad (7.24)$$

Here P_1 is path-1 flow and P_t the prescribed total. Inductive X_{inj} pushes flow to path 2; capacitive X_{inj} draws it into path 1. For $X_1 = 0.3$, $X_2 = 0.4$ and $P_t = 1.6$ pu, the natural $P_1 = 0.9143$ pu exceeds 0.85 pu. Solving $0.85 = 1.6(0.4)/(0.7 + X_{inj})$ gives $X_{inj} = 0.05294$ pu. With voltage near 1 pu, current near 0.85 pu gives a voltage-drop magnitude about 0.045 pu. This is an approximate routing calculation, not a solved AC OPF.

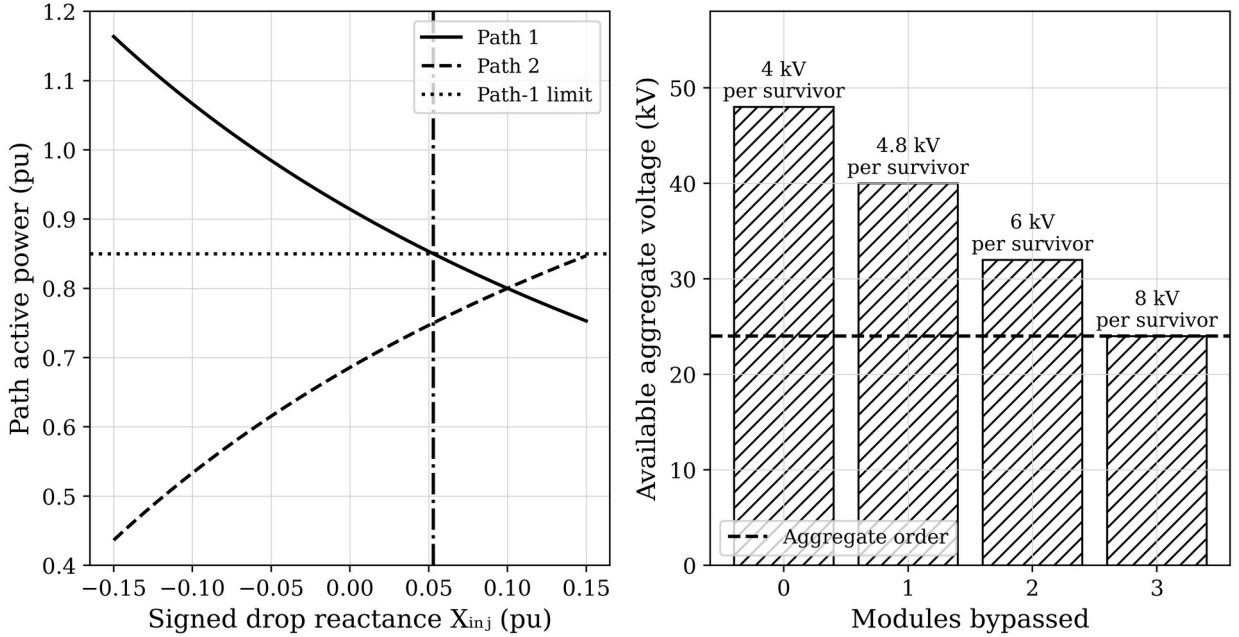


Figure 7.29. Parallel-path routing for $X_1 = 0.3$, $X_2 = 0.4$ and $P_t = 1.6$ pu (left), and voltage headroom of six 8 kV modules with a 24 kV aggregate order (right). Inductive signed drop reactance reduces path-1 flow.

At system scale, the fleet is dispatched by an optimal power flow in which the branch equation of each compensated line contains the aggregate injection:

$$V_s - V_r = Z_\ell I + V_\Sigma, |I_\ell^{(c)}| \leq I_{\ell, \max} \quad \forall \ell, \forall c \in C \quad (7.25)$$

Equation (7.25) uses V_Σ as an **algebraic series drop** in the branch orientation. The common-reference module phasors add; their individual voltage, current and MVA constraints must also be enforced. Each contingency needs its own available-module set and dispatch or permitted corrective action. Thermal inequalities alone do not establish N-1 security: include bus voltage, generator limits, power balance, converter bypass and protection constraints. The objective can be redispatch cost or curtailment; placement and sizing add integer decisions [17].

7.7.2 Interaction with line protection

A series device sits inside the measurement loop of the distance relays protecting its line. For a fault at a fraction m of the line beyond the device, the relay voltage contains the injected voltage, and the apparent impedance measured at the relay is

$$Z_{app} = \frac{V_{relay}}{I} = m Z_L + \frac{V_{inj}}{I} \quad (7.26)$$

This is a simplified single-source loop with the device between relay and fault, negligible fault resistance and no remote infeed. V_{inj} is the **series drop**, positive in the relay-to-fault direction. Capacitive drop $-j X_q I$ reduces apparent reactance; inductive drop $+j X_q I$ increases it. Figure 7.30 shows geometric overreach/underreach in this model. Actual tripping also depends on directional polarization, loop compensation, timers, fault resistance, infeed, MOV and bypass behavior; the phasor geometry alone is not a complete relay-performance test.

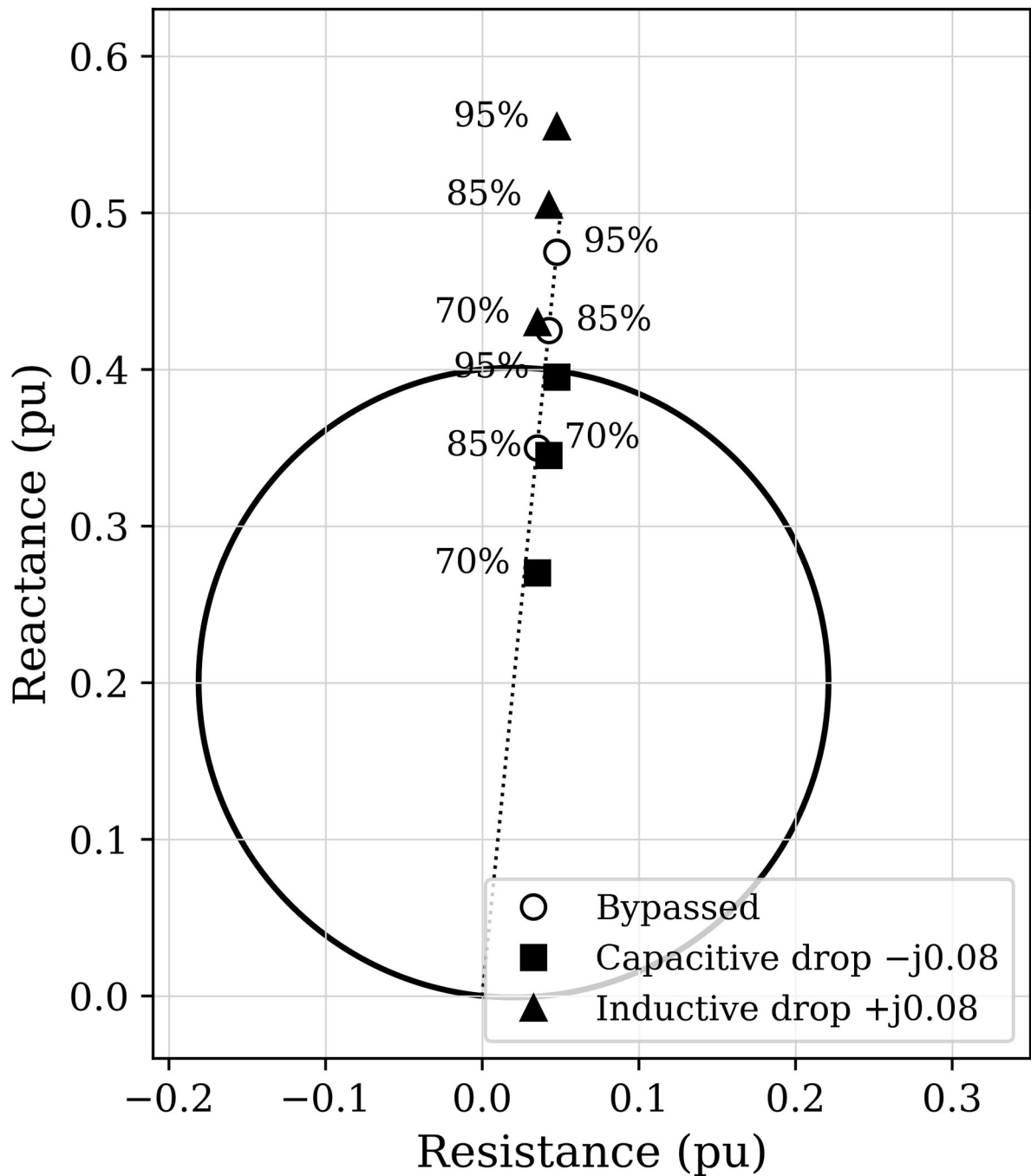


Figure 7.30. Apparent-impedance geometry for $Z_L = 0.05 + j0.5$ pu, an 80% zone-1 mho circle and series drops of $\pm j0.08$ pu. Open circles, filled squares and triangles remain distinct in black-and-white reproduction. This is a simplified single-source phasor screen.

Bypass timing matters because the injected voltage and measured impedance evolve during the relay window. Current limiting, blocking, MOV conduction, transformer leakage and bypass must be represented as applicable. Published SSSC operating experience and protection studies [15], [16] motivate EMT and relay-HIL assessment with the actual logic; replacing every converter by an instantaneous ideal short cannot establish zone-1 security.

7.7.3 SSR, SSCI and impedance-based stability analysis

Classical SSR (Section 7.1.5) couples an electrical series resonance with a turbine-generator torsional mode. Two further mechanisms dominate recent work. The *induction-generator effect* arises because a machine rotor turning near synchronous speed presents a negative resistance to subsynchronous stator currents. *Subsynchronous control interaction* (SSCI) arises when the current controllers of converter-interfaced generation, notably doubly fed

induction generators, add further negative resistance at subsynchronous frequencies. SSCI involves no mechanical mode at all, and it has caused damaging oscillations near series capacitors in several networks [19]. Grid-forming inverters connected radially through series capacitors can also exhibit SSR, so the problem does not disappear with the change of control paradigm [18].

A first-order screening criterion follows from the series resonant circuit. At the electrical resonance $f_e = f_0 \sqrt{k}$ (Equation (7.5)) the network reactance vanishes, and the oscillation is damped only if the net series resistance at that frequency is positive:

$$R_{net}(f_e) = R_\ell + R_s + \frac{R_r' f_e}{f_e - f_0} + R_c(f_e) > 0 \quad (7.27)$$

The screening model treats the machine as a simple induction equivalent with a nearly synchronous rotor, neglecting the magnetizing branch and frequency-dependent parameters. At stator frequency f_e the slip is $s = (f_e - f_0)/f_e$, so the rotor contributes $R_r'/s = R_r' f_e / (f_e - f_0)$, which is negative for $0 < f_e < f_0$. The converter term is an illustrative scalar resistance rather than a property of any particular controller. A positive R_{net} at one resonance is a screening result only; torsional modes, frequency coupling and contingencies require coupled models. Figure 7.31 uses $R_c(f) = -0.004(f/50)^4$ pu so that the example can be reproduced, and Example 7.11 gives the boundary without the converter term.

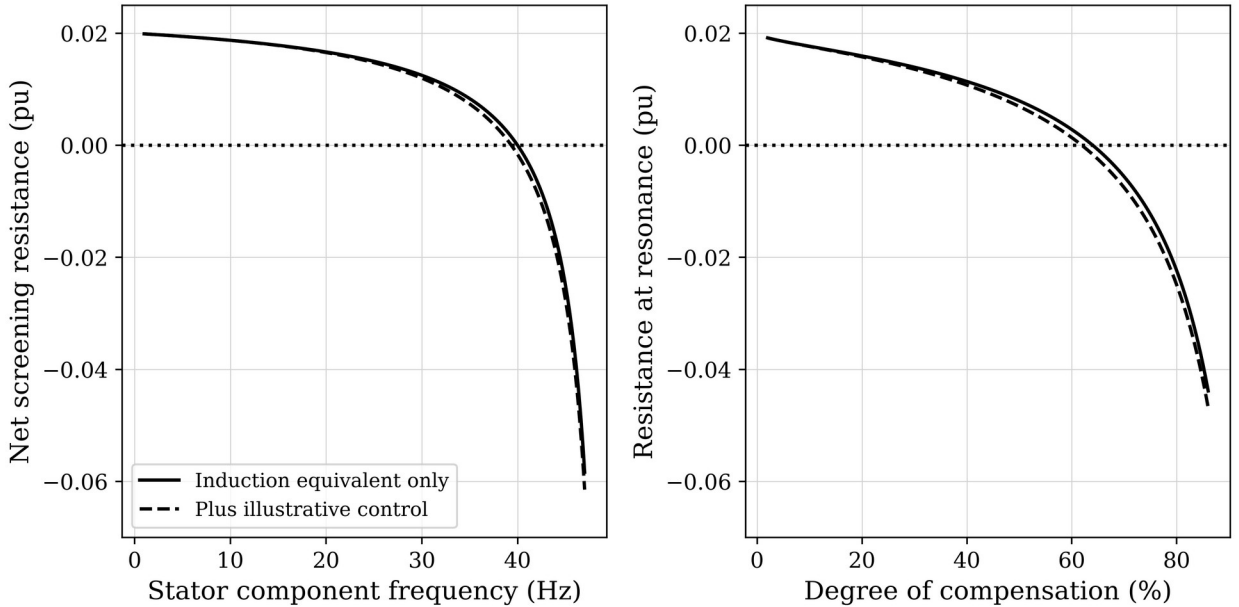


Figure 7.31. Scalar resistance screen with $R_{line} + R_s = 0.02$ pu, $R_r' = 0.005$ pu and optional illustrative $R_c(f) = -0.004(f/50)^4$ pu. The zero-resistance boundary is a screening flag, not a complete torsional or converter stability verdict.

For converter-dominated systems the scalar criterion generalizes to an impedance-ratio test. The grid and the converter are linearized around an operating point, and their terminal impedances $Z_g(s)$ and $Z_c(s)$ are obtained in the sequence or dq frame. The interconnection is stable if the minor-loop gain

$$L(s) = Z_g(s) Z_c^{-1}(s) \quad (7.28)$$

Equation (7.28) is a source-load interconnection test with consistent voltage/current orientation and a defined partition. Its Nyquist conclusion requires counting unstable open-loop poles; it is not simply a plot that must avoid -1 . A dq or coupled-sequence converter model is generally a matrix, so use the determinant/eigenlocus formulation with the correct pole count. Series capacitance changes the frequency-dependent grid impedance, but a magnitude crossing or negative resistance alone does not prove closed-loop instability. Use eigenvalues or EMT disturbances to verify candidate modes and to study saturation, bypass and large disturbances [20].

7.7.4 Wide-area damping and communication delay

A series controller damps an inter-area mode most effectively when it sits on a corridor with high modal observability and controllability, and remote synchrophasor signals often observe the mode better than local ones [21]. The price is communication delay. In the damping loop

$$\Delta V_q(s) = G_{POD}(s) e^{-sT_d} \Delta y(s) \quad (7.29)$$

where Δy is the measured oscillatory signal, G_{POD} the washout and phase-compensation network and T_d the end-to-end delay, the delay adds a phase lag that grows linearly with frequency:

$$\phi_d = -360^\circ f_m T_d, e^{-sT_d} \approx \frac{1 - sT_d/2}{1 + sT_d/2} \quad (7.30)$$

where f_m is the modal frequency; the second expression is the first-order Padé approximation used to include the delay in linear models. Delay compensation in G_{POD} restores the phase at one frequency only, and network conditions shift the modal frequency, so robust designs are evaluated over a range of delays and operating points [22].

The effect can be seen in a single-mode model. Representing an electromechanical mode of natural frequency $\omega_n = 2\pi f_m$ and open-loop damping ζ_0 , with the series controller providing speed-proportional damping of gain K through the delayed channel, the closed-loop characteristic equation is

$$s^2 + 2\zeta_0\omega_n s + \omega_n^2 + K s e^{-sT_d} = 0 \quad (7.31)$$

Use $K = 2(0.15 - \zeta_0)\omega_n$ with $\zeta_0 = 0.02$. Figure 7.32 plots the tracked electromechanical root of the **exact exponential-delay** equation, not a Padé approximation. At 200 ms, the 0.4 Hz mode has about 14.0% damping, while the 1 Hz root falls below 5% near 190 ms and crosses into instability before 250 ms. A delay equation has infinitely many roots: this tracked branch illustrates phase sensitivity, and an engineering stability assessment must also search for other unstable roots. Loss-of-signal fallback and delay robustness are controller requirements, not consequences of this single-mode plot.

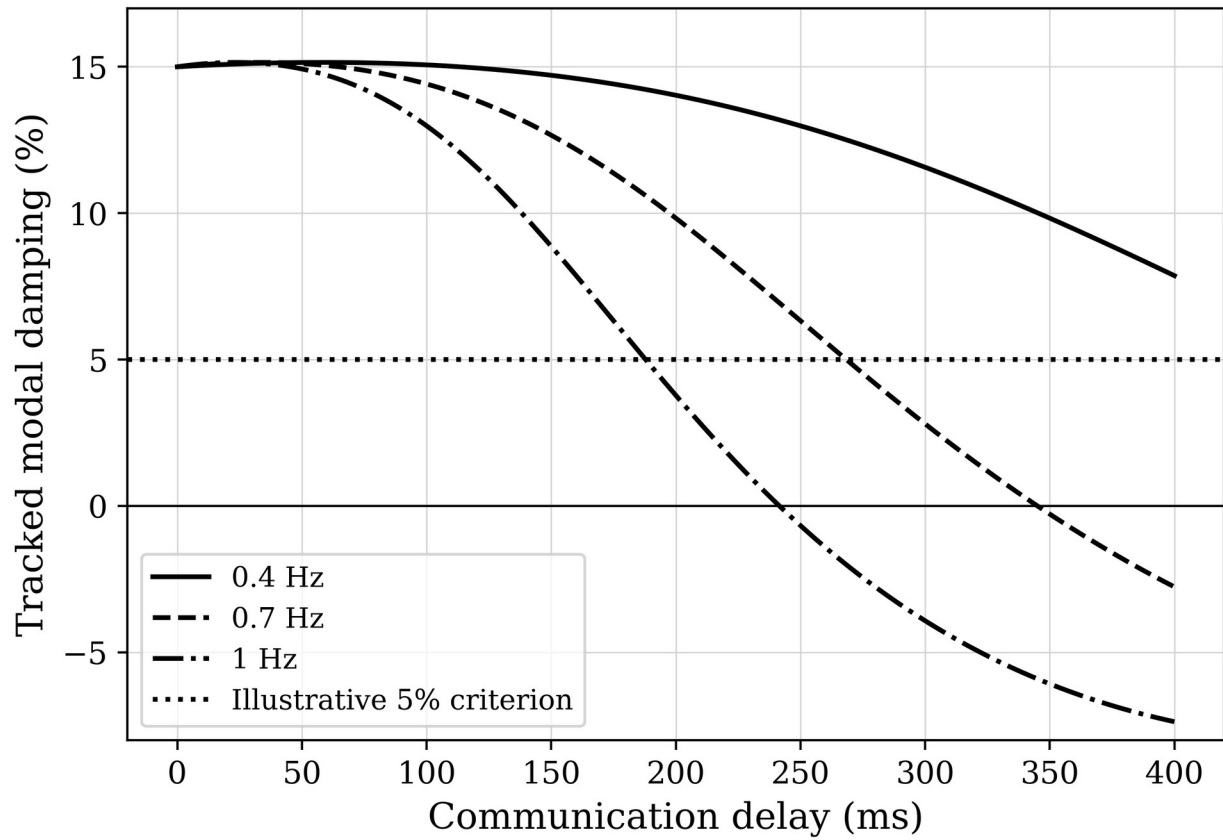


Figure 7.32. Tracked electromechanical-root damping of the exact exponential-delay equation, $\zeta_0 = 0.02$ and $K = 2(0.15 - \zeta_0)\omega_n$. The 5% line is an illustrative criterion; other delay-equation roots must also be assessed.

7.7.5 Model fidelity and validation

Model fidelity follows from the phenomenon being claimed (Figure 7.33 and Table 7.7). IEC 60143-4:2023 replaces the earlier network-simulator test of TCSC control and protection with a hardware-in-the-loop test [13]. Published M-SSSC EMT models support offline and real-time studies [14], although a published model is not automatically a validated replica of a given commercial unit. Controller HIL and protection HIL answer different questions and are complementary rather than ranked.

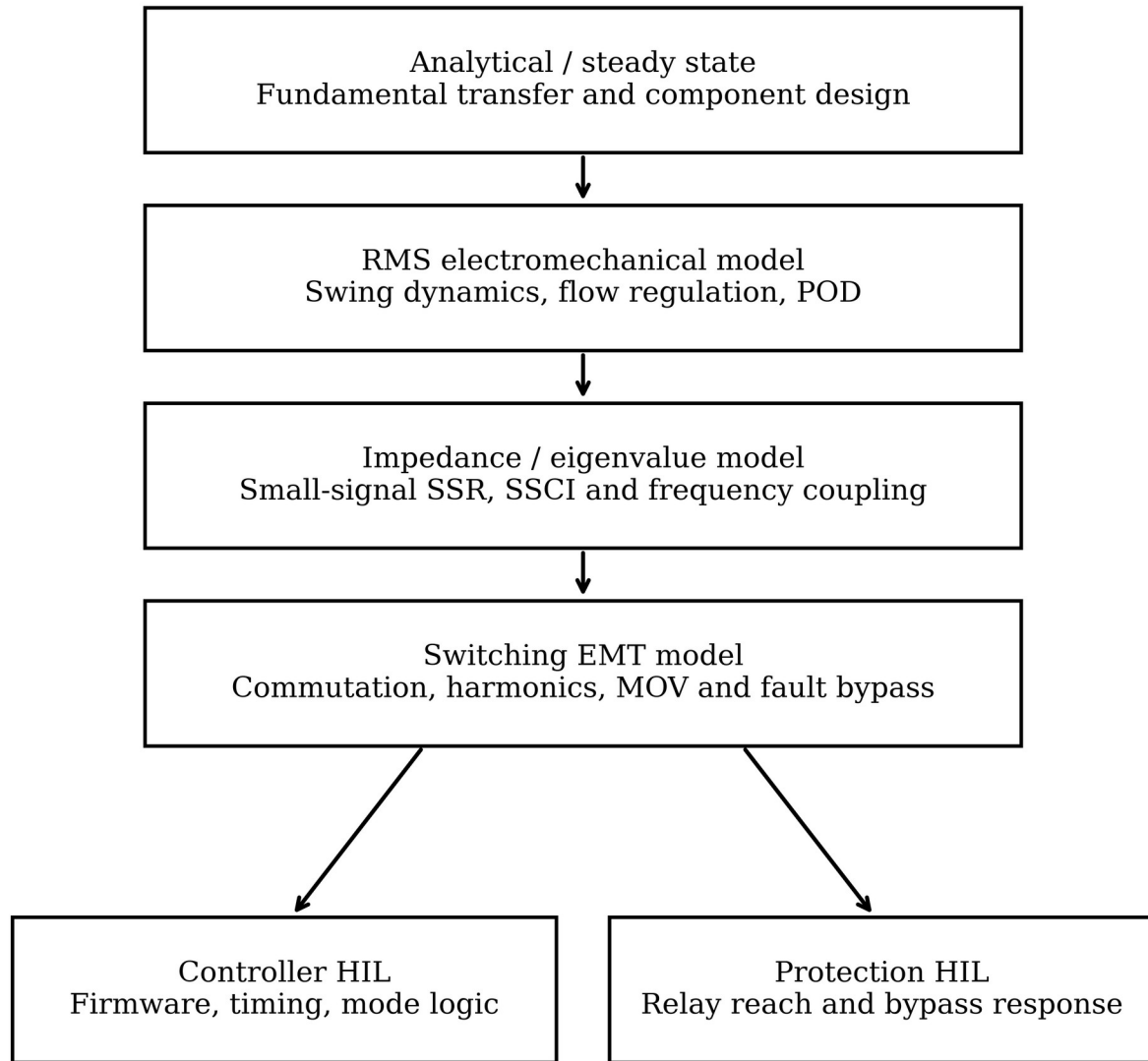


Figure 7.33. Study and validation paths. Controller HIL and protection HIL are complementary tests of different real hardware; combined tests may use the same validated EMT plant.

Table 7.7 Model fidelity and the claims each level can support.

Model level	Represents	Suitable claims	Cannot establish
Analytical / steady state	Fundamental-frequency phasors	Transfer capability, ratings, resonance boundary	Dynamics, harmonics, protection
Phasor (RMS) dynamic	Electromechanical dynamics, averaged controls	POD tuning, transient stability	SSR/SSCI, switching events
Small-signal impedance	Linearized converter and network over a specified validated frequency band	SSR, SSCI, PLL interaction screening	Limiter action, large disturbances
Switching EMT	Valves, firing or modulation, MOV, bypass	Harmonics, fault recovery, protection duty	Behaviour of the real controller code
Controller HIL	Real control hardware with simulated plant	Timing, mode logic, limits, firmware	Relay response
Protection HIL	Real relays with simulated converter and line	Reach, directionality, bypass timing	Long-term planning metrics

7.7.6 Open research questions

The most productive research questions follow from limitations established earlier in this chapter. Table 7.8 lists representative questions together with a suggested method and a measurable outcome. A strong contribution

identifies one such limitation, proposes a controller, model or topology that addresses it, and demonstrates the improvement over several operating conditions and at more than one level of the validation ladder.

Table 7.8 Representative research questions in series compensation.

Research question	Why it matters	Suggested method	Measurable outcome
Reduced-order TCSC models that keep the exact resonance boundary	Simplified models misplace resonance (Section 7.4.3)	Dynamic phasors, harmonic-domain models	Error in β_r and in subsynchronous impedance
Robust POD for series devices under changing grid strength and delay	Damping varies with T_d and operating point (Figure 7.32)	H_∞ , gain scheduling, data-driven tuning	Minimum damping ratio over the operating set
Coordinated dispatch of M-SSSC fleets with N-1 security	Headroom is consumed by module outages (Figure 7.29b)	Security-constrained OPF with fleet constraints	Congestion relief per installed MVA
Protection-aware series converter control	Injection shifts relay reach (Figure 7.30)	EMT and protection HIL, adaptive bypass	Zone-1 overreach and underreach margins
SSCI-robust converter and series-device control	Negative resistance near f_e (Figure 7.31)	Impedance shaping, passivity-based design	Net resistance or passivity margin across 5 to 45 Hz

Worked examples for Section 7.7

Example 7.8. Modular SSSC loss of one module

A four-module SSSC must inject 24 kV line-to-line. The modules share voltage equally and each is rated 8 kV. If one module is bypassed, determine the voltage required from each remaining module and state whether the original command remains feasible.

Solution. Initially each module supplies $24/4=6$ kV and the headroom from Equation (7.23) is $1 - 24/32 = 25\%$. With one module bypassed, the remaining three must each supply $24/3=8$ kV. The command is still feasible, but the headroom is now zero and the survivors operate exactly at their rating.

Note. A single module outage can consume all control headroom even when the aggregate command remains achievable, which is why fleet dispatch must include module availability.

Example 7.9. Communication delay in a wide-area POD loop

A remote damping signal contains a 0.7 Hz inter-area oscillation. Estimate the additional phase lag caused by communication delays of 50 ms and 150 ms.

Solution. From Equation (7.30), $\phi_d = -360 \times 0.7 \times 0.05 = -12.6^\circ$ for 50 ms and -37.8° for 150 ms.

Note. A delay that looks small in milliseconds can consume a large part of the phase compensation designed for an inter-area mode.

Example 7.10. Distance-relay reach with series injection

A line has $Z_L = 0.05 + j0.5$ pu and a zone-1 mho relay set at 80% of the line, with its characteristic passing through the origin. An SSSC between the relay and the line injects $X_q = 0.08$ pu. Determine whether zone 1 operates for (a) a fault at 85% with capacitive injection and (b) a fault at 70% with inductive injection.

Solution. The mho circle has centre $0.201 \angle 84.3^\circ = 0.020 + j0.200$ pu and radius 0.201 pu. (a) $Z_{app} = 0.0425 + j0.425 - j0.08 = 0.0425 + j0.345$ pu, at a distance of 0.147 pu from the centre: inside zone 1, so the relay trips instantaneously for a fault beyond its reach (overreach). (b) $Z_{app} = 0.035 + j0.35 + j0.08 = 0.035 + j0.43$ pu, at 0.230 pu from the centre: outside zone 1, so a fault that zone 1 should clear is delayed to zone 2 (underreach).

Note. Bypass removes the commanded injection, but transformer/leakage impedance, fault-loop compensation and relay filtering can remain. The measurement window and the complete bypass transition must be evaluated together.

Example 7.11. Compensation limit from the induction-generator effect

For $R_\ell + R_s = 0.02$ pu and $R_r' = 0.005$ pu at 50 Hz, find the compensation level at which the net resistance of Equation (7.27) becomes zero, ignoring converter control.

Solution. In the stated scalar equivalent, $0.02 + 0.005 f_e / (f_e - 50) = 0$ gives $f_e = 40$ Hz and $k = (40/50)^2 = 0.64$. Above 64%, this model has negative net series resistance at its electrical resonance; that is a screening flag, not a complete shaft or converter stability result.

Note. Converter control can move this boundary; with the illustrative contribution of Figure 7.31 it falls to about 62%.

Example 7.12. Delay tolerance of a wide-area damping loop

For the single-mode model of Equation (7.31) with $\zeta_0 = 2\%$ and the gain tuned for 15% damping without delay, find the damping ratio of a 0.7 Hz mode for delays of 150 ms and 200 ms.

Solution. Solving the exact exponential-delay equation for the tracked electromechanical root gives damping ratios of 12.7% at 150 ms and 9.8% at 200 ms. The delay phases are -37.8° and -50.4° . These values describe the tracked root; they do not replace a search for other roots of the delay equation.

Note. A lead chosen to cancel the delay phase at 0.7 Hz does not necessarily restore a 15% root damping ratio: the closed-loop root shifts and the compensator changes gain and phase away from that frequency. Recompute the roots across the expected frequency, delay and operating-point range.

Key Equations Recap

Table 7.9 Key equations of Chapter 7.

Equation	Expression	Conditions
(7.3)	$I = \frac{2V}{(1-k)X} \sin \frac{\delta}{2}, P = \frac{V^2}{(1-k)X} \sin \delta$	Symmetric lossless line, $k = X_c/X$
(7.4)	$Q_c = \frac{2V^2}{X} \frac{k}{(1-k)^2} (1 - \cos \delta)$	Series capacitor reactive power
(7.5)	$f_e = f_0 \sqrt{k}$	Electrical resonance; rotor sees $f_0 - f_e$
(7.6)	$v_c = \frac{I}{\omega C} (\sin \omega t - \sin \gamma)$	GCSC, $\gamma \leq \omega t \leq \pi - \gamma$
(7.8)	$X_c(\gamma) = X_c \left(1 - \frac{2\gamma}{\pi} - \frac{\sin 2\gamma}{\pi} \right)$	GCSC fundamental reactance
(7.9)	$V_{c,rms} = \frac{I}{\omega C} \sqrt{\frac{(\pi - 2\gamma) \left(\frac{1}{2} + \sin^2 \gamma \right) - \frac{3}{2} \sin 2\gamma}{\pi}}$	GCSC total RMS voltage
(7.11)	$v_c = I_m X_c (1 - \cos \omega t)$	TSSC insertion at current zero

Equation	Expression	Conditions
(7.14)	$X_{TCSC} = \frac{X_C X_L(\alpha)}{X_C - X_L(\alpha)}$	Simplified model only
(7.16)	$X_{TCSC} = -X_C + C_1(2\beta + \sin 2\beta) - C_2 \cos^2 \beta (\omega \tan \omega \beta - \tan \beta)$	Ideal periodic sinusoidal-current model; β half conduction; exclude poles and check natural commutation
(7.18)	$\beta_r = \pi / (2\omega)$	TCSC resonance (first)
(7.20)	$P_q = \frac{V^2}{X} \sin \delta + \frac{V V_q}{X} \cos \frac{\delta}{2}$	Fixed-axis signed V_q ; per-phase or common pu base; reversal needs mode/synchronization logic
(7.21)	$P = \frac{V^2}{X^2 + R^2} [X \sin \delta - R(1 - \cos \delta)]$	Line with resistance

Common Misconceptions

“Series compensation controls power flow.” Fixed series compensation raises transfer capability, but the actual flow is still set by the angle and by the network. Control of flow requires a controllable device.

“A frequency screen establishes SSR safety.” The electrical frequency and its rotor-frame complement locate possible coincidences. They do not establish modal coupling, damping or the response of a controlled converter. Repeat the screen for network contingencies, then use the appropriate coupled model.

“The simplified TCSC formula is good enough for design.” It misplaces the resonance and can put an operating point on the wrong side of it. Use the exact characteristic, Equation (7.16), or a validated time-domain model.

“A TCSC is automatically SSR-neutral.” Favorable subsynchronous behavior is a property of a particular controller, operating band and network. Fundamental-frequency synchronization is important, but closed-loop impedance and coupled torsional validation are still required.

“A TSSC can be inserted at any instant.” Thyristors commute only at current zero, which inserts the capacitor with a DC offset unless it is pre-charged. Bypassing should be done at capacitor voltage zero.

“The SSSC power term needs a unit conversion factor.” With line-to-line kilovolts and ohms, both V^2/X and $V V_q/X$ come out directly in megawatts for the three-phase system. An extra factor, as in some worked solutions, overstates the SSSC contribution tenfold (1108.7 MW instead of 110.9 MW in Example 7.40).

“An SSSC’s DC capacitor supplies its balanced reactive power.” An ideal balanced three-phase common link need not buffer fundamental reactive power, but it buffers real-power mismatch, losses and ripple. Individual floating cells and converter arms can have substantial internal energy ripple even during reactive operation. Sustained intentional active power needs a source/sink or other DC power path.

Chapter Summary

Series compensation reduces transfer reactance and can raise ideal transfer limits, subject to current, voltage, protection and stability constraints. The GCSC controls a current-driven capacitor through turn-off timing; the TSSC switches complete stages with natural commutation and possible insertion offsets. The TCSC boosts capacitor voltage through finite-time charge reversal, requiring an exact waveform model for resonance and component stress. Branch current and individual-thyristor current must be distinguished. The SSSC provides either series-voltage polarity without a literal fixed-capacitor resonance; current reversal needs explicit

reference and synchronization logic. Storage enables sustained active exchange subject to power and energy bounds. Every family requires a model appropriate to its control, subsynchronous and protection claims.

Solved Numerical Practice Problems

Unless specified otherwise, $f_0=50$ Hz and $\omega=314.159$ rad/s. Line-to-line RMS kilovolts and ohmic reactances give balanced three-phase megawatts through $P=V_{LL}^2 \sin \delta / X$. For ideal TCSC periodic pulses, β is half conduction and $\alpha=90^\circ - \beta$ is the complementary peak-referenced delay; neither is the zero-referenced TCR angle of Chapter 6. Peak and RMS current are stated in each example. A computed fundamental var rating is not a complete harmonic, fault or thermal rating.

Example 7.13 Rating a series capacitor for a 50% power increase

A 400 kV line has $X=80 \Omega$ and equal end voltages and operates at $\delta=30^\circ$. Find (a) the line current; (b) the transmitted power; (c) the reactive power drawn by the line from each end; (d) the degree of compensation for a 50% increase in power at the same angle; (e) the new current; (f) the capacitor reactance and three-phase rating.

Solution. (a) $V_{ph}=230.94$ kV; $I=2 V_{ph} \sin 15^\circ / X=1494.3$ A.

(b) $P=400^2 \sin 30^\circ / 80=1000$ MW (three-phase).

(c) $Q=V^2(1 - \cos \delta) / X=267.9$ Mvar at each end.

(d) $1/(1 - k)=1.5$, so $k=1/3$.

(e) $I'=I/(1 - k)=2241.4$ A.

(f) $X_C=26.67 \Omega$; $Q_C=3 I'^2 X_C=401.9$ Mvar, which agrees with Equation (7.4).

Note. The capacitor must be rated for the boosted current, not the original one, which is why a 50% transfer gain at the same angle costs about 400 Mvar of series capacitors.

Example 7.14 Current, power and capacitor duty at 40% compensation

A 220 kV line ($X=50 \Omega$) has $k=0.4$ and operates at $\delta=45^\circ$. Find (a) X_{eff} ; (b) the current; (c) the power; (d) the capacitor voltage; (e) the capacitor rating.

Solution. (a) $X_{eff}=30 \Omega$.

(b) $I=2(127.02 \text{ kV}) \sin 22.5^\circ / 30=3240.5$ A.

(c) $P=220^2 \sin 45^\circ / 30=1140.8$ MW.

(d) $V_C=I X_C=3240.5 \times 20=64.81$ kV per phase.

(e) $Q_C=3 I^2 X_C=630.0$ Mvar.

Note. Differential capacitor voltage governs segment insulation and MOV duty. Platform-to-ground insulation is coordinated separately with its location on the transmission line and the line-to-ground voltage; it is not fixed solely by the 64.8 kV capacitor differential voltage.

Example 7.15 Midpoint voltage and power of a compensated line

A symmetric lossless line has $V_s=V_r=1.0$ pu, $X=0.5$ pu and $k=0.3$. Find its electrical-midpoint reference voltage, transmitted power and uncompensated power at $\delta=30^\circ$, 60° and 90° , and then its maximum ideal power and percentage increase.

Solution. $X_{eff}=0.35$ pu.

(a) 30° : $V_m=0.966$ pu, $P=1.429$ pu, uncompensated 1.000 pu.

(b) 60° : $V_m=0.866$ pu, $P=2.474$ pu, uncompensated 1.732 pu.

(c) 90° : $V_m=0.707$ pu, $P=2.857$ pu, uncompensated 2.000 pu.

- (d) $P_{max}=2.857$ pu, 42.9% above 2.0 pu. At a fixed angle the symmetric electrical-midpoint reference $V \cos(\delta/2)$ is unchanged; the physical capacitor terminals are distinct and must be found from the half-line voltage drops.

Note. The ideal electrical-midpoint relation is a useful reference, not a statement that every physical midpoint bus or capacitor-terminal voltage is unaffected by series compensation.

Example 7.16 Voltage at the capacitor terminal

Two buses with $V_1=1 \angle 30^\circ$ and $V_2=1 \angle 0^\circ$ pu are connected through a generator reactance $X_g=0.1$ pu at each end and a line of $X_L=1.0$ pu. A series capacitor is placed next to the sending-end machine. For $X_C=0, 0.3, 0.4, 0.5, 0.6$ and 0.7 pu find (a) the net reactance; (b) the current; (c) the voltage at the line-side capacitor terminal F; (d) the degree of compensation; (e) the practical conclusion.

Solution. (a) $X_{net}=1.2 - X_C$.

(b) $I=|V_1 - V_2|/X_{net}=0.5176/X_{net}$, at 15° : 0.431, 0.575, 0.647, 0.739, 0.863 and 1.035 pu.

(c) $V_F=V_1 - jI(X_g - X_C)$: 0.990, 1.036, 1.067, 1.114, 1.187 and 1.307 pu.

(d) 0 to 70%.

(e) At 70% compensation of the line alone the current rises by 140% and the line-side capacitor terminal reaches 1.307 pu. These figures flag the insulation and current duty; the permissible compensation still has to come from a full study. Relative to the entire 1.2 pu path, the compensation is only 58.3%.

Note. The overvoltage is local to the capacitor terminal: with both bus voltages at 1.0 pu, the line-side terminal of a heavily compensated segment can exceed 1.3 pu.

Example 7.17 SSR frequency screening

A path of total reactance $X=60 \Omega$ at 50 Hz is compensated 40%. A turbine-generator has a torsional mode at 32 Hz. Find (a) X_C ; (b) f_e ; (c) $f_0 - f_e$; (d) whether that complement coincides with 32 Hz; (e) the compensation for exact 32 Hz coincidence; (f) the compensation for 20 Hz coincidence. Explain the limitation of this screen.

Solution. (a) 24Ω .

(b) $f_e=50\sqrt{0.4}=31.62$ Hz.

(c) 18.38 Hz.

(d) The 18.38 Hz rotor-frame complement does not coincide with 32 Hz. This reduces concern about that particular coincidence, but says nothing conclusive about excitation or damping without the coupled machine/network model.

(e) $f_e=50 - 32=18$ Hz, so $k=(18/50)^2=12.96\%$.

(f) For a 20 Hz complement, electrical resonance is 30 Hz and $k=36\%$. The 40% operating point is near that frequency coincidence; modal damping and contingencies must determine whether it is acceptable.

Note. Screening must use the complementary frequency seen by the rotor, and it must be repeated for every outage that changes the total series reactance of the path.

Example 7.18 Critical clearing angle with series compensation

$P_m=0.6$ pu is delivered to an infinite bus. Without compensation $P_{max}=1.0$ pu; with $k=1/3$ it is 1.5 pu. During the fault $P_{max,f}=0.4$ pu; the post-fault network equals the pre-fault network. Find (a) δ_0 and δ_{max} without compensation; (b) the critical clearing angle; (c) δ_0 and δ_{max} with compensation; (d) the critical clearing angle; (e) the critical clearing angle if the controlled compensation keeps the fault curve at 0.6 pu.

Solution. The equal-area condition gives

$$\cos \delta_{cr} = \left[P_m (\delta_{max} - \delta_0) + P_{max} \cos \delta_{max} - P_{max,f} \cos \delta_0 \right] / (P_{max} - P_{max,f}).$$

- (a) $\delta_0 = 36.87^\circ$, $\delta_{max} = 143.13^\circ$.
- (b) $\cos \delta_{cr} = [0.6(1.8546) - 0.8 - 0.32] / 0.6 = -0.0121$: $\delta_{cr} = 90.7^\circ$.
- (c) $\delta_0 = 23.58^\circ$, $\delta_{max} = 156.42^\circ$.
- (d) $\delta_{cr} = 108.6^\circ$.
- (e) $\delta_{cr} = 126.4^\circ$. This is an ideal comparison in which the controlled device is assumed to keep the fault transfer maximum at 0.6 pu; real protection may instead bypass it. The result must not be read as demonstrated fault ride-through capability.

Note. The larger critical clearing angle follows directly from the raised post-fault curve; a controlled device that also lifts the fault-on curve gains a further 18 degrees.

Example 7.19 Series versus shunt support for a 30% increase

A 220 kV line ($X = 50 \Omega$) operates at $\delta = 45^\circ$. Find (a) the compensation for a 30% increase at the same angle; (b) X_C ; (c) the new current; (d) the capacitor rating; (e) the largest increase an ideal midpoint shunt compensator holding 1.0 pu could give at this angle; (f) the midpoint voltage a shunt device would need.

Solution. (a) $k = 1 - 1/1.3 = 0.2308$.

- (b) 11.54 Ω .
- (c) 2527.6 A.
- (d) 221.1 Mvar.
- (e) $P_{mid}/P = 2 \sin 22.5^\circ / \sin 45^\circ = 1.082$: only 8.2%.
- (f) $V_m/V = 1.3 \cos 22.5^\circ = 1.201$ pu. This exceeds a usual nominal-voltage support objective; permissibility depends on the stated voltage limits. At this fixed angle, an ideal midpoint shunt device constrained to 1.0 pu can give only an 8.2% increase.

Note. The comparison holds the end voltages, the angle and the shunt-voltage target fixed. It explains the series advantage for this transfer objective, not a general economic preference.

Example 7.20 GCSC reactance at several delays

A GCSC has $C = 1500 \mu\text{F}$. Find (a) X_C ; (b) $X_C(\gamma)$ at 20° , 50° and 70° ; (c) the equivalent capacitances; (d) the blocking angles.

Solution. (a) 2.1221 Ω .

- (b) Equation (7.8): factors 0.5732, 0.1310 and 0.0176, so 1.2163, 0.2779 and 0.0374 Ω .
- (c) $1/(\omega X)$: 2617, 11453 and 85147 μF .
- (d) $\xi = \pi - 2\gamma$: 140° , 80° , 40° .

Note. The equivalent capacitance becomes very large as the turn-off delay approaches 90 degrees, which is another way of saying that the capacitor is bypassed for almost the whole cycle.

Example 7.21 GCSC voltages, harmonics and THD

A GCSC with $C = 1000 \mu\text{F}$ carries a line current of 20 A peak. For $\gamma = 10^\circ$, 25° , 45° and 60° find (a) the base voltage $I/\omega C$; (b) the fundamental RMS voltage; (c) the total RMS voltage; (d) the 3rd, 5th and 7th harmonics; (e) the THD; (f) a check of the RMS formula.

Solution. (a) 63.66 V.

- (b) Equation (7.7) divided by $\sqrt{2}$: 35.11, 21.53, 8.18 and 2.60 V.

- (c) Equation (7.9): 35.33, 22.45, 9.56 and 3.58 V.
- (d) 10° : 3.17, 1.75, 1.09 V; 25° : 6.01, 1.89, 0.17 V; 45° : 4.78, 0.96, 0.68 V; 60° : 2.07, 1.24, 0.44 V.
- (e) 11.1%, 29.4%, 60.4% and 95.2%.
- (f) Integrating Equation (7.6) independently reproduces these RMS values. Replacing $\sin^2 \gamma$ by $\sin^2(2\gamma)$ would give about 41.2 V instead of 22.45 V at 25° . The incorrect expression agrees only at the endpoints and at the interior angle 60° ; one check there cannot validate it.

Note. THD rises because the fundamental decreases faster than the harmonics. Capacitor RMS-current/voltage, peak voltage and harmonic losses must be checked; semiconductor blocking is principally a peak-voltage and transient-sharing duty, not just a total-RMS voltage rating.

Example 7.22 GCSC delay for a required reactance

A GCSC with $C = 1000 \mu\text{F}$ must insert 1.5Ω . Find (a) X_C ; (b) the required ratio; (c) γ ; (d) the blocking angle.

Solution. (a) 3.1831Ω .

- (b) $X_C(\gamma)/X_C = 0.4712$.
- (c) Solving Equation (7.8) numerically: $\gamma = 25.39^\circ$.
- (d) $\xi = 129.2^\circ$.

Note. Because the reactance law is nonlinear, the controller stores this inverse relation as a look-up table or solves it numerically at each control step.

Example 7.23 GCSC in voltage compensating mode

A GCSC must maintain 5 kV (peak fundamental) for line currents from 100 A to 400 A peak. Find (a) X_C ; (b) C ; (c) γ at 400 A; (d) the converter-style rating; (e) γ at 100 A.

Solution. (a) $X_C = V_{C_{max}}/I_{min} = 50 \Omega$.

- (b) $63.66 \mu\text{F}$.
- (c) $400 \times 50 F(\gamma) = 5000$, $F = 0.25$: $\gamma = 39.40^\circ$.
- (d) Fundamental apparent rating: $V_{CF,peak} I_{peak}/2 = 1.0$ MVA per phase. This is not the complete capacitor or valve thermal/voltage rating; use the chopped-waveform RMS and peak values as well.
- (e) 0° , full insertion.

Note. At maximum current the capacitor is bypassed for most of each half-cycle, so the valve conduction loss is highest exactly where the inserted reactance is smallest.

Example 7.24 GCSC design for a 400 kV line

A GCSC is to provide 10%–40% compensation of an 80Ω corridor. **Case A:** equipment duty is evaluated at a prescribed line-current peak of 1000 A. Find (a) X_C and C ; (b) γ for 10%; (c) the fundamental rating at full insertion; (d) maximum capacitor voltage; and (f) third harmonic at 10%. **Case B:** independently, equal 400 kV terminals and $\delta = 30^\circ$ are imposed. Find (e) the ideal power and resulting currents at 0%, 10% and 40%; explain why these are not Case A ratings.

Solution. (a) $X_C = 32 \Omega$; $C = 99.47 \mu\text{F}$.

- (b) $F = 0.25$: $\gamma = 39.40^\circ$.
- (c) Case A: $I_{rms} = 1000/\sqrt{2} = 707.11$ A. At full insertion, $Q_C = I_{rms}^2 X_C = 16.0$ Mvar per phase, or 48.0 Mvar three-phase.
- (d) 32 kV, reached at full insertion ($I X_C$).

- (e) Case B: ideal powers are 1000, 1111.1 and 1666.7 MW. The corresponding line RMS currents are 1494.29, 1660.32 and 2490.49 A, all above Case A. At 40% the capacitor RMS voltage is $2490.49(32)=79.70$ kV, its peak is 112.71 kV and its three-phase duty is 595.44 Mvar. Thus Case A equipment cannot be declared adequate for Case B. At partial insertion, the chopped waveform also needs separate peak/RMS stress checks.
- (f) Case A at 10%: $\gamma=39.40^\circ$, third-harmonic peak ratio $F_3=0.1243$ on $I_{peak}X_C$. Hence $V_{C3,rms}=1000(32)(0.1243)/\sqrt{2}=2.81$ kV. The fundamental RMS is $1000(8)/\sqrt{2}=5.66$ kV, so the third harmonic is about 49.7% of that fundamental.

Note. At 10% compensation the third harmonic is about half of the fundamental capacitor voltage, which is why single-module phase control is avoided and sequential modular control is used.

Example 7.25 Sequential control of a GCSC

A GCSC bank of total bare-capacitor reactance $3.0\ \Omega$ carries 400 A peak. Compare (a) four equal modules and their fundamental ratings; (b) a **single unsegmented** $3.0\ \Omega$ bank controlled at $\gamma=45^\circ$; (c) a four-module sequential realization of the same effective reactance; (d) third-harmonic RMS voltages.

Solution. (a) $0.75\ \Omega$; $(400/\sqrt{2})^2 \times 0.75 = 60$ kvar per module.

(b) $3.0 \times 0.1817 = 0.545\ \Omega$.

(c) No module fully inserted; one module supplies $0.545/0.75 = 72.7\%$ of its reactance, at $\gamma = 12.49^\circ$.

(d) Unsegmented bank: 90.0 V RMS. Four-module sequential control: 18.1 V RMS, a factor of 4.97 lower. This exceeds four for this particular command because the one controlled module operates near full insertion. A lone $0.75\ \Omega$ module at 45° would give only $0.1363\ \Omega$; it is not the comparison in part (b).

Note. The fundamental reactance is the same in both realizations; only the harmonic content and the voltage that each valve must block change.

Example 7.26 TSSC stages and levels

A TSSC has four modules of $5\ \Omega$ and carries 800 A peak. For 0 to 4 modules inserted, find (a) the reactance; (b) the peak voltage; (c) the reactive power.

Solution. (a) 0, 5, 10, 15, 20 Ω .

(b) 0, 4, 8, 12, 16 kV.

(c) $I_{rms}^2 X$: 0, 1.6, 3.2, 4.8, 6.4 Mvar per phase.

Note. The reactive rating grows linearly with the number of inserted stages because every stage carries the same current; the valve current rating does not change.

Example 7.27 TSSC in voltage compensating mode

A TSSC has five $8\ \Omega$ modules. Treat the given line currents, 400–1200 A, and capacitor voltages as RMS fundamental quantities. Find (a) the 16 kV voltage ceiling set by all stages at 400 A; (b) the number of stages chosen to remain at or below that ceiling and the voltages at 400, 600, 800, 1000 and 1200 A.

Solution. (a) $V_{Cmax} = m X_C I_{min} = 16$ kV.

(b) 400 A: 5, 16.0 kV; 600 A: 3, 14.4 kV; 800 A: 2, 12.8 kV; 1000 A: 2, 16.0 kV; 1200 A: 1, 9.6 kV. The stepped approximation leaves gaps of up to 40% at some currents, which is the price of switching.

Note. Equal stages cannot hold the voltage closely over the whole current range; finer or binary-weighted stages narrow the gaps at the cost of more valves.

Example 7.28 TSSC insertion offset and bypass surge

A TSSC stage has $X_C = 12 \Omega$; the line current is 1000 A peak. Find (a) C ; (b) the AC peak capacitor voltage; (c) the maximum voltage after insertion at a current zero; (d) the characteristic impedance with a 0.5 mH valve reactor; (e) the surge current and initial di/dt if bypassed at 8 kV; (f) the remedy.

Solution. (a) 265.3 μF .

(b) 12 kV.

(c) 24 kV, Equation (7.11).

(d) $Z_0 = \sqrt{L_v/C} = 1.373 \Omega$; ring frequency 437 Hz.

(e) $I_{\text{surge}} = 8000/1.373 = 5.83 \text{ kA}$; $di/dt = V/L = 16 \text{ A}/\mu\text{s}$.

(f) Bypass at capacitor voltage zero, and size the valve reactor for the valve's di/dt capability.

Note. The offset doubles the peak capacitor voltage after insertion, and a bypass at non-zero voltage produces a surge of several times the peak line current.

Example 7.29 Power sharing between parallel paths with a TSSC

Two parallel paths share 600 MW: line 1 ($X_1 = 60 \Omega$) carries a TSSC of four 10Ω stages, line 2 has $X_2 = 40 \Omega$. Treating the angle across the corridor as adjusted to keep the total constant, find the sharing for 0 to 4 stages inserted.

Solution. Parallel paths share in inverse proportion to reactance: $P_1 = P_t X_2 / (X_1' + X_2)$, with $X_1' = 60 - 10n$.

$n=0$: 240/360 MW; $n=1$: 266.7/333.3; $n=2$: 300/300; $n=3$: 342.9/257.1; $n=4$: 400/200 MW. Two stages equalize the loading, and the full bank shifts 160 MW onto line 1. This is the power-flow-control use of series devices in meshed networks.

Note. Inserting stages draws power into the compensated path; the same principle underlies modular series compensation for congestion management (Section 7.7.1).

Example 7.30 SSR screening of a TSSC

A four-stage TSSC with 10Ω per stage is on a path of 80Ω total reactance, including generator and transformer. A torsional mode is at 20 Hz. Find electrical and complementary frequencies for 1–4 stages, identify settings with at least 5 Hz frequency separation, and distinguish the maximum permitted steady setting from a contiguous staged operating range.

Solution. For 1–4 stages, the electrical/complementary frequencies are 17.68/32.32, 25.00/25.00, 30.62/19.38 and 35.36/14.64 Hz. Settings 1, 2 and 4 pass the specified 5 Hz frequency-separation screen. The largest steady setting is therefore four stages; if every intermediate sequential setting must also pass, the contiguous range ends at two. Three stages fail the screen. A protected transition to four stages must be evaluated dynamically; brief passage through three stages is not by itself proof of instability or safety.

Note. Frequency separation is a screening requirement. Before the switching protocol is chosen, the unsuitable steady settings are excluded and the transition duration, control, torsional damping and contingencies are assessed.

Example 7.31 Simplified versus exact TCSC reactance

A TCSC has $C = 1000 \mu\text{F}$ and $L = 1.5 \text{ mH}$. Find (a) X_C , X_L and ω ; (b) the exact resonance; (c) the simplified-model resonance; (d) the reactance at $\alpha = 20^\circ$ by both models; (e) at $\alpha = 50^\circ$; (f) at $\alpha = 60^\circ$; (g) the conclusion.

Solution. (a) $X_C = 3.1831 \Omega$, $X_L = 0.4712 \Omega$, $\omega = 2.599$.

(b) $\beta_r = 90^\circ / \omega = 34.63^\circ$, that is $\alpha_r = 55.37^\circ$.

(c) $X_L(\alpha) = X_C$ gives $\alpha_r = 48.21^\circ$.

(d) $\beta = 70^\circ$: simplified +1.108 Ω , exact +0.745 Ω (both inductive).

- (e) $\beta = 40^\circ$: simplified -27.60Ω (capacitive), exact $+5.128 \Omega$ (inductive).
- (f) $\beta = 30^\circ$: simplified -5.214Ω , exact -8.111Ω ; the exact value was confirmed by direct time-domain solution.
- (g) Between 48° and 55° the two models disagree on the *sign* of the reactance. The exact characteristic must be used to set the vernier limits.

Note. The two models disagree on the sign of the reactance between about 48 and 55 degrees, so vernier limits derived from the simplified model are unsafe.

Example 7.32 Current division in the TCSC

In the simplified model, the controlled reactor has $X_L(\alpha) = 1.5 X_C$. Find (a) X_{TCSC}/X_C ; (b) the mode; (c) I_{TCR}/I ; (d) I_C/I ; (e) the exact half conduction angle giving the same boost for $X_L/X_C = 0.15$.

Solution. (a) $1.5/(1 - 1.5) = -3$.

(b) Capacitive, boost factor 3.

(c) $I_{TCR}/I = X_C/(X_C - X_L(\alpha)) = -2$: the reactor current is twice the line current and opposite in direction.

(d) $I_C = I - I_{TCR}$, so $|I_C| = 3I$.

(e) Equation (7.16) with $X_L = 0.15 X_C$: $\beta = 30.99^\circ$ ($\alpha = 59.0^\circ$), close to resonance at 34.86° . A boost of 3 is therefore an aggressive operating point.

Note. A boost of 3 leaves only about 4 degrees of margin to resonance for this reactor; practical boost limits are set with a wider margin.

Example 7.33 TCSC design for a 220 kV line

A 220 kV line ($X = 50 \Omega$) carrying 1200 A RMS needs controllable compensation from 20% to 40%. With the TCSC capacitor equal to the minimum compensation ($X_C = 10 \Omega$, blocked mode), find (a) the required boost; (b) C ; (c) the angle for 40% for $X_L/X_C = 0.1, 0.15$ and 0.2 ; (d) L in each case; (e) the resonance margin; (f) the fundamental voltage and rating at maximum compensation; (g) the peak capacitor voltage and thyristor current for $X_L = 1.5 \Omega$.

Solution. (a) $20/10 = 2$.

(b) $318.3 \mu\text{F}$.

(c) Equation (7.16) $\hat{i} - 20 \Omega$: $\beta = 23.90^\circ, 28.53^\circ$ and 32.32° .

(d) 3.18, 4.77 and 6.37 mH.

(e) The first poles are at $28.46^\circ, 34.86^\circ$ and 40.25° , leaving margins of $4.6^\circ, 6.3^\circ$ and 7.9° . **At the same boost 2**, branch RMS currents are 1852, 1707 and 1614 A; capacitor-voltage THD is 16.20%, 14.77% and 13.63%. Thus a larger reactor reduces these current and harmonic duties while increasing angular margin and reversal duration. For ratio 0.10 the second formal pole is near 85.4° ; transitions to bypass need validated protective mode logic, not an instruction to sweep quickly through it.

(f) $V_1 = 1200 \times 20 = 24 \text{ kV RMS}$; $Q = I^2 \times 20 = 28.8 \text{ Mvar per phase}$.

(g) For $X_L/X_C = 0.15$, the exact waveform gives capacitor peak 30.62 kV, valve-branch RMS 1706.6 A, individual-thyristor RMS $1706.6/\sqrt{2} = 1206.7 \text{ A}$, and pulse peak 4.24 kA. The line peak is 1.70 kA. Both devices share the branch waveform on alternate half-cycles.

Note. The branch RMS exceeds line RMS by 42.2%, while one thyristor's RMS is about 1207 A. Their peak pulse current is the same 4.24 kA. Use the complete waveform and thermal duty for device selection.

Example 7.34 TCSC reactance for constant power during a swing

A 500 kV line ($X = 100 \Omega$) with a TCSC at -15Ω operates at $\delta = 30^\circ$. Find (a) the net reactance; (b) the power; (c) the net reactance that holds the power if δ rises to 35° ; (d) the new TCSC setting.

Solution. (a) $X_{net} = 85 \Omega$. (b) $P = 1470.6$ MW. (c) Holding that power at 35° requires $X'_{net} = 500^2 \sin 35^\circ / P = 97.51 \Omega$. (d) $X'_{TCSC} = -2.49 \Omega$. This is an ideal **system-level reactance order**. A single TCSC with a larger bare-capacitor reactance cannot necessarily realize it; bank segmentation or another actuator is needed if the order lies in its forbidden gap. Power-flow regulation reduces compensation in this calculation. A POD channel instead seeks an electrical-power component in phase with speed deviation, with sign and phase determined from the modal sensitivity.

Note. The numerical reactance target alone does not establish that a specified TCSC bank can realize it or that a damping controller should oppose the power-flow loop at every instant. Define actuator range, bandwidth and priority.

Example 7.35 SSR and the TCSC

A turbine-generator feeds a line of total reactance 60Ω compensated 45% by a fixed capacitor. Find (a) X_C ; (b) f_e and the rotor frequency; (c) the reactances of a TCSC module ($X_C = 10 \Omega$, $X_L = 1.333 \Omega$ at 50 Hz) at 33 Hz; (d) the qualitative effect of replacing part of the fixed bank by that TCSC with capacitor-voltage-synchronized firing.

Solution. (a) 27Ω .

(b) $f_e = 33.54$ Hz; $f_0 - f_e = 16.46$ Hz.

(c) $X_C(33) = 15.15 \Omega$, $X_L(33) = 0.88 \Omega$.

(d) These component reactances do not determine the switched module's subsynchronous terminal impedance. Suitable capacitor-voltage-synchronized control can suppress capacitor-like response and add damping, as demonstrated for particular designs; the replacement must be checked using its closed-loop frequency-dependent or EMT model.

Note. A firing-reference choice is not sufficient evidence of SSR neutrality; operating band, filters, phase coupling and the connected machines determine the result.

Example 7.36 TCSC currents and losses

A TCSC with $X_C = 5 \Omega$, $X_L = 0.667 \Omega$ and lumped valve-reactor branch resistance 0.05Ω carries 800 A RMS at $\beta = 25^\circ$. Use the lossless exact waveform to estimate (a) resonance; (b) fundamental reactance; (c) fundamental and total RMS voltage; (d) branch, individual-thyristor and capacitor RMS currents; (e) branch resistive loss; (f) loss as a fraction of fundamental module rating. The resistance is a perturbative loss estimate, not included in the waveform solution.

Solution. (a) 32.87° .

(b) -8.00Ω (boost 1.6).

(c) 6.40 kV RMS; total RMS 6.44 kV.

(d) $I_{T,branch,rms} = 723.45$ A; $I_{T,device,rms} = 511.55$ A; pulse peak = 1.915 kA. Capacitor RMS current is 1389.41 A. In symmetric operation each thyristor carries one pulse per full cycle, giving the factor $1/\sqrt{2}$ in its RMS rating.

(e) Branch loss $\dot{I}_{T,branch,rms}^2 R = 723.45^2 (0.05) = 26.17$ kW per phase. The lumped resistance represents the stated branch loss only; a physical valve-loss model also uses on-state voltage, average current and switching/recovery duty.

(f) 5.12 Mvar rating per phase: losses 0.51%. Using the simplified model, which ignores the large reversal current pulses, would underestimate the reactor current.

Note. Losses of about 0.5% of the module rating are modest, but they rise steeply as the boost approaches resonance.

Example 7.37 TCSC in the admittance matrix

A 400 kV line of 60Ω connects buses k and m with $\theta_k - \theta_m = 10^\circ$ and 1.0 pu voltages; base 100 MVA. A TCSC at -20Ω is added. Find (a) the line reactance in pu; (b) the branch admittance without and with the TCSC; (c) B_{kk} and B_{km} ; (d) the flows; (e) the equivalent power injections.

Solution. (a) $Z_B = 1600 \Omega$: 0.0375 pu, TCSC -0.0125 pu.

(b) $-j26.67$ pu and $-j40.0$ pu.

(c) With the TCSC, $B_{kk} = -40$ and $B_{km} = +40$ pu.

(d) $P = \sin 10^\circ / X$: 463.1 MW and 694.6 MW.

(e) With the original line admittance retained, define equivalent **fictitious bus injections** as the negative of the incremental compensated-branch power drawn at each terminal. They are $P_k^{inj} = -231.5$ MW and $P_m^{inj} = +231.5$ MW, with $Q_k^{inj} = Q_m^{inj} = -20.26$ Mvar. These are load-flow bookkeeping terms, not energy sources in the lossless TCSC. Actual sending-terminal branch power increases by 231.5 MW; state the injection convention when assembling the bus mismatch.

Note. The power-injection form is convenient in Newton-Raphson iterations because the admittance matrix stays constant while the TCSC setting changes.

Example 7.38 SSSC power and equivalent capacitor

A two-machine system has $V_s = V_r = 1$ pu, $X = 1$ pu and an SSSC with $V_q = 0.6$ pu at $\delta = 20^\circ$. Find (a) the uncompensated power; (b) the power with the SSSC; (c) the current; (d) the capacitor giving the same power at this angle; (e) the power at $\delta = 0$.

Solution. (a) 0.342 pu. (b) Equation (7.20): $0.342 + 0.6 \cos 10^\circ = 0.933$ pu. (c) Equation (7.19): 0.947 pu. (d) $X_C = V_q / I = 0.633$ pu, $k = 63\%$. (e) 0.6 pu with the SSSC, but zero with any capacitor.

Note. The SSSC contribution is largest at small angles, exactly where a series capacitor contributes least.

Example 7.39 SSSC range and power reversal

$V = 1$ pu, $X = 1.2$ pu, $\delta = 40^\circ$, $V_{qmax} = 0.5$ pu. Find (a) the uncompensated power; (b) the maximum power; (c) the minimum power; (d) the percentage control range; (e) the angle below which power reverses at $V_q = -0.5$ pu.

Solution. $P_0 = 0.536$ pu; $P_{max} = 0.927$ pu; $P_{min} = 0.144$ pu, a symmetric change of about 73% of P_0 . For the fixed-axis order $V_q = -0.5$ pu, reversal occurs when $2 \sin(\delta/2) < 0.5$, hence $\delta < 28.96^\circ$. At the boundary current is zero. Below it, the phasor has reversed, so the voltage's current-relative mode label and controller synchronization must be reconsidered.

Note. The ideal voltage-source equation permits reversal at small angles; a practical current-synchronized SSSC needs a defined transition around zero current and sufficient voltage, current and DC-energy capability.

Example 7.40 SSSC on a 400 kV line

A 400 kV line ($X = 100 \Omega$) operates at 45° ; the SSSC injects up to 30 kV (line-to-line). Find (a) P_0 ; (b) the SSSC term; (c) the power range; (d) the current with full capacitive injection; (e) the SSSC MVA rating.

Solution. (a) 1131.4 MW. (b) $400 \times 30 \times \cos 22.5^\circ / 100 = 110.9$ MW, not 1108.7 MW. (c) 1020.5 to 1242.2 MW. (d) 1940.7 A (against 1767.5 A without injection). (e) $3(30/\sqrt{3})(1.941) = 100.8$ MVA.

Note. With line-to-line kilovolts and ohms the SSSC term is already in megawatts; no further unit factor is needed.

Example 7.41 SSSC with storage compensating R and X

A 275 kV line has $X = 75 \Omega$, $R = 10 \Omega$ and operates at $\delta = 35^\circ$. An SSSC with storage must raise power by 25% while keeping $X/R = 7.5$. Find (a) the base power, reactive power and current; (b) the target power; (c) R_{eff} and X_{eff} ; (d) the new current; (e) the voltage components; (f) the MVA rating and the power drawn from storage.

Solution. (a) Using forward sending-to-receiving current, $S_R = 3 \underline{V}_r \underline{I}^i = 544.4 - j254.9$ MVA. Thus receiving-end active power is 544.4 MW and **received** reactive power is -254.9 Mvar; equivalently the receiving bus supplies 254.9 Mvar into the line. Line current magnitude is 1262 A. The sign depends on the declared terminal-power direction, not on taking an absolute value.

(b) 680.5 MW.

(c) Solving with $X_{eff} = 7.5 R_{eff}$: $R_{eff} = 8.0 \Omega$, $X_{eff} = 60 \Omega$.

(d) 1577 A.

(e) $V_q = (75 - 60) \times 1577 = 23.66$ kV per phase; $V_p = (10 - 8) \times 1577 = 3.15$ kV per phase.

(f) $3 \times 23.87 \times 1.577 = 113.0$ MVA; the storage supplies $3 V_p I = 14.9$ MW, which is the share of the line's $I^2 R$ loss now carried by the SSSC.

Note. The 14.9 MW supplied by the store is continuous, so it must come from a real energy source rather than from the DC capacitor.

Example 7.42 SSSC in impedance compensating mode

A 500 kV line of 200Ω is to behave as 150Ω ; $V_{qmax} = 80$ kV line-to-line. Find (a) the reactance to emulate; (b) the maximum current for which this holds; (c) the MVA rating; (d) the current at $\delta = 30^\circ$; (e) the largest angle at which the emulation holds.

Solution. (a) $X_q = 50 \Omega$. (b) The voltage ceiling allows emulation up to $I = 46.19 \text{ kV} / 50 \Omega = 923.76$ A. (c) Fundamental rating there is 128 MVA. (d) At 30° , unconstrained emulation would require 996.19 A and exceeds the voltage ceiling. Holding 80 kV line-to-line instead gives the consistent saturated current $I = [2 V_{ph} \sin 15^\circ + 80 \text{ kV} / \sqrt{3}] / 200 \Omega = 978.09$ A. This also requires adequate converter current capability. (e) Emulation remains below its voltage ceiling for $\delta \leq 27.77^\circ$.

Note. The 923.76 A value is the constant-reactance emulation boundary imposed by voltage capability; it need not be a separate hardware current limit. Both limits must be declared.

Example 7.43 Capacitor versus SSSC for a 50% increase

A 110 kV line ($X = 40 \Omega$) operates at 45° . Compare a fixed capacitor and an SSSC that each raise power by 50% at this angle: (a) P_0 ; (b) the capacitor reactance; (c) its rating; (d) the SSSC voltage; (e) its rating; (f) the behaviour at other angles.

Solution. (a) $P_0 = 213.9$ MW. (b) $k = 1/3$, $X_C = 13.33 \Omega$. (c) The compensated current is 1822.8 A and the capacitor duty 132.9 Mvar. (d) The required SSSC voltage is 42.10 kV line-to-line. (e) Its fundamental rating is $\sqrt{3}(42.10)(1.8228) = 132.9$ MVA, equal at this operating point because both devices produce the same fundamental voltage and current. (f) At smaller angles a constant-voltage SSSC contributes relatively more, and it can also command the opposite polarity. Which solution is more economical depends on installed equipment, losses, protection and service value. A hybrid has to be sized separately.

Note. Equal fundamental ratings at one operating point do not imply equal harmonic, transient, control or lifecycle performance. The SSSC removes the literal fixed-capacitor resonance mechanism but still has controlled impedance and interaction limits.

Numerical consistency note. All SSSC examples use three-phase quantities. With line-to-line RMS voltage and line current the converter rating is $S = \sqrt{3} V_{LL} I$; with per-phase voltage it is $S = 3 V_{ph} I$. Per-unit calculations use one declared three-phase base and need no additional $\sqrt{3}$.

Unsolved Practice Problems

P7.1 A 345 kV line of $100\ \Omega$ operates at 35° . Find the degree of compensation for a 40% increase in power, the capacitor reactance, the new current and the three-phase capacitor rating.

P7.2 A path of $70\ \Omega$ total reactance is compensated 35%. Find the capacitor reactance, the electrical and complementary frequencies, and the compensation for exact frequency coincidence with a 25 Hz torsional mode. Explain why this is not a complete excitation test.

P7.3 With $P_m = 0.7$ pu, $P_{max} = 1.2$ pu (uncompensated) or 1.6 pu (compensated) and a fault curve of 0.3 pu, find the initial angle and critical clearing angle in both cases.

P7.4 A GCSC has $C = 800\ \mu\text{F}$. Find X_C , the reactance at $\gamma = 30^\circ$ and 60° , and the delay for $1.0\ \Omega$.

P7.5 A GCSC with $C = 600\ \mu\text{F}$ carries 300 A peak at $\gamma = 40^\circ$. Find the fundamental RMS voltage, the total RMS voltage, the third harmonic RMS voltage and the THD.

P7.6 A GCSC must hold 8 kV (peak) from 200 A to 600 A peak. Find X_C , C and γ at 600 A.

P7.7 A TSSC has six $4\ \Omega$ stages and carries 1000 A RMS. Find the maximum voltage and reactive power. In voltage compensating mode with $V_{Cmax} = 18$ kV, find the stages in service at 750, 900 and 1500 A.

P7.8 A TSSC stage ($X_C = 10\ \Omega$) carries 1500 A peak. Find the maximum voltage after insertion at a current zero with a discharged capacitor, and the surge current if it is bypassed at 6 kV through 0.3 mH.

P7.9 A TCSC has $X_C = 12\ \Omega$ and $X_L = 1.6\ \Omega$. Find the resonance (β_r and α_r), the exact reactance and boost at $\beta = 20^\circ$, and the exact reactance at $\beta = 60^\circ$.

P7.10 For the TCSC of Problem P7.9, compare simplified and exact reactance at $\beta = 60^\circ$ ($\alpha = 30^\circ$).

P7.11 A line of $60\ \Omega$ needs 15% to 30% controllable compensation. With $X_C = 9\ \Omega$ and $X_L/X_C = 0.15$, find the angle for 30%, C , L and the resonance angle.

P7.12 An SSSC on a line with $V = 1$ pu, $X = 0.9$ pu, $\delta = 30^\circ$ injects ± 0.3 pu. Find the power range and the currents.

P7.13 A 500 kV line of $150\ \Omega$ at 40° has an SSSC of ± 50 kV (line-to-line). Find the power range.

P7.14 An SSSC must make a 1.0 pu line behave as 0.75 pu with $V_{qmax} = 0.3$ pu. Find the current limit of the emulation and the rating at 1.2 pu current.

P7.15 A 400 kV, $60\ \Omega$ line between buses at 10° apart receives a TCSC of $-12\ \Omega$. Find the flows before and after (100 MVA base) and the percentage increase.

P7.16 Compare a TSSC, a TCSC and an SSSC for a meshed 400 kV network where flows must be both raised and lowered, a nearby thermal plant has a 22 Hz torsional mode, and the site is space-limited.

Answer Key (Unsolved Problems)

P7.1 $k = 0.2857$; $28.57\ \Omega$; 1677 A; 241.1 Mvar.

P7.2 $24.5\ \Omega$; 29.58 Hz; 20.42 Hz; $k = 25\%$.

P7.3 Uncompensated 35.69° , 83.06° ; compensated 25.94° , 96.33° .

P7.4 $3.979\ \Omega$; $1.556\ \Omega$; $0.229\ \Omega$; 39.30° .

P7.5 272.4 V; 306.6 V; 138.0 V; 51.6%.

P7.6 $40\ \Omega$; $79.58\ \mu\text{F}$; 33.59° .

P7.7 24 kV; 24 Mvar per phase; 6, 5 and 3 stages (18 kV each).

P7.8 30 kV; $Z_0 = 0.971\ \Omega$, 6.18 kA.

P7.9 $\beta_r = 32.86^\circ$, $\alpha_r = 57.14^\circ$; -14.51Ω (boost 1.21); $+2.74 \Omega$.

P7.10 Simplified $+6.21 \Omega$; exact $+2.74 \Omega$.

P7.11 $\beta = 28.53^\circ$ ($\alpha = 61.47^\circ$); $353.7 \mu\text{F}$; 4.30 mH ; $\beta_r = 34.86^\circ$.

P7.12 0.556 pu uncompensated; 0.878 to 0.234 pu ; 0.908 and 0.242 pu .

P7.13 1071.3 MW ; 914.7 to 1227.9 MW .

P7.14 1.2 pu ; 0.36 pu .

P7.15 463.1 MW ; 578.8 MW ; 25% .

P7.16 Open-ended. Compare bidirectional voltage range, physical capacitor resonance, closed-loop SSR/SSCI behavior, space, losses and fault bypass. The SSSC is a strong candidate for bidirectional routing; a TCSC may also be feasible with suitable bank range and validated damping. A TSSC retains capacitor-related risk, but its suitability cannot be decided from the 22 Hz mode alone without network and damping data. None of the three should be declared automatically safe.

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